Martingales in Financial Mathematics



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These lecture notes are based on the *Martingales in Financial Mathematics* course given at EPFL for the academic year 2024/2025 by Prof. Michael Schmutz.

Chapter 1

Classical Conventions and Deterministic Interest Rates

Let r be the interest rates per annum and C_0 be the initial capital at $t=0$. If compounded once per annum then of course:

$$C_1 = C_0 + C_0 r = C_0(1+r). \tag{1.1}$$

If m-times compounded we obtain:

$$C_1 = C_0 \prod_{i=1}^{m} \left(1+\frac{r_m}{m}\right) = C_0 \left(1+\frac{r_m}{m}\right)^m. \tag{1.2}$$

Thus in generality if we compound it n times then $C_n = C_0\left(1+\frac{r_m}{m}\right)^{mn}$. And thus, if we let $m \to \infty$ we obtain the following continuous compounding:

$$\left(1+\frac{r_c}{m}\right)^m \xrightarrow{m\to\infty} e^{r_c}. \tag{1.3}$$

In this case, $C_t = C_0 e^{r_c t}$ solves the differential equation

$$\frac{dC_t}{dt} = r_c \frac{C_0 e^{r_c t}}{C_t} = r_c C_t. \tag{1.4}$$

We note that there are also other classical conventions, however if the interest rate convention is clear from the context we often omit the subscript. Finally remark that we can obtain price change factors, given by:

$$e^{r_c} = \left(1+\frac{r_m}{m}\right)^m \Longrightarrow \begin{Bmatrix} r_c = m\log\left(1+\frac{r_m}{m}\right) \\ r_m = m\left(e^{\frac{r_c}{m}}-1\right) \end{Bmatrix} - \text{equivalent interest rates.}$$

Example 1.0.1. *If we use the same nominal interest rate, e.g. 5%, then we obtain the following scenario:*
- Annual compounding given by $100(1+0.05) = 105$.
- Continuous compounding given by $100e^{0.05} \approx 105.127 > 105$.

Chapter 2

Introduction

In general, investors can invest in various financial products comprising of bonds (compensation is usually given by a fixed coupon), shares (compensation is given by dividends), or diverse hybrid instruments. Note that fixed income instruments are also issued by the Federal Government and are the so-called government bonds. In Switzerland, Swiss government bonds may be considered as being risk-free (i.e. free of counterparty risk).

As a first classical approach of valuation, the Net Present Value (NPV) is a rather popular method for valuation of real estate and for analysis of investment opportunities, but it is also sometimes used for bond valuation. Its formulation is as follows:

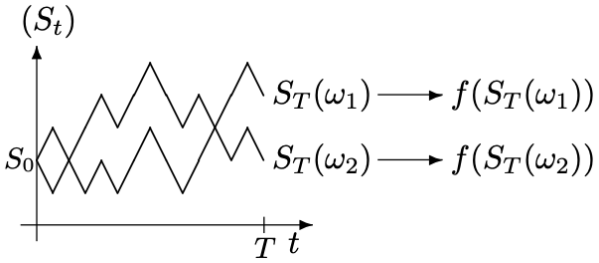
$$NPV = \sum_{t=0}^{N} \frac{CF_t}{(1+i_t)^t}, \tag{2.1}$$

where t is the date of the corresponding cash flow (CF), i_t is the interest rate reflecting the cost of tying up capital/risks, and CF_t is the net cash flow at t.



Definition 2.0.1. *A **financial derivative** is an instrument whose value depends on, or is derived from, the value of another asset (aka underlying assets or underlying(s) for short).*

Types of underlyings comprise shares, indices, currencies, interest rates, bonds, commodities, electricity, etc... Mathematically speaking, we can often define a derivative by a function $f : \mathbb{R}_+ \longrightarrow \mathbb{R}$ on a quote S_T in the future, or by a functional F on the price process.



Definition 2.0.2. *The **payoff** of the derivative at maturity $T > 0$ is given by $f(S_T)$, or in terms of the functional F by $F((S_t)_{t\in[0,T]})$.*

Example 2.0.3 (Important types of payoffs)**.** *Typical examples of payoffs are:*
(i) A forward payoff is $f(S_T(\omega)) = S_T(\omega) - k$.
(ii) The payoff of a European call option is $f(S_T(\omega)) = \max(S_T(\omega) - k, 0)$, for $k \geq 0$.
(iii) The payoff of a European put option is $f(S_T(\omega)) = \max(k - S_T(\omega), 0)$, for $k \geq 0$.

Now, if we have payoffs of the form $f(S_T(\omega))$ where $\Omega : \xrightarrow{S_T} \mathbb{R}_+ : \xrightarrow{f} \mathbb{R}$, then we have that $S_T(\omega)$ and $f(S_T(\Omega))$ depend on the same $\omega \in \Omega$, where the payoff of the derivative is given by a deterministic function on S_T. Thus the price of the derivative and the price of the underlying asset should be in a certain relation.
Let us now think about how we could approach valuation for financial derivatives. The crucial point for this is to exclude all arbitrage opportunities. Roughly speaking, an arbitrage opportunity occurs when one can make money from nothing without risks. Hence the general idea for deriving a price of a new product is to fix it such that no arbitrage opportunity is created. We will call this process an exclusion of arbitrage or also the no-arbitrage assumption (NA assumption).

Example 2.0.4 (Forward Price)**.** *Consider the payoff at maturity $T > 0$ to be $f(S_T) = S_T - k$. At $t = 0$ fix $k = F_{0,T}$ such that no premium is the price/payment at $t = 0$ (i.e. there are no cash flows at $t = 0$ in a usual situation, although other situations may be considered). We assume that the underlying asset is given by a non-dividend paying share (until maturity $T > 0$). Furthermore, we assume that the risk-free interest rate r is constant.*

Assume $F_{0,T} > S_0 e^{rT}$.

Then follow the following strategy:

At $t = 0$	CF
(1) Borrow money	S_0
(2) Buy spot the (cheap) share	$-S_0$
(3) Short the forward (short forward)	0
Total Cash flows at $t = 0$	0

At $t = T$	CF
Sell (2) for $F_{0,T}$ (based on (3))	$F_{0,T}$
Pay back the money (credit) incl. interest payment (1)	$-S_0 e^{rT}$
Total cash flow at $t = T$	$F_{0,T} - S_0 e^{rT}$

Since $F_{0,T} - S_0 e^{rT} > 0$ we have that this is an arbitrage strategy.

Assume that $F_{0,T} < S_0 e^{rT}$.

Then follow the following strategy:

At $t = 0$	CF
(1) Sell the (expensive) share (e.g. borrowed from a pension found)	S_0
(2) Pay the money into an account	$-S_0$
(3) Enter into a long position in the forward (long forward)	0
Total cash flow at $t = 0$	0

At $t = T$	CF
Close your bank account (2)	$S_0 e^{rT}$
Buy with the help of (3) a share for $F_{0,T}$ and settle (1)	$-F_{0,T}$
Total cash flow at $t = T$	$S_0 e^{rT} - F_{0,T}$

Since $S_0 e^{rT} - F_{0,T} > 0$ we observe that this is an arbitrage opportunity.

Thus for $F_{0,T} < S_0 e^{rT}$ and $F_{0,T} > S_0 e^{rT}$ we have an arbitrage, and the only option for a no-arbitrage opportunity is $F_{0,T} = S_0 e^{rT}$.

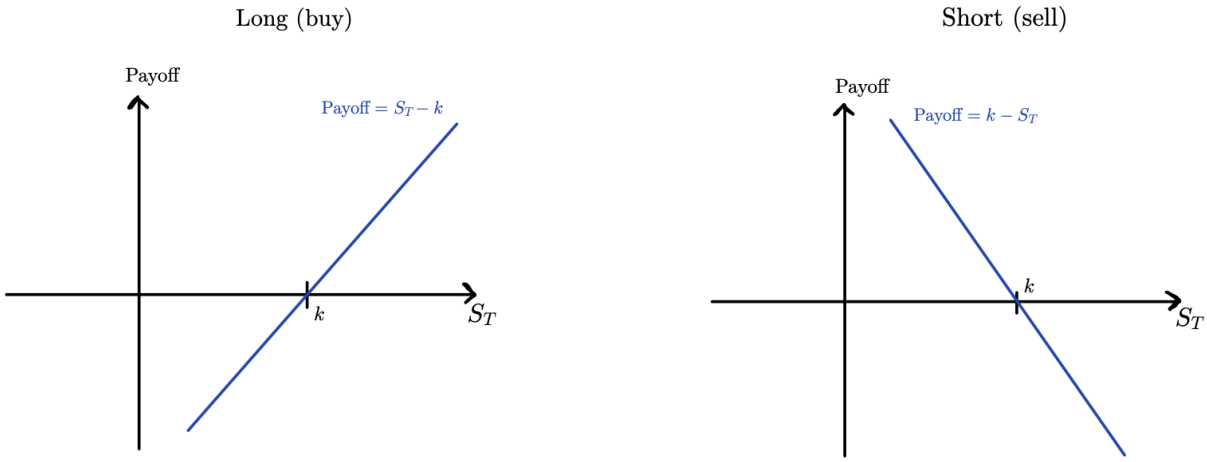
In the above example, if r is the expected yield of the share under $\mathbb{Q}$ (which we assume it exists) and if we consider the price process as being a process indexed by $t \in \{0, T\}$, we have:
(i) $\mathbb{E}_{\mathbb{Q}}(S_T) = S_0 e^{rT}$. This will not be the "real-world" measure otherwise risk-neutral investors would be needed.
(ii) $e^{-rT}\mathbb{E}_{\mathbb{Q}}(S_T - F_{0,T}) = 0$ is the price to be paid at $t = 0$ (NPV-philosophy).
(iii) $\mathbb{E}_{\mathbb{Q}}(S_T/e^{rT}) = S_0$ implies that the discounted price process is a martingale.

There is a general translation of the economic assumption of NA in the language of mathematics. Very roughly speaking, the Fundamental Theorem of Asset Pricing states that a model of a financial market is free of arbitrage if and only if there is a probability measure $\mathbb{Q}$, equivalent to the real-world measure $\mathbb{P}$ (i.e. $\mathbb{P}$ and $\mathbb{Q}$ vanish on the same events), such that the discounted asset price processes are martingales under $\mathbb{Q}$. The measure $\mathbb{Q}$ is called the equivalent martingale measure. In this case, taking discounted expectations with respect to $\mathbb{Q}$ in order to price contingent claims yields arbitrage-free rules where $\mathbb{Q}$ runs through all equivalent martingale measures; i.e.:

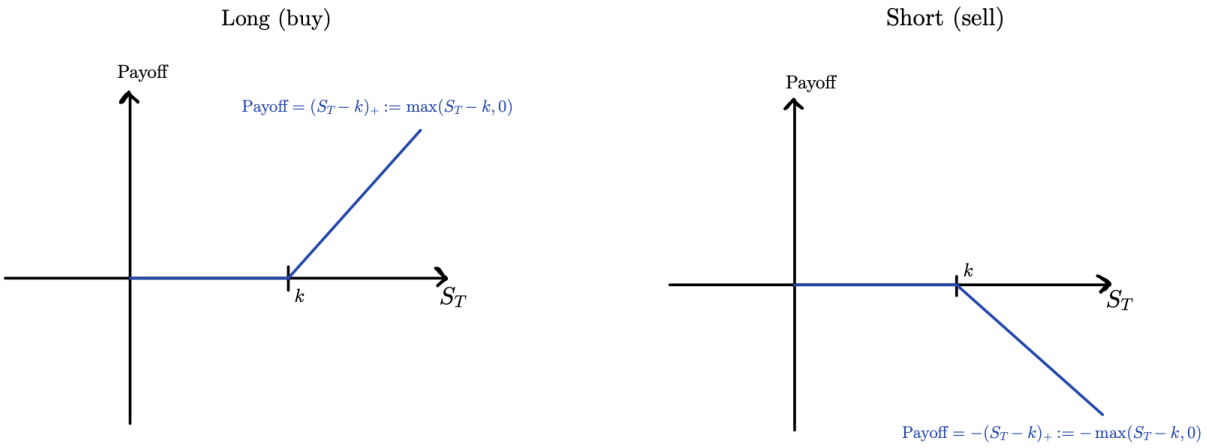
$$P_0^{\mathbb{Q}} = e^{-rT}\mathbb{E}_{\mathbb{Q}}(f(S_T)), \quad \text{for a } \mathbb{Q}\text{-equivalent martingale measure.} \tag{2.2}$$

Note that in general, $\mathbb{Q}$ is often not unique, where this approach is often called risk-neutral valuation. Let us now formally define a European option. The general idea is that a European option may be exercised at maturity, in contrast to an American option that may be exercised until maturity.

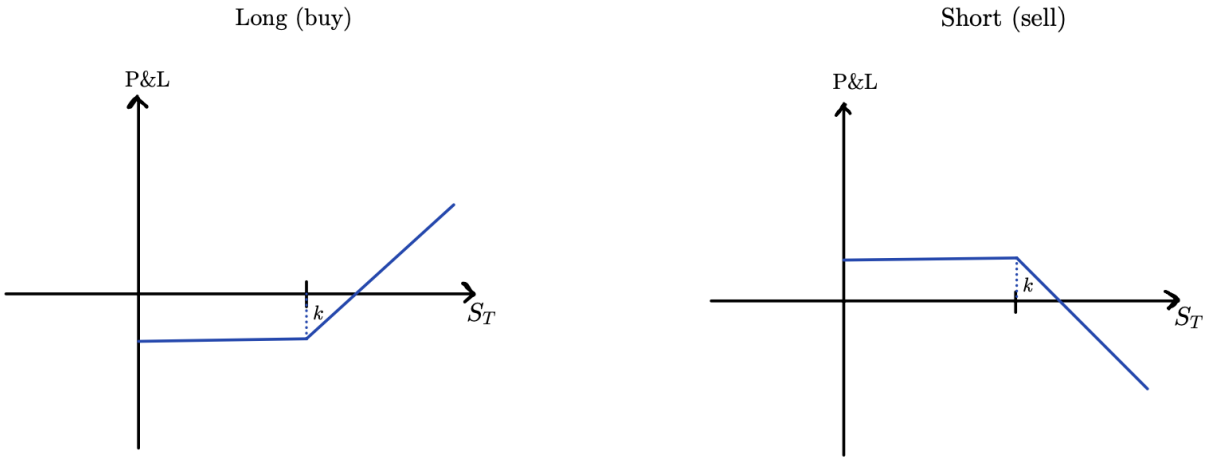
Definition 2.0.5. *A **forward contract** is one in which an investor has the right and the obligation to buy the asset (S_t) at maturity T for the price k.*



Definition 2.0.6. *A **European call option** is one in which an investor has the right but not the obligation to buy the asset (S_t) at maturity T for the price k.*

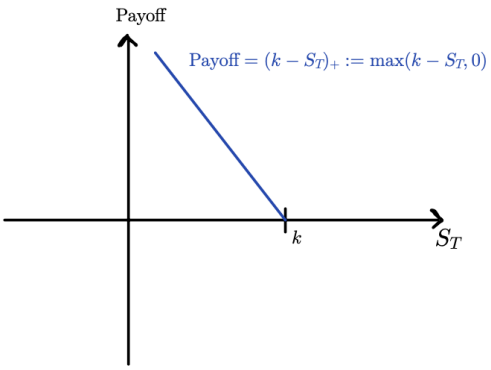


For options, a premium is paid at $t = 0$ (the beginning of the contract). Premium and payoff result in the profit and losses (P& L).

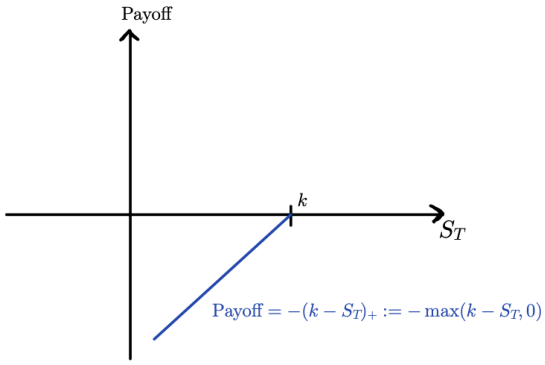


Definition 2.0.7. *A **European put option** is one in which an investor has the right but not the obligation to sell the asset (S_t) at maturity T for the price k.*

Long (buy)

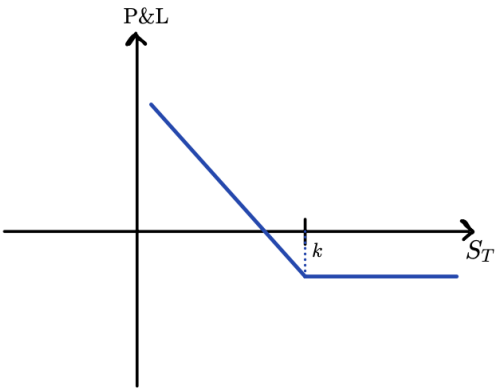


Short (sell)

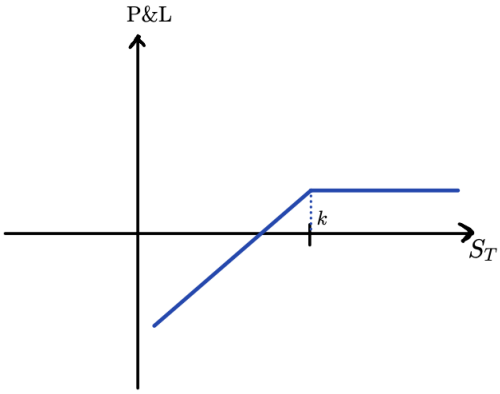


Also here, premium and payoff result in the profit and losses (P& L).

Long (buy)



Short (sell)



Perfect market assumption

Market structure characterized by a very large number of buyers and sellers of a homogeneous (non-differentiated) product. Entry and exit from the industry is costless, or nearly so. Information is freely available to all market participants. The following concrete simplifying assumptions are often used:

(1) No transaction costs or taxes, no bid-ask spread.

(2) No limitations on the quantities of transactions, securities are perfectly divisible and there are no short-selling restrictions.

(3) The market is liquid.

(4) The risk-free lending and borrowing rates are the same.

Chapter 3

Fundamentals of Stochastic Processes

In this chapter, we present some important concepts in stochastic processes. In particular, we will go through the theory of martingales for discrete-time models and through stochastic calculus for continuous-time models.

3.1 Martingales

Definition 3.1.1. *A sequence $\phi = (\phi_n^0, ..., \phi_n^d)_{n=1,...,N}$ is said to be **predictable** if ϕ_j^i is $\mathcal{F}_{j-1}$-measurable for all $i, j \geq 1$ and ϕ_0^i is $\mathcal{F}_0$-measurable for all i.*

Definition 3.1.2. *An adapted sequence $\{M_n\}_{n=0,...,N}$ of real-valued random variables is a **martingale** if $M_n = \mathbb{E}[M_{n+1}|\mathcal{F}_n]$ for all n.*

Definition 3.1.3. *An adapted sequence $\{M_n\}_{n=0,...,N}$ of real-valued random variables is a **submartingale** if $M_n \leq \mathbb{E}[M_{n+1}|\mathcal{F}_n]$ for all n.*

Definition 3.1.4. *An adapted sequence $\{M_n\}_{n=0,...,N}$ of real-valued random variables is a **supermartingale** if $M_n \geq \mathbb{E}[M_{n+1}|\mathcal{F}_n]$ for all n.*

Remark 3.1.5. *An $\mathbb{R}^d$-valued sequence of random vectors is a martingale if each component is a real-valued martingale.*

Lemma 3.1.6. *$(M_n)_{0 \leq n \leq N}$ is a martingale if and only if $\mathbb{E}[M_{n+j}|\mathcal{F}_n] = M_n$ for all j, n.*

Proof. ($\Longrightarrow$) We simply use the Tower property and find:

$$\mathbb{E}[M_{n+j}|\mathcal{F}_n] = \mathbb{E}[\mathbb{E}[...\underbrace{\mathbb{E}[M_{j+n}|\mathcal{F}_{n+j-1}]}_{=M_{n+j-1}}|...|\mathcal{F}_n] = ... = \mathbb{E}[M_{n+1}|\mathcal{F}_n] = M_n. \tag{3.1}$$

($\Longleftarrow$) Choosing $j = 1$, we obtain the classical definition of a martingale (3.1.2). □

Lemma 3.1.7. *If $(M_n)_{0\leq n\leq N}$ is a martingale then we have $\mathbb{E}(M_n) = \mathbb{E}(M_0)$.*

Proof. We again use the Tower property:

$$\mathbb{E}[\mathbb{E}[...\underbrace{\mathbb{E}[M_n|\mathcal{F}_{n-1}]}_{=M_{n-1}}|...|\mathcal{F}_0] = ... = \mathbb{E}[M_1] = \mathbb{E}[\mathbb{E}[M_1|\mathcal{F}_0]] = \mathbb{E}(M_0). \tag{3.2}$$

Remark that if $\mathcal{F}_0 = \{\varnothing, \Omega\}$ then $\mathbb{E}(M_n) = \mathbb{E}(M_0) = M_0$ (however more generally, M_0 is random).

□

We also have the following two properties:
(i) If M_n is a martingale and H_n is a predictable process, then $X_n = \sum_{i=1}^n H_i(M_i - M_{i-1})$ defines a martingale (see Martingale Transform).
(ii) The sum of two martingales is also a martingale (holds by linearity of the conditional expectation).

Proposition 3.1.8 (Martingale Transform)**.** *For $(M_n)_{0\leq n\leq N}$ a martingale and $(H_n)_{0\leq n\leq N}$ a predictable sequence both with respect to a filtration $(\mathcal{F}_n)_{0\leq n\leq N}$, we have that the sequence $(X_n)_{0\leq n\leq N}$ defined by:*

$$X_n := \begin{cases} H_0 M_0 & \text{for } n = 0 \\ H_0 M_0 + \sum_{i=1}^n H_i(M_i - M_{i-1}) & \text{for } n \geq 1 \end{cases} \tag{3.3}$$

is a martingale with respect to the filtration $(\mathcal{F}_n)_{0\leq n\leq N}$.

Remark 3.1.9. *The martingale property also depends on the probability measure.*

Proof. Since (X_n) is a composition of measurable functions, it is clear that it is adapted. Let us now verify the martingale property:

$$\mathbb{E}(X_{n+1} - X_n|\mathcal{F}_n) = \mathbb{E}(H_{n+1}(M_{n+1} - M_n)|\mathcal{F}_n) = \mathbb{E}(H_{n+1}M_{n+1}|\mathcal{F}_n) - \mathbb{E}(H_{n+1}M_n|\mathcal{F}_n).$$

Since H_{n+1} and M_n are measurable with respect to $\mathcal{F}_n$, and also that M_n is a martingale satisfying $\mathbb{E}(M_{n+1}|\mathcal{F}_n) = M_n$, we finally find:

$$\mathbb{E}(X_{n+1} - X_n|\mathcal{F}_n) = H_{n+1}(\mathbb{E}(M_{n+1}|\mathcal{F}_n) - \mathbb{E}(M_n|\mathcal{F}_n)) = 0 \Longrightarrow \mathbb{E}(X_{n+1}|\mathcal{F}_n) = \mathbb{E}(X_n|\mathcal{F}_n) = X_n.$$

□

We now present an important Proposition for the characterization of a martingale, where we first recall a result from Probability Theory without proof.

Lemma 3.1.10. *For any integrable random variable X (real-valued) there exists a real valued integrable $\widetilde{\mathcal{F}}$-measurable random variable Y such that for all $B \in \widetilde{\mathcal{F}}$, where $\widetilde{\mathcal{F}}$ is a sub-sigma algebra of $\mathcal{F}$, we have that $\mathbb{E}(X\mathbb{1}_B) = \mathbb{E}(Y\mathbb{1}_B)$ and Y is a.s. unique.*

Remark 3.1.11. *This Lemma may be seen as the conditional expectation* $\mathbb{E}(X|\widetilde{\mathcal{F}}) = Y$.

Proposition 3.1.12. *Set* $\Delta M_n := M_n - M_{n-1}$. *An adapted sequence of real-valued random variables* (M_n) *is a martingale if and only if for any predictable sequence* (H_n) *we have:*

$$\mathbb{E}\left(\sum_{n=1}^{N} H_n \Delta M_n\right) = 0. \tag{3.4}$$

Proof. ($\Longrightarrow$) Let (M_n) be a martingale. We use the Martingale Transform (3.1.8) and define (X_n) by:

$$X_n := \begin{cases} 0 & \text{for } n = 0 \\ \sum_{i=1}^{n} H_i \Delta M_i & \text{for } n \geq 1 \end{cases} \tag{3.5}$$

For any predictable (H_n) we have that (X_n) is a martingale. Thus $\mathbb{E}(X_N) = \mathbb{E}(X_0) = 0$ but also $\mathbb{E}(X_N) = \mathbb{E}\left(\sum_{i=1}^{N} H_i \Delta M_i\right)$. Therefore $\mathbb{E}\left(\sum_{i=1}^{N} H_i \Delta M_i\right) = 0$ as required.

($\Longleftarrow$) Suppose that $(*)$ $\mathbb{E}\left(\sum_{i=1}^{N} H_i \Delta M_i\right) = 0$ for all predictable (H_n). Now, for any $\mathcal{F}_i$-measurable A, for the sequences defined by

$$H_n = \begin{cases} 0 & \text{for } n \neq i+1 \\ \mathbb{1}_A & \text{for } n = i+1 \end{cases} \tag{3.6}$$

we have that (H_n) is predictable. Thus from $(*)$ we have that $\mathbb{E}[\mathbb{1}_A(M_{i+1} - M_i)] = 0$ for all $A \in \mathcal{F}_i$. Now from Lemma (3.1.10), setting $X := M_{i+1} - M_i$ and $Y := 0$, for all $A \in \mathcal{F}_i$ we find:

$$\mathbb{E}[\mathbb{1}_A(M_{i+1} - M_i)] = 0 = \mathbb{E}[\mathbb{1}_A Y] \Longrightarrow \mathbb{E}(M_{i+1} - M_i|\mathcal{F}_i) = 0 \Longrightarrow \mathbb{E}(M_{i+1}|\mathcal{F}_i) = \mathbb{E}(M_i|\mathcal{F}_i) = M_i.$$

□

Theorem 3.1.13 (Doob Decomposition Theorem)**.** *Every supermartingale* (X_n) *has the following unique decomposition,*

$$X_n = M_n - A_n, \tag{3.7}$$

where (M_n) *is a martingale and* (A_n) *is a non-decreasing, predictable process and null at 0.*

3.2 Stopping Times

Definition 3.2.1. *A random variable* $\tau : \Omega \to \{0, 1, ..., N\}$ *is a* ***stopping time*** *if for all* $n \in \{0, 1, ..., N\}$ *we have that* $\{\tau \leq n\} \in \mathcal{F}_n$.

Remark 3.2.2. *In the discrete time case* $\{\tau \leq n\} \in \mathcal{F}_n \iff \{\tau = n\} \in \mathcal{F}_n$ *for all* n*. Indeed, for* $(\Longleftarrow)$*,* $\{\tau \leq n\} = \bigcup_{i=1}^{n} \{\tau = i\} \in \mathcal{F}_n$ *since a* σ*-algebra is closed under countable unions. Hence* $\{\tau \leq n\} \in \mathcal{F}_n$*. For* $(\Longrightarrow)$*,* $\{\tau = n\} = \{\tau \leq n\} \cap \{\tau \leq n-1\}^c$*. Since* $\{\tau \leq n\} \in \mathcal{F}_n$ *and* $\{\tau \leq n-1\}^c \in \mathcal{F}_{n-1} \subset \mathcal{F}_n$ *it holds that* $\{\tau = n\} \in \mathcal{F}_n$*.*

Definition 3.2.3. *Let* (X_n) *be a stochastic process (sequence) adapted to the filtration* $(\mathcal{F}_n)$ *and let* τ *be a stopping time. The* ***process stopped at time*** $\boldsymbol{\tau}$ *is defined by,*

$$X_n^{\tau}(\omega) = X_{n \wedge \tau(\omega)}(\omega). \tag{3.8}$$

Thus, on the set $\{\tau = j\}$ *we have,*

$$X_n^{\tau} = \begin{cases} X_j, & \text{if } j \leq n \\ X_n, & \text{if } j > n \end{cases} . \tag{3.9}$$

Proposition 3.2.4. *Let* (X_n) *be an adapted process with respect to the filtration* $(\mathcal{F}_n)$ *and* τ *be a stopping time. The stopped sequence* (X_n^{τ}) *is adapted. Moreover, if* (X_n) *is a* $(\mathcal{F}_n)$*-martingale (resp. sub- or supermartingale) then* (X_n^{τ}) *is also a martingale (resp. sub- or supermartingale).*

Proof. • For $n \geq 1$, we have the following telescoping sum,

$$X_{\tau \wedge n} = X_0 + \sum_{j=1}^{n} \mathbb{1}_{\{j \leq \tau\}} (X_j - X_{j-1}), \tag{3.10}$$

where $\mathbb{1}_{\{j \leq \tau\}}$ is predictable $\Big(\{j \leq \tau\}^c = \{j > \tau\} = \{\tau \leq j-1\} \in \mathcal{F}_{j-1}\Big)$. Hence, $(X_{\tau \wedge n})$ is adapted to $(\mathcal{F}_n)$ and it is a martingale since it has the form of a martingale transform [Proposition (3.1.8)].

• For sub- or supermartingales, we use an analogous formulation of the martingale transform and use similar arguments as above. Let us prove it for a supermartingale (for the submartingale the proof is similar). Let,

$$Y_n := \begin{cases} \phi_0 X_0 & \text{for } n = 0 \\ \phi_0 X_0 + \sum_{j=1}^{n} \phi_j (X_j - X_{j-1}) & \text{for } n \geq 1 \end{cases} \tag{3.11}$$

where (X_n) is a supermartingale with respect to $(\mathcal{F}_n)$ and (ϕ_n) is a non-negative predictable sequence with respect to $(\mathcal{F}_n)$. Now, by measurability,

$$\mathbb{E}(Y_{n+1} - Y_n | \mathcal{F}_n) = \mathbb{E}(\phi_{n+1}(X_{n+1} - X_n) | \mathcal{F}_n) = \phi_{n+1}\big(\mathbb{E}(X_{n+1} | \mathcal{F}_n) - X_n\big). \tag{3.12}$$

Since $\phi_{n+1} \geq 0$ and for (X_n)-supermartingale, $\mathbb{E}(X_{n+1} | \mathcal{F}_n) \leq X_n \Longrightarrow \big(\mathbb{E}(X_{n+1} | \mathcal{F}_n) - X_n\big) \leq 0$, we finally have that $\mathbb{E}(Y_{n+1} - Y_n | \mathcal{F}_n) \leq 0$ and thus $\mathbb{E}(Y_{n+1} | \mathcal{F}_n) \leq Y_n$ for all n. □

Definition 3.2.5. *Denote by* $\boldsymbol{\mathcal{T}_{n,N}}$ *the set of all stopping times taking values in* $\{n, n+1, ..., N\}$*.*

Definition 3.2.6. *A stopping time τ^* is said to be **optimal** for the sequence $(Z_n)_{0\leq n\leq N}$ if,*

$$\mathbb{E}_{\mathbb{Q}}(Z_{\tau^*}) = \sup_{\tau\in\mathcal{T}_{0,N}} \mathbb{E}_{\mathbb{Q}}(Z_\tau). \tag{3.13}$$

3.3 Stochastic Calculus

In this section, we present the tools that will be used for continuous time financial models. From a first course in Stochastic Calculus, we know that if a process (X_t) is given to us as a stochastic integral that we may write in shorthand as,

$$dX_t = a(\omega,t)dt + b(\omega,t)dB_t, \tag{3.14}$$

where (B_t) is a Brownian Motion, then it is natural to define a dX_t-integral of $f(\omega,t)$ by setting,

$$\int_0^t f(\omega,s)dX_s = \int_0^t f(\omega,s)a(\omega,s)ds + \int_0^t f(\omega,s)b(\omega,s)dB_s, \tag{3.15}$$

provided that $f(\omega,t)$ is an integrand for which the last two integrals make sense; i.e. (i) $f(\omega,t)$ is adapted, (ii) $f(\omega,s)a(\omega,s)\in L^1(ds)$ for all $\omega\in\overline{\Omega}$ with $\mathbb{P}(\overline{\Omega})=1$ and, (iii) $f(\omega,s)b(\omega,s)\in L^2_{\text{loc}} = L^2_{\text{loc}}[0,T]$ that is the set of adapted, measurable processes such that,

$$\mathbb{P}\left(\int_0^T \left(f(\omega,s)b(\omega,s)\right)^2 ds < \infty\right) = 1. \tag{3.16}$$

Definition 3.3.1. *We say that a process $(X_t)_{t\in[0,T]}$ is a **standard Itô process** if for all $t\in[0,T]$ the process (X_t) has the integral representation ,*

$$X_t = X_0 + \int_0^t a(\omega,s)ds + \int_0^t b(\omega,s)dB_s, \tag{3.17}$$

where $a(\cdot,\cdot)$ and $b(\cdot,\cdot)$ are adapted, measurable processes that satisfy the integrability conditions,

$$\mathbb{P}\left(\int_0^T |a(\omega,s)|ds < \infty\right) = 1 \quad \text{and,} \quad \mathbb{P}\left(\int_0^T |b(\omega,s)|^2 ds < \infty\right) = 1. \tag{3.18}$$

Theorem 3.3.2 (Itô Formula)**.** *Let $f\in C^{1,2}(\mathbb{R}_+\times\mathbb{R})$ and $(X_t)_{t\in[0,T]}$ a standard process with integral representation $X_t = X_0 + \int_0^t a(\omega,s)ds + \int_0^t b(\omega,s)dB_s$, where $0\leq t\leq T$. Then,*

$$\begin{aligned} f(t,X_t) &= f(0,X_0) + \int_0^t \left(\frac{\partial f}{\partial s}(s,X_s) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(s,X_s)b^2(\omega,s)\right)ds + \int_0^t \frac{\partial f}{\partial x}(s,X_s)dX_s \\ &= f(0,X_0) + \int_0^t \left(\frac{\partial f}{\partial s}(s,X_s) + \frac{\partial f}{\partial x}(s,X_s)a(\omega,s) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(s,X_s)b^2(\omega,s)\right)ds \\ &\qquad + \int_0^t \frac{\partial f}{\partial x}(s,X_s)b(\omega,s)dB_s. \end{aligned}$$

For convenience, with a process $dX_t = a(\omega,t)dt + b(\omega,t)dB_t$ such that $g \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R})$ we will often use the abbreviation,

$$df(t,X_t) = \left(\frac{\partial f}{\partial s}(s,X_s) + \frac{\partial f}{\partial x}(s,X_s)a(\omega,s) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(s,X_s)b^2(\omega,s)\right)dt + \frac{\partial f}{\partial x}(s,X_s)b(\omega,s)dB_t. \quad (3.19)$$

Now, as in the discrete case, we introduce the *filtration*.

Definition 3.3.3. *Let $(\Omega,\mathcal{F},\mathbb{P})$ be a probability space. Then a **filtration** $(\mathcal{F}_t)$ is an increasing family of sub-σ-algebras of $\mathcal{F}$.*

We can in fact construct a σ-algebra generated by the random variables $\{X_i\}_i$ as $\overline{\mathcal{F}} = \sigma(X_s, s \leq t)$. However, from now on, we assume that the filtration satisfies the usual conditions:

- If $A' \subset A \in \mathcal{F}$ and if $\mathbb{P}(A) = 0$ then for any t, $A' \in \mathcal{F}_t$.
- $\mathcal{F}_t = \bigcap_{s>t} \mathcal{F}_s$.

Example 3.3.4 (The standard Brownian filtration)**.** *If (X_t) is a Brownian Motion and if $\mathcal{F}_t$ is the σ-algebra generated by both, $\overline{\mathcal{F}}_t$ and $\mathcal{N}$ (the σ-algebra generated by all the $\mathbf{P}$-null sets of $\mathcal{F}$), we obtain a filtration satisfying the usual conditions (called the standard Brownian filtration).*

These usual conditions are convenient for several situations. For example: (i) if $\mathcal{F}_t$ contains all $\mathbf{P}$-null sets of $\mathcal{F}$ and if $X = Y$ $\mathbf{P}$-a.s. with Y being $\mathcal{F}_t$-measurable, then X is also $\mathcal{F}_t$-measurable. Also (ii) if the filtration $(\mathcal{F}_t)_{t\in[0,T]}$ is right-continuous it is possible to show that every martingale has a càdlàg version. Finally we may also mention that (iii) it has some advantages when working with stopping times.

Definition 3.3.5. *On some time interval T, a process ξ is a **version** of η if,*

$$\mathbb{P}\big(\xi(t) = \eta(t)\big) = 1, \quad \forall t \in T. \quad (3.20)$$

Furthermore, a version is **indistinguishable** if,

$$\mathbb{P}\big(\xi(t) = \eta(t), \forall t \in T\big) = 1. \quad (3.21)$$

Remark 3.3.6. *Continuity of the sample paths does not imply continuity of the filtrations.*

Definition 3.3.7. *A probability measure $\mathbb{Q}$ on $(\Omega,\mathcal{F})$ is **absolutely continuous** with respect to $\mathbb{P}$ if for all $A \in \mathcal{F}$, $\mathbb{P}(A) = 0 \Longrightarrow \mathbb{Q}(A) = 0$, in which case we write $\mathbb{Q} \ll \mathbb{P}$.*

Two probability measures $\mathbb{P}$ and $\mathbb{Q}$ are equivalent if $\mathbb{P} \ll \mathbb{Q}$ and $\mathbb{Q} \ll \mathbb{P}$.

Remark 3.3.8. *If $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$ with density Z, then $\mathbb{P}$ and $\mathbb{Q}$ are equivalent if and only if $\mathbb{P}(Z > 0) = 1$.*

Theorem 3.3.9 (Radon-Nikodym). *A probability measure $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$ if and only if there exists a non-negative random variable Z on $(\Omega, \mathcal{F})$ such that, $\forall A \in \mathcal{F}$,*

$$\mathbb{Q}(A) = \int_A Z d\mathbb{P}. \tag{3.22}$$

The random variable Z is the density of $\mathbb{Q}$ with respect to $\mathbb{P}$ and denoted by $\frac{d\mathbb{Q}}{d\mathbb{P}}$.

Theorem 3.3.10 (Girsanov Theorem). *Let $\theta = (\theta_t)_{t\in[0,T]}$ be an adapted process satisfying $\int_0^T \theta_s^2 ds < \infty$ a.s. and such that the process $(L_t)_{t\in[0,T]}$ defined by,*

$$L_t = e^{-\int_0^t \theta_s dB_s - \frac{1}{2}\int_0^t \theta_s^2 ds} \tag{3.23}$$

is a martingale. Then, under the probability $\mathbb{P}^{(L)}$ with density L_T with respect to $\mathbb{P}$, the process $(W_t)_{t\in[0,T]}$ defined by,

$$W_t = B_t + \int_0^t \theta_s ds, \tag{3.24}$$

is an $(\mathcal{F}_t)$-standard Brownian motion.

Remark that here we have $\frac{d\mathbb{P}^{(L)}}{d\mathbb{P}}|_{\mathcal{F}_t} = \mathbb{E}(L_T|\mathcal{F}_t) = L_t$.

Remark 3.3.11. *A sufficient condition for $(L_t)_{t\in[0,T]}$ to be a martingale is,*

$$\mathbb{E}\left(\exp\left(\frac{1}{2}\int_0^T \theta_t^2 dt\right)\right) < \infty. \tag{3.25}$$

*This is known as the **Novikov's Condition**.*

Theorem 3.3.12 (Martingale Representation Theorem). *Let $M = (M_t)_{t\in[0,T]}$ be a square-integrable martingale with respect to the filtration $(\mathcal{F}_t)_{t\in[0,T]}$ generated by a standard Brownian Motion $(W_t)_{t\in[0,T]}$. There exists an adapted process $(K_t)_{t\in[0,T]}$ such that $\mathbb{E}\left(\int_0^T K_u^2 du\right) < \infty$ and for all $t \in [0,T]$,*

$$M_t = M_0 + \int_0^t K_u dW_u \quad a.s. \tag{3.26}$$

Note that this representation only applies to martingales relative to the augmented filtration of the Brownian Motion. Also, from the Martingale Representation Theorem, it follows that if U is an $\mathcal{F}_T$-measurable and square-integrable random variable, it can be written as,

$$U = \mathbb{E}(U) + \int_0^T H_s dB_s \quad \text{a.s.} \tag{3.27}$$

where (H_t) is an adapted process such that $\mathbb{E}\left(\int_0^T H_s^2 ds\right) < \infty$. Indeed, considering $M_t = \mathbb{E}(U|\mathcal{F}_t) = \mathbb{E}(U) + \int_0^t H_s dB_s$ a.s. one has that $M_0 = \mathbb{E}(U|\mathcal{F}_0) = \mathbb{E}(U)$ and so that at time T,

$$M_T = \mathbb{E}(U|\mathcal{F}_T) = U = \mathbb{E}(U) + \int_0^T H_s dB_s. \tag{3.28}$$

Remark 3.3.13. *It can also be shown that if $(M_t)_{t\in[0,T]}$ is a (not necessarily square-integrable) martingale, then there is a similar representation with a process satisfying only $\int_0^T H_s^2 ds < \infty$ a.s.*

In this section, we have approached ideas in Stochastic Calculus by assuming the existence and the construction of the stochastic integral. In the next section, we present such a construction in the case of vector stochastic integration.

3.4 Stochastic Analysis - Construction of Vector Stochastic Integrals

In this section, we delve into the Analysis to describe concretely the construction of the vector stochastic integral. We first define some important and basic definitions before presenting the construction.

Definition 3.4.1. *Let $(\Omega,\mathcal{F},\mathbb{P})$ be a probability space. Then a **filtration** $(\mathcal{F}_t)_{t\geq 0}$ is an increasing family of sub-σ-algebras of $\mathcal{F}$, that is assumed to satisfy the usual conditions of:*
- right continuity: $\mathcal{F}_t = \bigcap_{s>t}\mathcal{F}_s$
- completeness: $\mathcal{F}_0$ contains all $\mathbb{P}$-null sets of $\mathcal{F}$.

The role a filtration is to describe the information available at any time t.

Definition 3.4.2. *A **stochastic process** X on $(\Omega,\mathcal{F},(\mathcal{F}_t)_{t\geq 0},\mathbb{P})$ is a collection of $\mathbb{R}$-or-$\mathbb{R}^n$-valued random variables (vectors) $X=(X_t)_{t\geq 0}$.*

Definition 3.4.3. *A process X is said to be **adapted** if X_t is $\mathcal{F}_t$-measurable for each t (i.e. X_t is known if $\mathcal{F}_t$ is known).*

Definition 3.4.4. *A process X with sample paths being a.s. right-continuous with left limits are called **càdlàg**.*

Definition 3.4.5. *A map $\tau:\Omega\mapsto[0,\infty]$ is a **stopping time** if $\{\tau\leq t\}\in\mathcal{F}_t$ for all $t\geq 0$.*

Definition 3.4.6. *Let $\tau:\Omega\mapsto[0,\infty]$ be a stopping time, then the **stopping time σ-algebra** $\mathcal{F}_\tau$ is defined as,*

$$\mathcal{F}_\tau = \{A\in\mathcal{F} : A\cap\{\tau\leq t\}\in\mathcal{F}_t \text{ for all } t\geq 0\}. \tag{3.29}$$

Intuitively, $\mathcal{F}_\tau$ represents the events known at time τ.

Definition 3.4.7. *For a stochastic process X and a stopping time τ, the process X^τ defined by $X_t^\tau := (X_{t\wedge\tau})_{t\geq 0}$ is the **stopped process (at τ)**, where $a\wedge b=\min(a,b)$ for $a,b\in\mathbb{R}$.*

As we will see later on, **martingales** are of particular interest, where we recall that these are $\mathbb{R}$-valued, adapted stochastic processes $M = (M_t)_{t\geq 0}$ such that M_t is integrable for all $t \geq 0$ and $\mathbb{E}(M_t|\mathcal{F}_s) = M_s$ a.s. for all $s \leq t$.

Definition 3.4.8. *An adapted process M is a **local martingale** if there exists an a.s. increasing sequence $(\tau_k)_{k=1}^{\infty}$ of stopping times such that $\tau_k \to \infty$ a.s. and, for each $k \geq 1$, the stopped process M^{τ_k} is a martingale.*

Definition 3.4.9. *A process $(A_t)_{t\geq 0}$ is of **bounded variation** if, for each $t < \infty$,*

$$\sup_{0\leq t_0<\ldots<t_n\leq t} \sum_{i=1}^{n} |A_{t_i} - A_{t_{i-1}}| < \infty \quad a.s. \tag{3.30}$$

Definition 3.4.10. *An $\mathbb{R}^n$-valued process $M = (M_1, ..., M_n)$ is said to be a **$\mathbb{R}^n$-(local) martingale** if $M_1, ..., M_n$ are one-dimensional (local) martingales.*

Definition 3.4.11. *An $\mathbb{R}^n$-valued process $A = (A_1, ..., A_n)$ is of **$\mathbb{R}^n$-bounded variation** if $A_1, ..., A_n$ are all of bounded variation.*

Definition 3.4.12. *For $\tau_1 \leq \tau_2$ being two stopping times, $[\![\boldsymbol{\tau_1}, \boldsymbol{\tau_2}]\!]$ denotes the set,*

$$\big\{(t,\omega) : t \in \mathbb{T}, \omega \in \Omega \ \text{ and } \ \tau_1(\omega) \leq t \leq \tau_2(\omega)\big\}. \tag{3.31}$$

The σ-algebra generated by all stochastic intervals of the form $[\![0, \tau]\!]$ where τ is a stopping time is called the **predictable σ-algebra** which is denoted by $\mathcal{B}$. Any process which is measurable with respect to $\mathcal{B}$ is said to be **predictable**. The σ-algebra $\mathcal{B}$ on $\Omega \times \mathbb{R}_+$ is also generated by the left-continuous $(\mathcal{F}_t)$-adapted processes.

Theorem 3.4.13 (Doob-Meyer Decomposition Theorem)**.** *A n-dimensional càdlàg process X is called a semimartingale if for all $t \geq 0$ it admits a decomposition,*

$$X_t = X_0 + M_t + A_t, \tag{3.32}$$

where $(M_t)_{t\geq 0}$ is a n-dimensional local martingale, $(A_t)_{t\geq 0}$ is a n-dimensional adapted process of bounded variation, and $A_0 = M_0 = 0$.

Definition 3.4.14. *A semimartingale X is said to be a **special semimartingale** if a decomposition in the form of Theorem (3.4.13) exists with A being also predictable.*

If such a decomposition exists, then it is unique. In particular, this is the case for continuous semimartingales.
For formulating the FTAP in the continuous time higher-dimensional setting, integrals with respect to vector-valued semimartingales have to be constructed (for integrable H, assumed to be predictable).

We now present a quite tricky construction (the difficult point being closedness of the space of integrals) of the sufficiently general so-called **vector stochastic integrals** denoted by,

$$H \cdot X = (H \cdot X)_t = \int_0^t \langle H_s, dX_s \rangle, \quad t \geq 0, \tag{3.33}$$

has been provided by Jacob (1980). An explicit approach to the vector stochastic integral and a discussion of its application in the (First and Second) FTAP has been provided by Cherny and Shiryaev (2002), based on (1998).

Consider $H = \{H_t(\omega) : t \geq 0, \omega \in \Omega\}$ a $\mathcal{B}(\mathbb{R}_+) \otimes \mathcal{F}$-measurable and $(\mathcal{F}_t)$-adapted with $\mathbb{E}\left(\int_0^t H_s^2 ds\right) < \infty$ for $t \geq 0$ (Δ). As usual with the construction of integrals in Probability Theory , we first consider simple functions,

$$H_t(\omega) = h_0(\omega)\mathbb{1}_{\{0\}}(t) + \sum_{i \geq 0} h_i(\omega)\mathbb{1}_{(t_i, t_{i+1}]}(t), \tag{3.34}$$

where individual terms are $\mathcal{F}_{t_i}$-measurable and square integrable. There is a canonical definition of the stochastic integral given by,

$$(H \cdot W)_t = \sum_{i \geq 0} h_i(\omega)\mathbb{1}_{(t_i, t_{i+1}]}(t)(W_{t_{i+1}} - W_{t_i}). \tag{3.35}$$

Then, one succeeds to construct simple functions $(H^n)_{n \geq 0}$ such that for $t > 0$ fixed,

$$\mathbb{E}\left[\int_0^t (H_s - H_s^n)^2 ds\right] \xrightarrow{n \to \infty} 0, \tag{3.36}$$

and therefore,

$$\mathbb{E}\left[\int_0^t (H_s^n - H_s^m)^2 ds\right] \xrightarrow{n,m \to \infty} 0. \tag{3.37}$$

A little calculation yields the Itô-Isometry for simple processes where for $t > 0$ we have,

$$\mathbb{E}\left[\left(\int_0^t H_s^n dW_s\right)^2\right] = \mathbb{E}\left[\int_0^t (H_s^n)^2 ds\right]. \tag{3.38}$$

As a result for $t > 0$ we obtain,

$$\mathbb{E}\left[\left(\int_0^t H_s^n dW_s - \int_0^t H_s^m dW_s\right)^2\right] = \mathbb{E}\left[\int_0^t (H_s^n - H_s^m)^2 ds\right] \xrightarrow{n,m \to \infty} 0. \tag{3.39}$$

Hence $\left(\int_0^t H_s^n dW_s\right)_{n \geq 0}$ for $t > 0$ is a Cauchy sequence in the space L^2 (square integrable random variables). Since L^2 is complete we may define, for $t > 0$ fixed,

$$\int_0^t H_s dW_s := \lim_{n \to \infty} \int_0^t H_s^n dW_s \text{ in } L^2, \tag{3.40}$$

which is often denoted by $(H \cdot W)_t$ for the stochastic integral on $[0, t]$. The next task is to extend this idea to processes and localization so that we have an integral for H-measurable, adapted, and satisfying $\mathbb{P}\left(\int_0^t H_s^2 ds < \infty\right) = 1$ for $t > 0$. In essence, the same construction works if instead of a Wiener process W one takes a square integrable martingale M. We thus replace the condition (Δ) by $\mathbb{E}\left(\int_0^t H_s^2 d\langle M \rangle_s\right) < \infty$ for $t > 0$ $(\widetilde{\Delta})$. Note that in this new condition, by Doob, for a square

integrable martingale there exists a unique increasing predictable process (starting at 0) $\langle M\rangle$ such that $M^2-\langle M\rangle \in \mathcal{M}_{\text{Loc}}$ (set of local martingales). A natural question would be to know for which functions is there a sequence of simple functions $(H^n)_{n\geq 0}$ such that for $t>0$,

$$\mathbb{E}\left[\int_0^t (H_s - H_s^n)^2 d\langle M\rangle_s\right] \xrightarrow{n\to\infty} 0. \tag{3.41}$$

The answer depends on properties of $\langle M\rangle$ where it is always fine for H-predictable such that $(\widetilde{\Delta})$ holds.

Remark 3.4.15. *Recall that all processes we consider are at least measurable and adapted:*
- *$\langle M\rangle$-absolutely continuous with respect to the Lebesgue measure, measurable and adapted is fine.*
- *$\langle M\rangle$-$\mathbb{P}$ a.s. continuous. Measurable and $H(\tau(\omega),\omega)$ $\mathcal{F}_\tau$-measurable for all finite stopping times τ is fine (in particular progressively measurable).*
- *$\langle M\rangle$- no extra regularity, it is fine for predictable processes.*

Analogously to the Brownian Motion case we show $\mathbb{E}\left[\left(\int_0^t H_s dM_s\right)^2\right] = \mathbb{E}\left(\int_0^t H_s^2 d\langle M\rangle_s\right)$ for simple functions, with $t>0$. The rest of the theory is similar by analogous arguments.

Let $X=(X_1,...,X_d)$ be a d-dimensional semimartingale admitting a decomposition [Theorem (3.4.13)] $X_t = X_0 + A_t + M_t$ where (A_t) is an adapted process of bounded variation and $(M_t)\in \mathcal{M}^d_{\text{Loc}}$ is a local martingale.

3.4.1 Integral with respect to a Local Martingale

There exists a non-decreasing adapted process $C=(C_t)_{t\geq 0}$ with $C_0=0$ and optional processes $C^{ij} = (C_t^{ij})_{t\geq 0}$ with $C_t^{ij}(\omega) = C_t^{ji}(\omega)$ for $i,j\in\{1,...,d\}$ such that $(C_t^{ij}(\omega))$ is positive definite. These optional processes are measurable with respect to the σ-field on $\Omega\times\mathbb{R}_+$ generated by the càdlàg $(\mathcal{F}_t)$-adapted (measurable) processes. The existence of these processes is such that the quadratic covariation $[M^i,M^j]$ is given by,

$$[M^i,M^j] = \int_0^\bullet C_s^{ij} dC_s = \sum_{s\leq \bullet} \Delta M_s^i \Delta M_s^j + \underbrace{[M^{ic},M^{jc}]}_{\text{continuous part}}. \tag{3.42}$$

For the above result we observe similarities with Itô such that for $M^1 = \int_0^\bullet b_s^1 dB_s$, $M^2 = \int_0^\bullet b_s^2 dB_s$ one has $[M^1,M^2]_t = \int_0^t b_s^1 b_s^2 ds$.

Definition 3.4.16. *We say that $\boldsymbol{H\in L^1(M)}$ if $H=(H^1,...,H^d)$ is predictable with,*

$$\mathbb{E}\left[\left(\int_0^\infty \left(\sum_{i,j=1}^d H_s^i C_s^{ij} H_s^j\right) dC_s\right)^{\frac{1}{2}}\right] < \infty. \tag{3.43}$$

Definition 3.4.17. *A **stochastically simple integrand** is a process H of the form,*

$$H_t = h_0 \mathbb{1}_{\{t=0\}} + \sum_{k=1}^{m} h_k \mathbb{1}_{\{\tau_k < t \leq \tau_{k+1}\}}, \quad t > 0, \tag{3.44}$$

where h_k is d-dimensional bounded $\mathcal{F}_{\tau_k}$-measurable random vector, and $0 \leq \tau_1 \leq ... \leq \tau_{m+1}$ are stopping times.

As a result, it turns out that:
(1) The function,

$$\|H\|_{L^1(M)} = \mathbb{E}\left[\left(\int_0^\infty \left(\sum_{i,j=1}^{d} H_s^i C_s^{ij} H_s^j\right) dC_s\right)^{\frac{1}{2}}\right] \tag{3.45}$$

is a norm on $L^1(M)$ that is defined with respect to (3.43).

(2) The set of simple integrands that belong to $L^1(M)$ is dense in $L^1(M)$ (by a monotone class argument).

Definition 3.4.18. *For a simple integrand we define the **integral** $(\boldsymbol{H \cdot M})$ as,*

$$(H \cdot M)_t = \sum_{i=1}^{d} \sum_{k=1}^{m} h_k^i (M^i_{t \wedge \tau_{k+1}} - M^i_{t \wedge \tau_k}) \in \mathbb{R}, \quad t > 0. \tag{3.46}$$

As a result it turns out that,

$$[H \cdot M] = \int_0^\bullet \left(\sum_{i,j=1}^{d} H_s^i C_s^{ij} H_s^j\right) dC_s, \tag{3.47}$$

where we may compare with $[\int_0^\bullet \varphi_s b_s dB_s]_t = \int_0^t \varphi_s^2 b_s^2 ds$. Now, let $H \in L^1(M)$ then there exists a sequence (H^n) of simple integrands that converges to H in $L^1(M)$. Consider the Banach space (i.e. normed and complete space),

$$H^1 = \left\{ M \in M_{\text{Loc}} : \mathbb{E}\left(\sup_{t \geq 0} |M_t|\right) < \infty \right\}, \tag{3.48}$$

where $\mathcal{M}_{\text{Loc}}$ is the set of local martingales and the norm is $\|M\|_{H^1} = \mathbb{E}\left(\sup_{t \geq 0} |M_t|\right)$.

Proposition 3.4.19 (Burkholder-Davis-Gundy Inequalities)**.** *There exists some constants $0 < c < C$ such that for any $M \in H^1$ with $M_0 = 0$,*

$$c\mathbb{E}\left([M]_\infty^{\frac{1}{2}}\right) \leq \mathbb{E}\left(\sup_{t \geq 0} |M_t|\right) \leq C\mathbb{E}\left([M]_\infty^{\frac{1}{2}}\right). \tag{3.49}$$

Proposition 3.4.20. *$(H^n \cdot M)$ is a Cauchy sequence in H^1.*

Proof. We will use Equation (3.47) along with BDG inequalities [Theorem (3.4.19)] yield that $(H^n \cdot M)$ is a Cauchy sequence in H^1. To this end, set (H^n) to be a sequence of stochastic simple integrands that converges to H in $L^1(M)$, i.e.,

$$\|H - H^n\|_{L^1(M)} = \mathbb{E}\left[\left(\int_0^\infty \left(\sum_{i,j=1}^d (H_s^i - H_s^{in}) C_s^{ij} (H_s^j - H_s^{jn})\right) dC_s\right)^{\frac{1}{2}}\right] \xrightarrow{n\to\infty} 0. \tag{3.50}$$

Consider now $\|H^n \cdot M - H^m \cdot M\|_{H^1}$. We bound this quantity using the BDG inequalities, and we find,

$$\begin{aligned}
\|H^n \cdot M - H^m \cdot M\|_{H^1} &= \mathbb{E}\left(\sup_{t\geq 0} \left|(H^n \cdot M)_t - (H^m \cdot M)_t\right|\right) \\
&= \mathbb{E}\left(\sup_{t\geq 0} \left|((H^n - H^m) \cdot M)_t\right|\right) \\
&\leq C\mathbb{E}\left[\left(\int_0^\infty \left(\sum_{i,j=1}^d (H_s^{in} - H_s^{im}) C_s^{ij} (H_s^{jn} - H_s^{jm})\right) dC_s\right)^{\frac{1}{2}}\right] \xrightarrow{n,m\to\infty} 0.
\end{aligned}$$

Hence, there exists a H^1-limit for the sequence $(H^n \cdot M)$. Therefore, for $H \in L^1(M)$, the vector stochastic integral $H \cdot M$ is defined as the H^1-limit of $(H^n \cdot M)$.

□

We now extend the construction by localization.

Definition 3.4.21. *The set $\boldsymbol{\mathcal{A}_{Loc}}$ is the set of processes of locally integrable variation, i.e. for $A \in \mathcal{A}_{Loc}$ there exists an increasing sequence of stopping times such that $\tau_n \to \infty$ a.s. and for all $n \in \mathbb{N}$, $\mathbb{E}(\mathrm{Var}(A)_{\tau_n}) < \infty$. Here, $\mathrm{Var}(A)_t(\omega)$ is the **total variation** of $s \mapsto A_s(\omega)$ on $[0,t]$ with,*

$$\mathrm{Var}(A)_t(\omega) := \sup_{0\leq t_0<\ldots<t_n\leq t} \sum_{i=1}^n |A_{t_i} - A_{t_{i-1}}|. \tag{3.51}$$

Definition 3.4.22. *We say that $\boldsymbol{H \in L^1_{Loc}(M)}$ if it is predictable and,*

$$\left(\int_0^\bullet \left(\sum_{i,j=1}^d H_s^i C_s^{ij} H_s^j\right) dC_s\right)^{\frac{1}{2}} \in \mathcal{A}_{Loc}. \tag{3.52}$$

Proposition 3.4.23. *Let $H \in L^1_{Loc}(M)$. Then there exists an increasing sequence $(\tau_n)_{n=1}^\infty$ of stopping times such that $\tau_n \to \infty$ a.s. and,*

$$\mathbb{E}\left[\left(\int_0^{\tau_n} \left(\sum_{i,j=1}^d H_s^i C_s^{ij} H_s^j\right) dC_s\right)^{\frac{1}{2}}\right] < \infty. \tag{3.53}$$

For each $n \in \mathbb{N}$, $H\mathbb{1}_{[\![0,\tau_n]\!]} \in L^1(M)$. Furthermore there exists a unique process $H \cdot M$ such that, for $n \in \mathbb{N}$,

$$(H \cdot M)^{\tau_n} = \left(H\mathbb{1}_{[\![0,\tau_n]\!]}\right) \cdot M, \tag{3.54}$$

where $H \cdot M$ does not depend on the choice of the localizing sequence $(\tau_n)_{n=1}^{\infty}$.

Definition 3.4.24. *For $H \in L^1_{Loc}(M)$, the process $H \cdot M$ satisfying (3.54) is called the **vector stochastic integral** of H with respect to M.*

We have $H \cdot M \in \mathcal{M}_{\text{Loc}}$ and,

$$[H \cdot M] = \int_0^{\bullet} \left(\sum_{i,j=1}^{d} H_s^i C_s^{ij} H_s^j \right) dC_s. \tag{3.55}$$

3.4.2 Integral with respect to a Process of Finite Variation

For $A \in \nu^d$, there exists an adapted, increasing process C and an optional process $(a^i)_{i=1}^d$ such that,

$$A^i = \int_0^{\bullet} a_s^i dC_s. \tag{3.56}$$

Definition 3.4.25. *We say that a predictable process $H = (H^1, ..., H^d)$ is in $\boldsymbol{L_{\text{Var}}(A)}$ if for any $t \geq 0$,*

$$\int_0^t \left| \sum_{i=1}^{d} H_s^i a_s^i \right| dC_s < \infty \quad a.s. \tag{3.57}$$

Remark that $L_{\text{Var}}(A)$ does not depend on the choice of (a^i) and C. As a result, for $H \in L_{\text{Var}}(A)$, the vector stochastic integral is defined by,

$$H \cdot A = \int_0^{\bullet} \left(\sum_{i=1}^{d} H_s^i a_s^i \right) dC_s. \tag{3.58}$$

In particular, for $H \cdot A \in \nu$, $\text{Var}(H \cdot A) = \int_0^{\bullet} \left| \sum_{i=1}^{d} H_s^i a_s^i \right| dC_s$.

3.4.3 Integral with respect to a Semimartingale

We denote the just defined integral with respect to $M \in \mathcal{M}^d_{\text{Loc}}$ by $(M)H \cdot M$ (local martingale sense) and the integral with respect to $A \in \nu^d$ (d-dimensional processes of bounded variation) by $(LS)H \cdot A$ (Lebesgue-Stieltjes integral sense).

Definition 3.4.26. *Let X be a d-dimensional semimartingale. A process H is **X-integrable** if there exists a decomposition $X = A + M$ with $A \in \nu^d$ and $M \in \mathcal{M}^d_{Loc}$ such that $H \in L_{Var}(A) \cap L^1_{Loc}(M)$.*

In this setting, the vector stochastic integral is defined by,

$$H \cdot X = (LS)H \cdot A + (M)H \cdot M. \tag{3.59}$$

Definition 3.4.27. *The space of X-integrable processes is denoted by $\boldsymbol{L(X)}$.*

We end this section by making several remarks.

- For $Z \in \nu^d \cap \mathcal{M}^d_{\text{Loc}}$ and $H \in L_{\text{Var}}(Z) \cap L^1_{\text{Loc}}(Z)$ we have that $(LS)H \cdot Z = (M)H \cdot Z$.

- If $H \in L(X)$ then it may not belong to $L_{\text{Var}}(A) \cap L^1_{\text{Loc}}(M)$ for all the decompositions $X = A + M$ with $A \in \nu^d$ and $M \in \mathcal{M}^d_{\text{Loc}}$.

- However, if $X = A + M = A' + M'$ with $A, A' \in \nu^d$, $M, M' \in \mathcal{M}^d_{\text{Loc}}$ and $H \in L_{\text{Var}}(A) \cap L^1_{\text{Loc}}(M) \cap L_{\text{Var}}(A') \cap L^1_{\text{Loc}}(M')$ then we have,

$$(LS)H \cdot A + (M)H \cdot M = (LS)H \cdot A' + (M)H \cdot M', \tag{3.60}$$

so that the integral is well-defined.

- Linearity in X holds,

$$H \cdot (\alpha X_1 + \beta X_2) = \alpha(H \cdot X_1) + \beta(H \cdot X_2). \tag{3.61}$$

We also have linearity in H,

$$(\alpha H_1 + \beta H_2) \cdot X = \alpha(H_1 \cdot X) + \beta(H_2 \cdot X). \tag{3.62}$$

For example, $\{H \cdot X : H \in L(X)\}$ is closed (in the every topology).

Remark 3.4.28. *If the componentwise stochastic integral $\sum_{i=1}^n (H_i \cdot X_i)_t$, $t \geq 0$, of one-dimensional stochastic integrals exists, then the vector stochastic integral also exists and the integrals coincide. However, it may happen that the componentwise stochastic integral does not exist while the vector stochastic integral is well defined (e.g. showing the necessity of using vector stochastic integrals rather than componentwise in the context of the FTAP).*

Definition 3.4.29. *A semimartingale $X = (X_{t1}, ..., X_{tn})$ is a **σ-martingale** if there exists an $\mathbb{R}^n$-valued local martingale M and a process $H = (H_{t1}, ..., H_{tn})_{t \geq 0}$ such that for each i, H_i is M_i-integrable and $X_1 = X_{0i} + H_i \cdot M_i$.*

Note that every càdlàg martingale is a local martingale (take $\tau_n = n$). Furthermore, a local martingale is a σ-martingale. The converse, however, is not true.

Chapter 4

Discrete-time Models

In this chapter, we present the discrete-time model and use some elements of martingale theory to study some interesting properties.

Definition 4.0.1. *A **discrete-time financial model** is built on a finite probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with a filtration $\{\mathcal{F}_n\}_{n=0,...,N}$ with $N \in \mathbb{N}^*$ (i.e. an increasing sequence of σ-algebras included in $\mathcal{F}$). The horizon N will often correspond to the maturity of the considered options and $\mathcal{F}_n$ can be interpreted as the information available at time n.*

From now on, we assume the following points:

- Standard filtration: $\mathcal{F}_0 = \{\Omega, \varnothing\}$ and $\mathcal{F}_N = \mathcal{F}$.
- Atoms: $\mathbb{P}(\omega) > 0$ for all $\omega \in \Omega$.
- Assets: The market consists of $d+1$ assets with price at time n being given by $S_n = (S_n^0, S_n^1, ..., S_n^d)$.
- Measurability: The positive random variables $S_n^0, S_n^1, ..., S_n^d$ are measurable with respect to $\mathcal{F}_n$.
- Risk-less asset: There is a risk-less asset in the market such that $S_0^0 = 1$ and $S_n^0 = (1+r)^n$ with r being the positive risk-less interest rate.

Remark 4.0.2. • *Working with a finite probability space avoids some technical problems, e.g. all real-valued random variables are integrable.*

• *The economical interpretation of $\mathcal{F}_n$ here is that the investors know past and present prices but quite reasonably they do not know the future ones.*

• *$S_n^0 = (1+r)^n$ is also the price change factor of a risk-less asset until time n and hence, $(1+r)^{-n}$ is the discount factor from time n to time 0. The assets indexed by $i = 1, ..., d$ are called risky assets.*

4.1 Strategies

We now present different trading strategies and some properties for a self-financing condition.

Definition 4.1.1. *A **trading strategy** is formally defined as a stochastic process $\phi = (\phi_n^0, \phi_n^1, ..., \phi_n^d)_{n=1,...,N}$ with values in $\mathbb{R}^{d+1}$, where ϕ_n^i denotes the number of shares of asset i held in the portfolio on the time interval between $n-1$ and n.*

Remark 4.1.2. *The sequence ϕ is assumed to be predictable in the sense of Definition (3.1.1).*

The value of the portfolio at time n, denoted by $V_n(\phi)$, is defined as the scalar product:

$$V_n(\phi) := \phi_n \cdot S_n = \sum_{i=0}^{d} \phi_n^i S_n^i. \tag{4.1}$$

Its discounted value is denoted by $\widetilde{V}_n(\phi) = V_n(\phi)(S_n^0)^{-1}$ and analogously $\widetilde{S}_n = S_n(S_n^0)^{-1}$.

Definition 4.1.3. *A strategy is **self-financing** if $\phi_{n+1} \cdot S_n = \phi_n \cdot S_n$ for all n.*

The important interpretation is that at time n, when the new prices $(S_n^0, ..., S_n^d)$ are quoted, the investor readjusts his position from ϕ_n to ϕ_{n+1} without bringing or consuming any wealth/money. The profit or loss realized by following a self-financing strategy is only due to the price movements.

Proposition 4.1.4. *For the self-financing condition, setting $\Delta S_{n+1} = S_{n+1} - S_n$, for all n it holds that:*

$$\phi_n \cdot S_n = \phi_{n+1} \cdot S_n \Longleftrightarrow V_{n+1}(\phi) = V_n(\phi) + \phi_{n+1} \cdot \Delta S_{n+1}. \tag{4.2}$$

Proof. We start from the self-financing condition and find:

$$\phi_n \cdot S_n = \phi_{n+1} \cdot S_n \Longleftrightarrow -\phi_n \cdot S_n = -\phi_{n+1} \cdot S_n \Longleftrightarrow \underbrace{\phi_{n+1} \cdot S_{n+1}}_{=V_{n+1}(\phi)} - \underbrace{\phi_n \cdot S_n}_{=V_n(\phi)} = \phi_{n+1} \cdot (S_{n+1} - S_n).$$

This reduces to:

$$V_{n+1}(\phi) = V_n(\phi) + \phi_{n+1} \cdot (S_{n+1} - S_n) =: V_n(\phi) + \phi_{n+1} \cdot \Delta S_{n+1}. \tag{4.3}$$

□

Proposition 4.1.5. *The following properties are all equivalent:*
(i) ϕ is self-financing.
(ii) For any $n \in \{1, ..., N\}$ and for $\Delta S_i = S_i - S_{i-1}$ we have:

$$V_n(\phi) = V_0(\phi) + \sum_{i=1}^{n} \phi_i \cdot \Delta S_i. \tag{4.4}$$

(iii) For any $n \in \{1, ..., N\}$ and for $\Delta \widetilde{S}_i = \widetilde{S}_i - \widetilde{S}_{i-1}$ we have:

$$\widetilde{V}_n(\phi) = V_0(\phi) + \sum_{i=1}^{n} \phi_i \cdot \Delta \widetilde{S}_i. \tag{4.5}$$

Remark 4.1.6. *Recall that S_i is a vector. Then we have that $\Delta \widetilde{S}_i = \widetilde{S}_i - \widetilde{S}_{i-1} = \frac{1}{S_i^0} S_i - \frac{1}{S_{i-1}^0} S_{i-1}$. Thus $\Delta \widetilde{S}_i^0 = \widetilde{S}_i^0 - \widetilde{S}_{i-1}^0 = \frac{1}{S_i^0} S_i^0 - \frac{1}{S_{i-1}^0} S_{i-1}^0 = 0$.*

Proof. • (i) $\Longrightarrow$ (ii) Let us assume that ϕ is self-financing and thus $\phi_{n+1}\cdot S_n = \phi_n \cdot S_n$. We prove this claim by induction on n. For the base case $n=1$, the self-financing condition reads $\phi_0\cdot S_0 = \phi_1\cdot S_0$ and thus $V_1(\phi) = V_0(\phi)+\phi_1\cdot\Delta S_1 = \phi_0\cdot S_0 - \phi_1\cdot S_0 + \phi_1\cdot S_1 = \phi_1\cdot S_1$ which corresponds to Definition (4.1) of $V_1(\phi)$. Let us now suppose that the claim (4.4) holds for the first n terms, and let's show that it still holds for the term $n+1$. We use the self-financing condition for a general term n and find:

$$\begin{aligned} V_{n+1}(\phi) = V_0(\phi) + \sum_{i=1}^{n+1}\phi_i\cdot\Delta S_i &= V_0(\phi) + \sum_{i=1}^{n}\phi_i\cdot\Delta S_i + \phi_{n+1}\cdot\Delta S_{n+1} \\ &= V_n(\phi) + \phi_{n+1}\cdot\Delta S_{n+1} \\ &= \underbrace{\phi_n\cdot S_n - \phi_{n+1}\cdot S_n}_{=0} + \phi_{n+1}\cdot S_{n+1} = \phi_{n+1}\cdot S_{n+1}. \end{aligned}$$

This corresponds precisely to Definition (4.1) of $V_{n+1}(\phi)$ and thus we are done.

• (ii) $\Longrightarrow$ (i) Let us assume that

$$V_n(\phi) = V_0(\phi) + \sum_{i=1}^{n}\phi_i\cdot\Delta S_i \tag{4.6}$$

holds for all n. Hence, for a fixed n we have that:

$$V_{n+1}(\phi) - V_n(\phi) = \phi_{n+1}\cdot\Delta S_{n+1} \Longleftrightarrow V_{n+1}(\phi) = V_n(\phi) + \phi_{n+1}\cdot\Delta S_{n+1}. \tag{4.7}$$

Hence by Proposition (4.1.4) the above equation holds if and only if $\phi_{n+1}\cdot S_n = \phi_n\cdot S_n$, which corresponds precisely to the definition of the self-financing condition.

• (i) $\Longleftrightarrow$ (iii) For all $n\in\{0,...,N-1\}$, setting $\beta_n := \frac{1}{S_n^0}$, from the self-financing condition we find:

$$\phi_n\cdot S_n = \phi_{n+1}\cdot S_n \Longleftrightarrow \phi_n\cdot\underbrace{(\beta_n S_n)}_{=\widetilde{S}_n} = \phi_{n+1}\cdot\underbrace{(\beta_n S_n)}_{=\widetilde{S}_n} \Longleftrightarrow \phi_{n+1}\cdot\widetilde{S}_{n+1} = \phi_n\cdot\widetilde{S}_n + \phi_{n+1}\cdot\underbrace{(\widetilde{S}_{n+1}-\widetilde{S}_n)}_{=\Delta\widetilde{S}_{n+1}}.$$

Hence similarly to (4.1.4), $\widetilde{V}_{n+1}(\phi) = \widetilde{V}_n(\phi) + \phi_{n+1}\cdot\Delta\widetilde{S}_{n+1}$. In an identical way as the proof (i) $\Longleftrightarrow$ (ii) replacing S_n by $\widetilde{S}_n$, it is not difficult to show that, using this equation, (i) $\Longleftrightarrow$ (iii).

□

Remark 4.1.7. *The above proposition tells us that we can equivalently make out considerations in discounted or in nominal (non-discounted) terms. However, it will turn out that the advantage of considering discounted terms, and hence, the power of this proposition is that discounted terms fit well into martingale related considerations. In view of the available machinery from martingale theory, this is a very considerable technical advantage.*
Furthermore note that, since $\Delta\widetilde{S}_j^0 = \beta_j S_j^0 - \beta_{j-1}S_{j-1}^0 = 0$, the value of the discounted portfolio is completely defined by the initial wealth and the strategy $(\phi_n^1,...,\phi_n^d)_{0\le n\le N}$.
[cf. Proposition (4.1.8)]

Proposition 4.1.8. *For any predictable process $(\phi_n^1, ..., \phi_n^d)_{n=0,...,N}$ and for any V_0, there exists a unique predictable process $(\phi_n^0)_{n=0,...,N}$ such that the strategy $\phi = (\phi_n^0, ..., \phi_n^d)$ is self-financing and its initial value is V_0.*

Proof. For all $n \in \{1, ..., N\}$, we use (4.5) given by $\widetilde{V}_n(\phi) = \sum_{i=0}^d \phi_n^i \cdot \widetilde{S}_n^i = V_0(\phi) + \sum_{i=1}^n \phi_i \cdot \Delta\widetilde{S}_i$ where $\phi_n^0 \widetilde{S}_n^0 = \phi_n^0$. Thus:

$$\begin{aligned}
\phi_n^0 &= V_0 + \sum_{i=1}^n \phi_i \cdot \Delta\widetilde{S}_i - \sum_{j=1}^d \phi_n^j \widetilde{S}_n^j \\
&= V_0 + \sum_{i=1}^{n-1} \phi_i \cdot \Delta\widetilde{S}_i + \phi_n^1(\widetilde{S}_n^1 - \widetilde{S}_{n-1}^1) + ... + \phi_n^d(\widetilde{S}_n^d - \widetilde{S}_{n-1}^d) - \phi_n^1 \widetilde{S}_n^1 - ... - \phi_n^d \widetilde{S}_n^d \\
&= V_0 + \underbrace{\sum_{i=1}^{n-1} \phi_i \cdot \Delta\widetilde{S}_i}_{(*)} - \phi_n^1 \widetilde{S}_{n-1}^1 - ... - \phi_n^d \widetilde{S}_{n-1}^d.
\end{aligned}$$

Now, the sum $(*)$ and the $\widetilde{S}_{n-1}^j$ are $\mathcal{F}_{n-1}$-measurable and the ϕ_n^j are all predictable. In other words, ϕ^0 is predictable and ϕ_n^0 is uniquely defined for all $n \in \{1, ..., N\}$. Furthermore:

$$\widetilde{V}_0(\phi) = V_0 = \phi_0^0 \cdot 1 + \phi_0^1 S_0^1 + ... + \phi_0^d S_0^d \iff \phi_0^0 = \underbrace{V_0 - \phi_0^1 S_0^1 - ... - \phi_0^d S_0^d}_{\mathcal{F}_0\text{- measurable + uniquely defined}}. \tag{4.8}$$

The strategy is also self-financing by the equivalences in Proposition (4.1.5).

□

Definition 4.1.9. *A strategy ϕ is **admissible** if it is self-financing and if $V_n(\phi) \geq 0$ for any $n \in \{0, ..., N\}$.*

Remark 4.1.10.
- *In the self-financing case we have that $\phi_0 \cdot S_0 = \phi_1 \cdot S_0$ where ϕ_1 is the investment for the period from 0 to 1, and in general the case ϕ_0 is not particularly interesting.*
- *We did not make any assumption on the sign of the quantities ϕ_n^i.*
- *If $\phi_n^0 < 0$ we have borrowed the amount $|\phi_n^0|$ in the risk-less asset.*
- *If $\phi_n^i < 0$ for $i \geq 1$ we have a short position in asset i where the amount is $|\phi_n^i|$.*
- *Here, short selling and borrowing are allowed.*
- *However, the value of the portfolio must remain non-negative at all times (able to pay back the debts at any time). This is the concept of admissibility [cf. Definition (4.1.9)].*

4.2 Arbitrage Opportunities

Definition 4.2.1. *An **arbitrage strategy** is an admissible strategy with zero initial value and non-vanishing probability of a non-vanishing final value, i.e. $V_0 = 0$ and $\mathbb{P}(V_N > 0) > 0$.*

Definition 4.2.2. *A market is **viable** if there is no arbitrage opportunity.*

We will shortly present the First Fundamental Theorem of Asset Pricing. Before we do so, we present one definition and one Theorem (without proof) that will be useful for the proof of FTAP1.

Definition 4.2.3. *A subset C of a vector space V is a **convex cone** if $\alpha X + \beta Y$ belongs to C for any $\alpha, \beta > 0$ and any $X, Y \in C$.*

Theorem 4.2.4 (Convex Separation Theorem - Linear Algebra version of Hahn-Banach)**.** *Consider $K \subset \mathbb{R}^n$ a convex and compact space along with V a subspace of $\mathbb{R}^n$. If $V \cap K = \varnothing$ then there is a linear function ξ defined on $\mathbb{R}^n$ satisfying: (i) for all $x \in K$, $\xi(x) > 0$ and (ii) for all $x \in V$, $\xi(x) = 0$. Hence, the subspace V is included in a hyperplane that does not intersect K.*

Theorem 4.2.5 (Fundamental Theorem of Asset Pricing 1 - FTAP1)**.** *The market is viable if and only if there exists a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ such that the discounted prices of assets are $\mathbb{Q}$-martingales.*

Remark 4.2.6. *The probability measure $\mathbb{Q}$ is not necessarily unique.*

Proof. ($\Longleftarrow$) Assume that $\mathbb{Q}$ is equivalent to $\mathbb{P}$ such that $\widetilde{S} - \mathbb{Q}$ (vector) martingale exists. For an admissible strategy ϕ, Proposition (4.1.5) part (iii) gives us that $\widetilde{V}_n(\phi) = V_0(\phi) + \sum_{j=1}^n \phi_j \cdot \Delta \widetilde{S}_j$. By the Martingale Transform (3.1.8) we have that $(\widetilde{V}_n(\phi))$ is a martingale and thus $\mathbb{E}_{\mathbb{Q}}(\widetilde{V}_N(\phi)) = \mathbb{E}_{\mathbb{Q}}(\widetilde{V}_0(\phi))$. Hence for an admissible strategy ϕ with $V_0(\phi) = \widetilde{V}_0(\phi) = 0$ we have that $\mathbb{E}_{\mathbb{Q}}(\widetilde{V}_N(\phi)) = 0$ with $\widetilde{V}_N(\phi) \geq 0$. Thus, for all $\omega \in \Omega$:

$$0 = \mathbb{E}_{\mathbb{Q}}(\widetilde{V}_N(\phi)) = \sum_{\omega \in \Omega} \underbrace{\widetilde{V}_N(\phi)(\omega)}_{\geq 0} \underbrace{\mathbb{Q}(\{\omega\})}_{>0} \Longrightarrow \widetilde{V}_N(\phi) = 0 \quad \mathbb{Q} \text{ a.s.}$$

Therefore, $\mathbb{Q}(\widetilde{V}_N(\phi) > 0) > 0$ is excluded, and since $\mathbb{Q}, \mathbb{P}$ are equivalent we finally obtain that $\mathbb{P}(\widetilde{V}_N(\phi) > 0) > 0$ is excluded. Hence we do not have any arbitrage, yielding a viable market.

($\Longrightarrow$) Showing the converse is a little bit more technical and will be done in six steps.

- (1) Define the convex cone Γ, in the sense of Definition (4.2.3), in the space of random variables

$$\Gamma := \{X : \text{non-negative random variables}, \mathbb{P}(X > 0) > 0\}. \tag{4.9}$$

This is indeed a convex cone: for $\alpha > 0$, $X \in \Gamma$ it holds that $\mathbb{P}(\alpha X > 0) > 0 \Longleftrightarrow \mathbb{P}(X > 0) > 0$. Hence, for $\alpha, \beta > 0$ and $X, Y \in \Gamma$, $\mathbb{P}(\alpha X + \beta Y > 0) \geq \mathbb{P}(\alpha X > 0) > 0$. Thus $\alpha X + \beta Y \in \Gamma$ for all $\alpha, \beta > 0$, $X, Y \in \Gamma$, which is precisely the definition of a convex cone.
Then, the market is viable if and only if for all admissible strategy ϕ with $V_0(\phi) = 0$ it implies that $\widetilde{V}_N(\phi) \notin \Gamma$.

• (2) Consider ϕ to be self-financing with $(\phi_n^1,...,\phi_n^d)$ a predictable process. Then, following the self-financing strategy given in Proposition (4.1.8) starting at $V_0 = 0$ [i.e. that there exists (ϕ_n^0) such that $(\phi_n^0,\phi_n^1,...,\phi_n^d)$ is self-financing with $V_0(\phi) = \widetilde{V}_0(\phi) = 0$], we may find the cumulative discounted gain process and it is given by:

$$\widetilde{G}_n(\phi) := \sum_{j=1}^{n} \phi_j \cdot \Delta\widetilde{S}_j. \tag{4.10}$$

For this strategy with $\widetilde{V}_n(\phi) = \widetilde{G}_n(\phi)$, since ϕ is admissible we obtain that $\widetilde{G}_n(\phi) \geq 0$ for all $n = \{1,...,N\}$ and with $\widetilde{G}_N(\phi) = 0$ by assumption/viable market.

• (3) This step concerns the statement and proof of the following Lemma which, in a sense, gets rid of the admissibility condition.

Lemma 4.2.7. *If a market is viable, any predictable process $(\phi^1,...,\phi^d)$ satisfies $\widetilde{G}_N(\phi) \notin \Gamma$.*

Proof. Seeking for a contradiction, suppose that $\widetilde{G}_N(\phi) \in \Gamma$. First, if $\widetilde{G}_N(\phi) \geq 0$ for all $n \in \{0,1,...,N\}$ the market is obviously not viable. Second, if the $\widetilde{G}_N(\phi)$'s are not all non-negative, consider the last n such that $\mathbb{P}(\widetilde{G}_n(\phi) < 0) > 0$. Hence the following three properties hold: (i) $n \leq N-1$, (ii) $\mathbb{P}(\widetilde{G}_n(\phi) < 0) > 0$ and (iii) for all $m > n$, $\widetilde{G}_m(\phi) \geq 0$ holds for this n. Now, define a new process ψ by:

$$\psi_j(\omega) := \begin{cases} 0 & \text{if } j \leq n \\ \mathbb{1}_A(\omega)\phi_j(\omega) & \text{if } j > n \end{cases} \tag{4.11}$$

for $A = \{\widetilde{G}_n(\phi) < 0\} \in \mathcal{F}_n$. Thus ψ is predictable with:

$$\widetilde{G}_j(\psi) := \begin{cases} 0 & \text{if } j \leq n \\ \mathbb{1}_A\Big(\widetilde{G}_j(\phi) - \widetilde{G}_n(\phi)\Big) & \text{if } j > n \end{cases} \tag{4.12}$$

where on A it holds that $\widetilde{G}_j(\phi) \geq 0$ and $\widetilde{G}_n(\phi) > 0$. Therefore, $\widetilde{G}_j(\psi) \geq 0$ for all $j \in \{0,...,N\}$ and $\widetilde{G}_N(\psi) > 0$ on A. Contradiction! □

• (4) Let $\mathbb{R}^\Omega := \{\Omega \longrightarrow \mathbb{R}, \text{random variable}\}$ be the space of random variables defined on our probability space. In our case, with a little bit of thought, the space of non-negative random variables corresponds to $\mathbb{R}^2_+$ and hence Γ corresponds to $\mathbb{R}^2_+ \backslash \{0\}$. Then,

$$K := \left\{ X \in \mathbb{R}^\Omega : X(\omega) \geq 0 \text{ for all } \omega, \sum_\omega X(\omega) = 1 \right\} \subset \Gamma \tag{4.13}$$

is the unit simplex in the Euclidean space, which is furthermore convex and compact. Note that any convex combination of elements $X, Y \in K$ with $c \in [0,1]$ is also in the simplex. Indeed, let $Z := cX + (1-c)Y$ then:

$$\sum_\omega Z(\omega) = c\sum_\omega X(\omega) + (1-c)\sum_\omega Y(\omega) = c + (1-c) = 1 \Longrightarrow Z \in K.$$

Finally, we define the space

$$\mathcal{V} := \{\text{random variables } \widetilde{G}_N(\phi)\colon\ \phi \text{ is predictable}\} \tag{4.14}$$

which is in fact a subspace of $\mathbb{R}^\Omega$. Indeed, we check that any linear combination of two elements of $\mathcal{V}$ is again in $\mathcal{V}$. To this end, let $\alpha, \beta \in \mathbb{R}$ and $\phi, \widehat{\phi} \in \mathcal{V}$, then:

$$\alpha\widetilde{G}_N(\phi) + \beta\widetilde{G}_N(\widehat{\phi}) = \alpha\sum_{j=1}^N \phi_j \cdot \Delta\widetilde{S}_j + \beta\sum_{j=1}^N \widehat{\phi}_j \cdot \Delta\widetilde{S}_j = \sum_{j=1}^N \left(\alpha\phi_j + \beta\widehat{\phi}_j\right)\cdot\Delta\widetilde{S}_j = \widetilde{G}_N\left(\alpha\phi + \beta\widehat{\phi}\right) \in \mathcal{V}.$$

By Lemma (4.2.7) and the assumption of the viable market it holds that $\mathcal{V}\cap\Gamma = \varnothing$ and therefore $\phi = \mathcal{V}\cap K$, where we recall that $K = \{X \in \Gamma : \sum_\omega X(\omega) = 1\}$.

• (5) We now use Theorem (4.2.4) with $\phi = \mathcal{V}\cap K$ and hence there is a random variable λ such that (i) $\sum_\omega \lambda(\omega)X(\omega) > 0$ for all $X \in K$ and (ii) $\sum_\omega \lambda(\omega)\widetilde{G}_N(\phi)(\omega) = 0$ for all $\widetilde{G}_N(\phi) \in \mathcal{V}$. Note now that $\mathbb{1}_{\omega_i}$ is in the unit simplex K. Hence, using the fact that $\sum_\omega \lambda(\omega)X(\omega) > 0$ for all $X \in K$ and then setting $X = \mathbb{1}_{\omega_i}$ for all ω_i we find that $\lambda(\omega) > 0$ for all $\omega \in \Omega$. Thus,

$$\mathbb{Q}(\{\omega\}) := \frac{\lambda(\omega)}{\sum_{\omega'\in\Omega}\lambda(\omega')} > 0 \qquad \text{for all } \omega \in \Omega \tag{4.15}$$

defines a probability measure equivalent to $\mathbb{P}$ satisfying $\sum_\omega \mathbb{Q}(\{\omega\}) = \frac{\sum_{\omega\in\Omega}\lambda(\omega)}{\sum_{\omega'\in\Omega}\lambda(\omega')} = 1$.

• (6) Since by (ii) in Theorem (4.2.4) we have that $\sum_\omega \lambda(\omega)\widetilde{G}_N(\phi)(\omega) = 0$ for any predictable ϕ then it holds that:

$$0 = \sum_\omega \frac{\lambda(\omega)}{\sum_{\omega'\in\Omega}\lambda(\omega')}\widetilde{G}_N(\phi)(\omega) = \mathbb{E}_\mathbb{Q}(\widetilde{G}_N(\phi)) = \mathbb{E}_\mathbb{Q}\left(\sum_{j=1}^N \phi_j\cdot\Delta\widetilde{S}_j\right) = \mathbb{E}_\mathbb{Q}\left(\sum_{j=1}^N\sum_{i=1}^d \phi_j^i\Delta\widetilde{S}_j^i\right)$$

for any predictable ϕ. Hence in particular, for $\phi = (0,...,0,\phi_j^k,0,...,0)$ predictable it holds that:

$$\mathbb{E}_\mathbb{Q}\left(\sum_{j=1}^N \phi_j^k\Delta\widetilde{S}_j^k\right) = 0 \qquad \text{for all } k \in \{1,...,d\}. \tag{4.16}$$

By Proposition (3.1.12) we have that $(\widetilde{S}_n^1),...,(\widetilde{S}_n^d)$ are $\mathbb{Q}$-martingales.

□

Remark 4.2.8 (**Important!**). *Assume that a market is viable. Hence,*

$$(\widetilde{S}_n^1,...,\widetilde{S}_n^d) = \left(\frac{S_n^1}{S_n^0},...,\frac{S_n^d}{S_n^0}\right)$$

is a $\mathbb{Q}$-martingale for at least one existing equivalent martingale measure $\mathbb{Q}$. Choose this $\mathbb{Q}$. Then, for X being the claim/payoff derivative set $(*)$ to be $V_n := S_n^0\mathbb{E}_\mathbb{Q}\left(\frac{X}{S_N^0}\middle|\mathcal{F}_n\right)$.

Hence for,

$$\widetilde{V}_n := \frac{V_n}{S_n^0} = \mathbb{E}_{\mathbb{Q}}\left(\frac{X}{S_N^0}\Big|\mathcal{F}_n\right) = \mathbb{E}_{\mathbb{Q}}(\widetilde{X}|\mathcal{F}_n) \tag{4.17}$$

the Tower property gives us that

$$\mathbb{E}_{\mathbb{Q}}(\widetilde{V}_{n+1}|\mathcal{F}_n) = \mathbb{E}_{\mathbb{Q}}\Big(\mathbb{E}_{\mathbb{Q}}(\widetilde{X}|\mathcal{F}_{n+1})|\mathcal{F}_n\Big) = \mathbb{E}_{\mathbb{Q}}(\widetilde{X}|\mathcal{F}_n) = \widetilde{V}_n \Longrightarrow (\widetilde{V}_n) \text{ is a } \mathbb{Q}\text{-martingale.}$$

Thus the vector $(\widetilde{S}_n^1, ..., \widetilde{S}_n^d, \widetilde{V}_n)$ is a $\mathbb{Q}$-martingale and hence,

$$(*) = V_n := S_n^0 \mathbb{E}_{\mathbb{Q}}\left(\frac{X}{S_N^0}\Big|\mathcal{F}_n\right) \tag{4.18}$$

is an **arbitrage free pricing rule**. Recall that $\mathbb{Q}$ is not necessarily unique!

4.3 Complete Markets and Option Pricing

Definition 4.3.1. *In Finance, a **contingent claim** is a non-negative $\mathcal{F}_N$-measurable random variable.*

Example 4.3.2. *All the following are contingent claims:*
(i) European digital given by $H = \mathbb{1}_{\{S_N > K\}}$.
(ii) European call given by $H = (S_N - K)_+$, for $K \geq 0$.
(iii) European put given by $H = (K - S_N)_+$, for $K \geq 0$.

Definition 4.3.3 (Attainable claim)**.** *The contingent claim defined by H is **attainable** if there exists an admissible strategy worth H at time N.*

Definition 4.3.4. *The market is **complete** if every contingent claim is attainable.*

Remark 4.3.5. *In a viable financial market, we just need to find a self-financing strategy worth H at maturity so that H is attainable. We then need a self-financing strategy ϕ and a $\mathbb{Q}$-equivalent martingale measure. By the Martingale Transform (3.1.8), $(\widetilde{V}_n(\phi))$ is a martingale under $\mathbb{Q}$. And thus:*

$$\widetilde{V}_n(\phi) = \mathbb{E}_{\mathbb{Q}}(\widetilde{V}_N(\phi)|\mathcal{F}_n) = \mathbb{E}_{\mathbb{Q}}\left(\frac{H}{S_N^0}\Big|\mathcal{F}_n\right) \geq 0, \tag{4.19}$$

i.e. there exists an admissible replicating strategy.

Theorem 4.3.6 (Fundamental Theorem of Asset Pricing 2 - FTAP2)**.** *A viable market is complete if and only if there exists a **unique** probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$, under which the discounted prices are martingales.*

Proof. ($\Longrightarrow$) Assume we have a viable and complete market. Then there exists an admissible strategy ϕ associated to a contingent claim $H = V_N(\phi)$. Thus, since ϕ is self-financing:

$$\frac{H}{S_N^0} = \widetilde{V}_N(\phi) = V_0(\phi) + \sum_{j=1}^{N} \phi_j \cdot \Delta \widetilde{S}_j. \tag{4.20}$$

Now, if $\mathbb{Q}_1$ and $\mathbb{Q}_2$ are two equivalent martingale measures then by the Martingale Transform (3.1.8) we have that $(\widetilde{V}_n(\phi))$ is a $\mathbb{Q}_1$ and $\mathbb{Q}_2$-martingales. Then, for $i \in \{1,2\}$, $\mathbb{E}_{\mathbb{Q}_i}(\widetilde{V}_N(\phi)) = \mathbb{E}_{\mathbb{Q}_i}(V_0(\phi)) = V_0(\phi)$, where the last equality holds in the case where $\mathcal{F}_0 = \{\varnothing, \Omega\}$. Hence,

$$\mathbb{E}_{\mathbb{Q}_1}\left(\frac{H}{S_N^0}\right) = \mathbb{E}_{\mathbb{Q}_2}\left(\frac{H}{S_N^0}\right) \tag{4.21}$$

is a H-arbitrary claim, i.e. in particular for $H = \mathbb{1}_A S_N^0$ with $A \in \mathcal{F}_N$. Thus, $\mathbb{Q}_1 = \mathbb{Q}_2$ on the whole σ-algebra $\mathcal{F}_N = \mathcal{F}$.

($\Longleftarrow$) Instead of the usual statement, we show its contrapositive (i.e. not complete implies not unique). Thus, assume that we have a viable and incomplete market. Hence there exists $H \geq 0$ such that H is not attainable. We first define

$$\widetilde{\mathcal{V}} := \left\{ \text{random variables } Z : Z = U_0 + \sum_{n=1}^{N} \phi_n \cdot \Delta \widetilde{S}_n \right\} \tag{4.22}$$

where U_0 is $\mathcal{F}_0$-measurable and $(\phi_n^1, ..., \phi_n^d)_{0 \leq n \leq N}$ is predictable. Let us first present a strategy for the proof:
- Use that there is a claim not in $\widetilde{\mathcal{V}}$ to construct a second measure (the first exists in a viable market).
- Show that this measure is equivalent but different to the first one.
- Show that it is a probability measure.
- Show that it is a martingale measure.

Since there exists $H \geq 0$ that is not attainable, then $\frac{H}{S_N^0} \notin \widetilde{\mathcal{V}}$ implies $\widetilde{\mathcal{V}} \not\subseteq \{\text{random variable on } (\Omega, \mathcal{F})\}$.
[Note that if a predictable strategy and a $\mathcal{F}_0$-measurable U_0 would exist such that $\frac{H}{S_N^0} \in \widetilde{\mathcal{V}}$ for all claims H, then by Remark (4.3.5) and Proposition (4.1.8) the market would be complete.]
Now, if $\mathbb{Q}_1$ is an equivalent martingale measure, define the scalar product

$$(X, Y) \longmapsto \mathbb{E}_{\mathbb{Q}_1}(XY). \tag{4.23}$$

Then there exists a non-zero random variable X orthogonal to $\widetilde{\mathcal{V}}$ (in the non-trivial orthogonal complement). Now, recall that the sup-norm is defined as $\|X\|_\infty := \sup_{\omega \in \Omega} |X(\omega)|$ and define:

$$\mathbb{Q}_2(\{\omega\}) := \underbrace{\left(1 + \frac{X(\omega)}{2\|X\|_\infty}\right)}_{>0} \underbrace{\mathbb{Q}_1(\{\omega\})}_{>0} > 0. \tag{4.24}$$

Remark that:

$$\sum_{\omega}\mathbb{Q}_2(\{\omega\})=\sum_{\omega}\left(1+\frac{X(\omega)}{2\|X\|_\infty}\right)\mathbb{Q}_1(\{\omega\})=\underbrace{\sum_{\omega}\mathbb{Q}_1(\{\omega\})}_{=1}+\frac{1}{2\|X\|_\infty}\underbrace{\mathbb{E}_{\mathbb{Q}_1}(X)}_{=0}=1.$$

Since $\phi=(0,...,0)$ is predictable we have that $0<U_0\in\widetilde{\mathcal{V}}$, and since X is in the orthogonal complement of $\widetilde{\mathcal{V}}$ then $\mathbb{E}_{\mathbb{Q}_1}(U_0X)=0=U_0\mathbb{E}_{\mathbb{Q}_1}(X)\Longrightarrow\mathbb{E}_{\mathbb{Q}_1}(X)=0$.
Hence $\mathbb{Q}_2$ is a new probability measure, satisfying $\sum_\omega\mathbb{Q}_2(\{\omega\})=1$, equivalent to $\mathbb{P}$ (since $\mathbb{Q}_2(\{\omega\})>0$) but different from $\mathbb{Q}_1$ (since X is non-trivial). Moreover, for $Z\in\widetilde{\mathcal{V}}$:

$$\mathbb{E}_{\mathbb{Q}_2}(Z)=\mathbb{E}_{\mathbb{Q}_2}\left(\sum_{n=1}^{N}\phi_n\cdot\Delta\widetilde{S}_n\right)=\sum_{\omega\in\Omega}Z(\omega)\mathbb{Q}_2(\{\omega\})=\underbrace{\sum_{\omega\in\Omega}Z(\omega)\mathbb{Q}_1(\{\omega\})}_{(*)}+\underbrace{\sum_{\omega\in\Omega}Z(\omega)\frac{X(\omega)}{2\|X\|_\infty}\mathbb{Q}_1(\{\omega\})}_{(**)}.$$

Using the Martingale Transform (3.1.8) under $\mathbb{Q}_1$ we have that $(*)=\mathbb{E}_{\mathbb{Q}_1}(Z)=0$. Furthermore, since X is orthogonal to $Z\in\widetilde{\mathcal{V}}$ we have that $\mathbb{E}_{\mathbb{Q}_1}(ZX)=0$ and thus $(**)=\mathbb{E}_{\mathbb{Q}_1}(ZX)\frac{1}{2\|X\|_\infty}=0$.
Therefore, for all predictable ϕ:

$$\mathbb{E}_{\mathbb{Q}_2}\left(\sum_{n=1}^{N}\phi_n\cdot\Delta\widetilde{S}_n\right)=0\Longrightarrow\left(\widetilde{S}_n\right)_{0\le n\le N}\text{ is a }\mathbb{Q}_2\text{-(vector) martingale.}$$

□

Remark 4.3.7. *The probability measure $\mathbb{Q}$ is a tool for deriving pricing formulas for options.*

Let us now look at pricing and hedging contingent claims in complete markets. For this, consider a viable and complete market with $\mathbb{Q}$ being the unique equivalent martingale measure. Furthermore, consider a contingent claim H. Under these assumptions there exists a self-financing ϕ_H with $V_N(\phi_H)=H$. We then have $\widetilde{V}_N(\phi_H)=\widetilde{H}$ and $\widetilde{V}_n(\phi_H)=\mathbb{E}_{\mathbb{Q}}[\widetilde{V}_N(\phi_H)|\mathcal{F}_n]=\mathbb{E}_{\mathbb{Q}}[\widetilde{H}|\mathcal{F}_n]$. And since $H\ge0$ implies that $V_n\ge0$ we have that the strategy ϕ_H is also admissible.
Furthermore, at any time, the value of an admissible strategy replicating H is completely determined by H. Thus $V_n(\phi_H)$ is the **value of the contingent claim at time** n. If at time 0 an investor sells the option for $\mathbb{E}_{\mathbb{Q}}(\widetilde{H})$, he can follow a replicating strategy ϕ in order to generate an amount H at time N and thus the investor is perfectly hedged.

Remark 4.3.8. *We note that the computation of the option price only requires the knowledge of $\mathbb{Q}$ but not of $\mathbb{P}$.*

4.4 Cox, Ross and Rubinstein Model

Definition of the Model

Trading dates$\longrightarrow n=0,1,...,N$ with $N\in\mathbb{N}^*$.

Risk-less asset$\longrightarrow S_n^0=(1+r)^n$ with $r>0$.

Risky asset$\longrightarrow$ there is only one risky asset with prices given by the vector $(S_0, S_1, ..., S_N)$.

Transition$\longrightarrow$ Assuming $-1 < a < b$ and for all $n = 1, ..., N$ we have:

$$S_n = \begin{cases} S_{n-1}(1+a) \\ S_{n-1}(1+b) \end{cases} . \tag{4.25}$$

Probability space$\longrightarrow$ $\Omega = \{(1+a), (1+b)\}^N$, $\mathcal{F} = \mathcal{P}(\Omega)$ with $\mathcal{F}_0 = \{\Omega, \varnothing\}$ and $\mathcal{F}_n = \sigma(S_1, S_2, ..., S_n)$.

Some results

$\longrightarrow$ It is necessary that $r \in (a, b)$ in order to end up with a viable market.

$\longrightarrow$ The price of a European call is given by:

$$C(0, S_0) = \frac{1}{(1+r)^N} \sum_{j=0}^{N} \frac{N!}{j!(N-j)!} p^j (1-p)^{N-j} \left[S_0 (1+a)^j (1+b)^{N-j} - K \right]_+ . \tag{4.26}$$

Chapter 5

The Snell Envelope, Optimal Stopping and American Options

Recall that an American option can be exercised at any time until maturity. Hence, at any n the buyer of an American option has to decide whether or not he wants to exercise the option. Some immediate questions arise: When should the long position stop/exercise? How can the short position hedge the exposure? What about arbitrage and the FTAP?
In view of this setup, it is reasonable to define a sequence (Z_n) representing at any n the realized payoff if the option is exercised at time n. Clearly, if the option is not exercised before N then its value U_N at N is Z_N.

Example 5.0.1. *All the following are examples of payoffs:*
(i) European call given by $H = (S_N - K)_+$, *for* $K \geq 0$.
(ii) European put given by $H = (K - S_N)_+$, *for* $K \geq 0$.

5.1 The Snell Envelope: Backward Induction and Optimal Stopping

Take the value of the (non-exercised) option at N to be known. If we consider the situation at $N-1$ we have two possibilities: (1) Wait and keep the option to obtain Z_N at N, or (2) exercise the option and we immediately obtain Z_{N-1}. While the buyer (with the long position) of the option can choose whether or not he wants to exercise the option, the seller (with the short position) has to be ready to settle the option in both cases.
From now on, we assume that the market is **viable and complete**. Thus, there exists a unique risk-neutral probability measure $\mathbb{Q}$ under which the discounted asset price processes are martingales. Furthermore, we recall that completeness means that claims may be replicated. Now, since the market is viable and complete we have that $\widetilde{Z}_n = \frac{Z_N}{S_N^0}$ where $S_n^0 = (1+r)^n$. We are looking for $\widetilde{U}_N$ such that ,

$$S_{N-1}^0 \mathbb{E}_{\mathbb{Q}}\left(\widetilde{U}_N | \mathcal{F}_{N-1}\right) = S_{N-1}^0 \mathbb{E}_{\mathbb{Q}}\left(\widetilde{Z}_N | \mathcal{F}_{N-1}\right). \tag{5.1}$$

For that, we want the maximum between Z_{N-1} and the value at time $N-1$ of an admissible strategy paying Z_N at time N. This is the value of the American option at time $N-1$,

$$U_{N-1} = \max\left(Z_{N-1}, S_{N-1}^0(\widetilde{Z}_N | \mathcal{F}_{N-1})\right). \tag{5.2}$$

By backward induction, for all $n \in \{1, ..., N\}$,

$$U_{n-1} = \max\left(Z_{n-1}, S^0_{n-1}\mathbb{E}_{\mathbb{Q}}\left(\frac{U_n}{S^0_n}\Big|\mathcal{F}_{n-1}\right)\right) \xrightarrow{\text{discount}} \widetilde{U}_{n-1} = \max\left(\widetilde{Z}_{n-1}, \mathbb{E}_{\mathbb{Q}}\left(\widetilde{U}_n|\mathcal{F}_{n-1}\right)\right). \tag{5.3}$$

The sequence $(\widetilde{U}_n)$ is the **Snell envelope** of $(\widetilde{Z}_n)$, where above $U_N = Z_N$. Remark that $\widetilde{U}_n$ is used in the conditional expectation, not $\widetilde{Z}_n$.

Proposition 5.1.1. *The sequence $(\widetilde{U}_n)$ is a supermartingale under $\mathbb{Q}$. Furthermore, it is the smallest $\mathbb{Q}$-supermartingale that dominates the sequence $(\widetilde{Z}_n)$ [i.e. $\widetilde{U}_n \geq \widetilde{Z}_n$ for all $n \in \{0, ..., N\}$].*

Proof. • We first prove that $(\widetilde{U}_n)$ is a supermartingale. We simply observe that,

$$\mathbb{E}_{\mathbb{Q}}\left(\widetilde{U}_n|\mathcal{F}_{n-1}\right) \leq \max\left(\widetilde{Z}_{n-1}, \mathbb{E}_{\mathbb{Q}}(\widetilde{U}_n|\mathcal{F}_{n-1})\right) = \widetilde{U}_{n-1}. \tag{5.4}$$

Thus for all n we must have that $\mathbb{E}_{\mathbb{Q}}(\widetilde{U}_n|\mathcal{F}_{n-1}) \leq \widetilde{U}_{n-1}$ which implies that $(\widetilde{U}_n)$ is a supermartingale under $\mathbb{Q}$.

• We next show the domination. Again we observe that,

$$\widetilde{Z}_{n-1} \leq \max\left(\widetilde{Z}_{n-1}, \mathbb{E}_{\mathbb{Q}}(\widetilde{U}_n|\mathcal{F}_{n-1})\right) = \widetilde{U}_{n-1}. \tag{5.5}$$

It is clear that $(\widetilde{U}_n)$ dominates $(\widetilde{Z}_n)$.

• We finally show that $(\widetilde{U}_n)$ is the smallest. Assume that $(\widetilde{T}_n)$ is another supermartingale that dominates $(\widetilde{Z}_n)$. Since $\widetilde{U}_N = \widetilde{Z}_N$ and by assumption $\widetilde{T}_N \geq \widetilde{Z}_N$ then it holds that $\widetilde{T}_N \geq \widetilde{U}_N$. Thus since $(\widetilde{T}_n)$ is a supermartingale we find,

$$\widetilde{T}_{N-1} \geq \mathbb{E}_{\mathbb{Q}}(\widetilde{T}_N|\mathcal{F}_{N-1}) \geq \mathbb{E}_{\mathbb{Q}}(\widetilde{U}_N|\mathcal{F}_{N-1}). \tag{5.6}$$

Since $(\widetilde{T}_n)$ dominates $(\widetilde{Z}_n)$, combining these results yields,

$$\widetilde{T}_{N-1} \geq \max\left(\mathbb{E}_{\mathbb{Q}}(\widetilde{U}_N|\mathcal{F}_{N-1}), \widetilde{Z}_{N-1}\right) = \widetilde{U}_{N-1}. \tag{5.7}$$

Now, by backward induction, for $\widetilde{T}_n \geq \widetilde{U}_n$ we find that,

$$\widetilde{T}_{n-1} \geq \mathbb{E}_{\mathbb{Q}}(\widetilde{T}_n|\mathcal{F}_{n-1}) \geq \mathbb{E}_{\mathbb{Q}}(\widetilde{U}_n|\mathcal{F}_{n-1}). \tag{5.8}$$

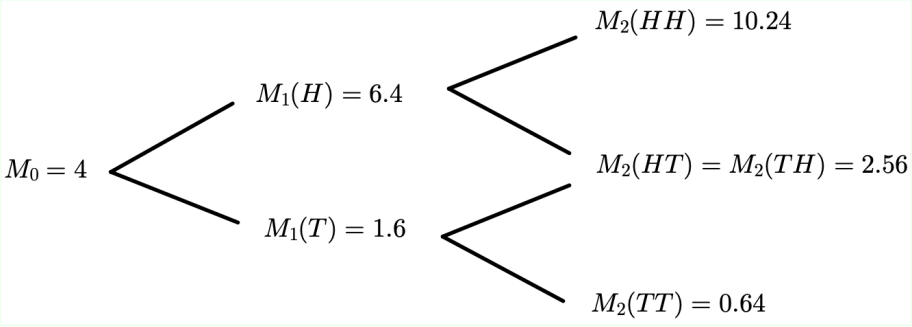
Since $(\widetilde{T}_n)$ dominates $(\widetilde{Z}_n)$, combining these results yields,

$$\widetilde{T}_{n-1} \geq \max\left(\mathbb{E}_{\mathbb{Q}}(\widetilde{U}_n|\mathcal{F}_{n-1}), \widetilde{Z}_{n-1}\right) = \widetilde{U}_{n-1}. \tag{5.9}$$

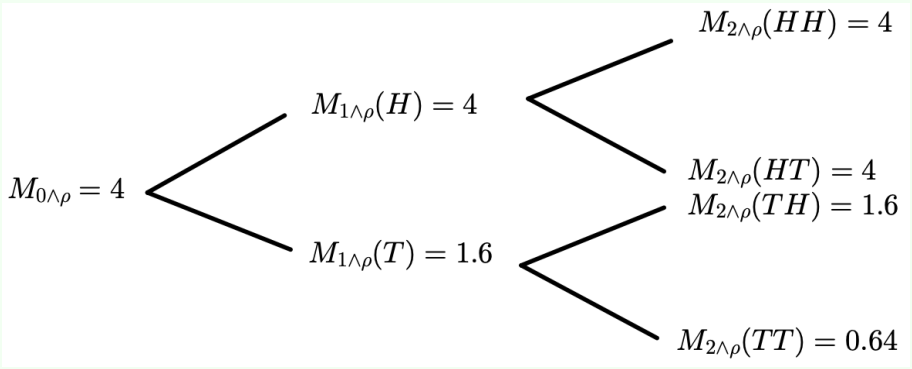
This proves that $(\widetilde{T}_n)$ dominates $(\widetilde{U}_n)$, hence $(\widetilde{U}_n)$ is the smallest $\mathbb{Q}$-supermartingale that dominates $(\widetilde{Z}_n)$.

□

Example 5.1.2 (Random Stopping)**.** *Consider two coin tosses such that* $\Omega = \{HH, HT, TH, TT\}$ *and* $H = \{H \text{ first}\} = \{HH, HT\}$ *and* $T = \{T \text{ first}\} = \{TH, TT\}$*. Define the following tree:*



Suppose we have independent increments such that $p = 1 - p = \frac{1}{2}$*. Then* $\mathbb{E}(M_2|H) = \frac{1}{2}(10.24 + 2.56) = 6.4$*,* $\mathbb{E}(M_2|T) = \frac{1}{2}(2.56 + 0.64) = 1.6$ *and* $\mathbb{E}(M_1|\mathcal{F}_0) = \frac{1}{2}(6.4 + 1.6) = 4 = M_0$ *where* $\mathcal{F}_0 = \{\varnothing, \Omega\}$*. Thus* M *is a martingale. Now define* $\rho(HH) = \rho(HT) = 0$*,* $\rho(TH) = 1$ *and* $\rho(TT) = 2$*. We now consider the following tree:*



Remark that $\mathbb{E}(M_{2\wedge\rho}) = \frac{1}{4}(4 + 4 + 1.6 + 0.64) = 2.56 < M_0 = 4$*. Thus* $(M_{n\wedge\rho})$ *is not a martingale, and in fact* ρ *is a random time but not a stopping time.*

Proposition 5.1.3. *The random variable defined by* $\nu = \inf\{n \geq 0 : \widetilde{U}_n = \widetilde{Z}_n\}$ *is a stopping time. Furthermore, the stopped sequence* $(\widetilde{U}_n^\nu)$ *is a martingale.*

Proof. • Since $\widetilde{U}_N = \widetilde{Z}_N$ is a well-defined element of $\{0, 1, ..., N\}$ with $\{\nu = 0\} = \{\widetilde{U}_0 = \widetilde{Z}_0\} \in \mathcal{F}_0$ and for $k \geq 1$, $\{\nu = k\} = \bigcap_{i=0}^{k-1}\{\widetilde{Z}_i < \widetilde{U}_i\} \cap \{\widetilde{U}_k = \widetilde{Z}_k\} \in \mathcal{F}_k$ we have that ν is a stopping time.

• We now show that $(\widetilde{U}_n^\nu)$ is a martingale. Similarly as before, define,

$$\widetilde{U}_n^\nu = \widetilde{U}_{n\wedge\nu} = \widetilde{U}_0 + \sum_{j=1}^{n} \mathbb{1}_{\{j \leq \nu\}} \Delta\widetilde{U}_j, \tag{5.10}$$

where $\mathbb{1}_{\{j\leq\nu\}}$ is a predictable sequence with $\{j \leq \nu\}^c = \{j > \nu\} = \{\nu \leq j - 1\}$. Hence $(\widetilde{U}_n^\nu)$ is adapted. We now look at the difference,

$$\widetilde{U}^{\nu}_{n+1} - \widetilde{U}^{\nu}_{n} = \mathbb{1}_{\{n+1\leq\nu\}}(\widetilde{U}_{n+1} - \widetilde{U}_n). \tag{5.11}$$

Now, on $\{n+1 \leq \nu\}$ one has $\widetilde{U}_n > \widetilde{Z}_n$ and recall that in general $\widetilde{U}_n = \max\Big(\widetilde{Z}_n, \mathbb{E}(\widetilde{U}_{n+1}|\mathcal{F}_n)\Big)$. Thus, combining these two facts, on $\{n+1 \leq \nu\}$ we have $\mathbb{E}(\widetilde{U}_{n+1}|\mathcal{F}_n) = \widetilde{U}_n$. Hence, equation (5.11) reads as,

$$\widetilde{U}^{\nu}_{n+1} - \widetilde{U}^{\nu}_{n} = \mathbb{1}_{\{n+1\leq\nu\}}(\widetilde{U}_{n+1} - \mathbb{E}(\widetilde{U}_{n+1}|\mathcal{F}_n)). \tag{5.12}$$

Thus taking $\mathbb{E}(\cdot|\mathcal{F}_n)$ on both sides yields,

$$\begin{aligned}\mathbb{E}(\widetilde{U}^{\nu}_{n+1} - \widetilde{U}^{\nu}_{n}|\mathcal{F}_n) &= \mathbb{E}\Big(\mathbb{1}_{\{n+1\leq\nu\}}(\widetilde{U}_{n+1} - \mathbb{E}(\widetilde{U}_{n+1}|\mathcal{F}_n))|\mathcal{F}_n\Big) \\ &= \mathbb{1}_{\{n+1\leq\nu\}}\Big(\mathbb{E}(\widetilde{U}_{n+1}|\mathcal{F}_n) - \mathbb{E}(\widetilde{U}_{n+1}|\mathcal{F}_n)\Big) = 0.\end{aligned}$$

Therefore, $\mathbb{E}(\widetilde{U}^{\nu}_{n+1} - \widetilde{U}^{\nu}_{n}|\mathcal{F}_n) = 0$ implies that $\mathbb{E}(\widetilde{U}^{\nu}_{n+1}|\mathcal{F}_n) = \mathbb{E}(\widetilde{U}^{\nu}_{n}|\mathcal{F}_n) = \widetilde{U}^{\nu}_{n}$ concluding that $(\widetilde{U}^{\nu}_{n})$ is a martingale.

□

Definition 5.1.4. *Denote by $\mathcal{T}_{n,N}$ the set of all stopping times taking values in $\{n, n+1, ..., N\}$.*

Corollary 5.1.5. *The stopping time ν satisfies,*

$$\widetilde{U}_0 = \mathbb{E}_{\mathbb{Q}}(\widetilde{Z}_\nu) = \sup_{\tau\in\mathcal{T}_{0,N}} \mathbb{E}_{\mathbb{Q}}(\widetilde{Z}_\tau). \tag{5.13}$$

Proof. Since $(\widetilde{U}^{\nu}_{n})$ is a martingale, then $\widetilde{U}_0 = \widetilde{U}^{\nu}_{0} = \mathbb{E}(\widetilde{U}^{\nu}_{N})$. Now, ν takes values in $\{0, 1, ..., N\}$ and thus,

$$\mathbb{E}(\widetilde{U}^{\nu}_{N}) = \sum_{k=0}^{N} \mathbb{E}(\widetilde{U}_{N\wedge\nu}|\nu = k)\mathbb{Q}(\nu = k) = \sum_{k=0}^{N} \mathbb{E}(\widetilde{U}_{k}|\nu = k)\mathbb{Q}(\nu = k) = \sum_{k=0}^{N} \mathbb{E}(\widetilde{U}_{\nu}|\nu = k)\mathbb{Q}(\nu = k) = \mathbb{E}(\widetilde{U}_{\nu}).$$

Therefore, $\widetilde{U}_0 = \widetilde{U}^{\nu}_{0} = \mathbb{E}(\widetilde{U}^{\nu}_{N}) = \mathbb{E}(\widetilde{U}_{\nu}) = \mathbb{E}(\widetilde{Z}_\nu)$ where we have used the definition of ν in the last equality. Finally, if $\tau \in \mathcal{T}_{0,N}$ then by Proposition (3.2.4) $(\widetilde{U}^{\tau})$ is a supermartingale. In particular, $\widetilde{U}_0 \geq \mathbb{E}(\widetilde{U}^{\tau}_{N}) = \mathbb{E}(\widetilde{U}_\tau) \geq \mathbb{E}(\widetilde{Z}_\tau)$ where we have used the fact that $\widetilde{U}$ dominates $\widetilde{Z}$ in the last inequality. Hence for all $\tau \in \mathcal{T}_{0,N}$ we have $\mathbb{E}(\widetilde{Z}_\tau) \leq \widetilde{U}_0 = \mathbb{E}(\widetilde{Z}_\nu)$.

□

Remark 5.1.6. *An immediate generalization of Corollary (5.1.5) yields,*

$$\widetilde{U}_n = \sup_{\tau\in\mathcal{T}_{n,N}} \mathbb{E}(\widetilde{Z}_\tau|\mathcal{F}_n) = \mathbb{E}(\widetilde{Z}_{\nu_n}|\mathcal{F}_n), \tag{5.14}$$

where $\nu_n = \inf\{j \geq n : \widetilde{U}_j = \widetilde{Z}_j\}$.

In light of Definition (3.2.6), we remark that **ν is an optimal stopping time!**

Proposition 5.1.7. *The stopping time τ^* is optimal (Definition (3.2.6)) if and only if $\widetilde{U}_{\tau^*} = \widetilde{Z}_{\tau^*}$ and $(\widetilde{U}_{\tau^*\wedge n})$ is a $(\mathcal{F}_n)$-martingale.*

Proof. • ($\Longleftarrow$) If $(\widetilde{U}_n^{\tau^*})$ is a martingale and $\widetilde{U}_{\tau^*} = \widetilde{Z}_{\tau^*}$ then $\widetilde{U}_0 = \mathbb{E}(\widetilde{U}_N^{\tau^*}) = \mathbb{E}(\widetilde{U}_{\tau^*})$. Thus the two facts $\widetilde{U}_0 = \mathbb{E}(\widetilde{U}_{\tau^*})$ and (by assumption) $\widetilde{U}_{\tau^*} = \widetilde{Z}_{\tau^*}$ gives $\widetilde{U}_0 = \mathbb{E}(\widetilde{Z}_{\tau^*})$. By Corollary (5.1.5) we conclude that τ^* is optimal.

• ($\Longrightarrow$) Conversely, if τ^* is optimal we have,

$$\widetilde{U}_0 = \mathbb{E}(\widetilde{Z}_{\tau^*}) \leq \mathbb{E}(\widetilde{U}_{\tau^*}) \leq \widetilde{U}_0, \tag{5.15}$$

where we have used the facts that $\widetilde{U}_n$ dominates $\widetilde{Z}_n$ in the first inequality and that $(\widetilde{U}_n^{\tau^*})$ is a supermartingale in the second inequality. Thus it follows immediately that $\mathbb{E}(\widetilde{Z}_{\tau^*}) = \mathbb{E}(\widetilde{U}_{\tau^*})$ implying that $\mathbb{E}(\widetilde{U}_{\tau^*} - \widetilde{Z}_{\tau^*}) = 0$. Since $\widetilde{U}_{\tau^*} \geq \widetilde{Z}_{\tau^*}$ we conclude that $\widetilde{U}_{\tau^*} = \widetilde{Z}_{\tau^*}$. It remains to show that $\widetilde{U}^{\tau^*}$ is a martingale. From Proposition (5.1.1) and Proposition (3.2.4) we know that $\widetilde{U}^{\tau^*}$ is a supermartingale. Thus,

$$\widetilde{U}_0 \geq \mathbb{E}(\widetilde{U}_{\tau^*\wedge n}) \geq \mathbb{E}(\widetilde{U}_{\tau^*\wedge(n+1)}) \geq \mathbb{E}(\widetilde{U}_{\tau^*\wedge N}) = \mathbb{E}(\widetilde{U}_{\tau^*}) = \mathbb{E}(\widetilde{Z}_{\tau^*}) = \widetilde{U}_0, \tag{5.16}$$

where in succession we have used the facts that $\widetilde{U}_{\tau^*} = \widetilde{Z}_{\tau^*}$ and that τ^* is optimal in the last equality. Therefore,

$$\mathbb{E}(\widetilde{U}_{\tau^*\wedge n}) = \mathbb{E}(\widetilde{U}_{\tau^*\wedge(n+1)}) = \mathbb{E}\Big(\mathbb{E}(\widetilde{U}_{\tau^*\wedge(n+1)}|\mathcal{F}_n)\Big) \Longrightarrow \mathbb{E}\Big(\widetilde{U}_{\tau^*\wedge n} - \mathbb{E}(\widetilde{U}_{\tau^*\wedge(n+1)}|\mathcal{F}_n)\Big) = 0. \tag{5.17}$$

Since $(\widetilde{U}_n^{\tau^*})$ is a supermartingale then for all n, $\mathbb{E}(\widetilde{U}_{\tau^*\wedge(n+1)}|\mathcal{F}_n) \leq \widetilde{U}_{\tau^*\wedge n}$. Thus, for all n, it holds that $\widetilde{U}_{\tau^*\wedge n} - \mathbb{E}(\widetilde{U}_{\tau^*\wedge(n+1)}|\mathcal{F}_n) \Longrightarrow \widetilde{U}_{\tau^*\wedge n} = \mathbb{E}(\widetilde{U}_{\tau^*\wedge(n+1)}|\mathcal{F}_n)$ concluding that $(\widetilde{U}^{\tau^*})$ is a martingale. □

In light of Proposition (5.1.7), we remark that ***ν* is the smallest optimal stopping time!**

Proposition 5.1.8. *The largest optimal stopping time for $(\widetilde{Z}_n)$ is given by,*

$$\nu_{\max} = \begin{cases} N, & \text{if } \widetilde{A}_N = 0 \\ \inf\{n : \widetilde{A}_{n+1} \neq 0\}, & \text{if } \widetilde{A}_N \neq 0 \end{cases}. \tag{5.18}$$

Remark 5.1.9. *Note that the result for the case $\widetilde{A}_n \neq 0$ comes from the Doob Decomposition (3.1.13).*

Proof. First note that $\nu_{\max}$ is a stopping time since $(\widetilde{A}_n)$ is predictable. Now, considering the Doob Decomposition $\widetilde{U}_n = \widetilde{M}_n - \widetilde{A}_n$ and the fact that $\widetilde{A}_j = 0$ for $j \leq \nu_{\max}$ the combining these two facts gives $\widetilde{U}^{\nu_{\max}} = \widetilde{M}^{\nu_{\max}}$. Thus, $\widetilde{U}^{\nu_{\max}}$ is a martingale. Hence in view of Proposition (5.1.7) it remains to show that $\widetilde{U}^{\nu_{\max}} = \widetilde{Z}^{\nu_{\max}}$ in order to see that $\nu_{\max}$ is optimal. We first rewrite $\widetilde{U}_{\nu_{\max}}$

in the following way,

$$\widetilde{U}_{\nu_{\max}} = \sum_{j=0}^{N-1} \mathbb{1}_{\{\nu_{\max}=j\}} \widetilde{U}_j + \mathbb{1}_{\{\nu_{\max}=N\}} \widetilde{U}_N = \sum_{j=0}^{N-1} \mathbb{1}_{\{\nu_{\max}=j\}} \max\left(\widetilde{Z}_j, \mathbb{E}(\widetilde{U}_{j+1}|\mathcal{F}_j)\right) + \mathbb{1}_{\{\nu_{\max}=N\}} \widetilde{Z}_N,$$

where we have used the definition of $\widetilde{U}_j$ and the fact that $\widetilde{U}_N = \widetilde{Z}_N$. Let us simplify the maximum by noting that,

$$\mathbb{E}(\widetilde{U}_{j+1}|\mathcal{F}_j) = \mathbb{E}(\widetilde{M}_{j+1} - \widetilde{A}_{j+1}|\mathcal{F}_j) = \widetilde{M}_j - \underbrace{\widetilde{A}_{j+1}}_{>0} < \widetilde{M}_j, \tag{5.19}$$

as $\{\nu_{\max} = j\}$ by definition of $\nu_{\max}$. Now remark that $\widetilde{M}_j - \widetilde{A}_j = \widetilde{M}_j$ on $\{\nu_{\max} = j\}$ and thus $\widetilde{U}_j = \widetilde{M}_j = \max\left(\widetilde{Z}_j, \mathbb{E}(\widetilde{U}_{j+1}|\mathcal{F}_j)\right) = \widetilde{Z}_j$ on $\{\nu_{\max} = j\}$. Indeed, the maximum is given by $\widetilde{Z}_j$ with $\mathbb{E}(\widetilde{U}_{j+1}|\mathcal{F}_j)$ being strictly smaller than $\widetilde{M}_j$ (5.19) and thus the maximum is $\widetilde{Z}_j$. Therefore,

$$\widetilde{U}_{\nu_{\max}} = \sum_{j=0}^{N-1} \mathbb{1}_{\{\nu_{\max}=j\}} \widetilde{Z}_j + \mathbb{1}_{\{\nu_{\max}=N\}} \widetilde{Z}_N = \widetilde{Z}_{\nu_{\max}}. \tag{5.20}$$

Thus in light of Proposition (5.1.7) we conclude that $\nu_{\max}$ is optimal. We finally show that $\nu_{\max}$ is the largest optimal stopping time. Since τ^* is optimal then by Proposition (5.1.7), $(\widetilde{U}_{\tau^* \wedge n})$ is a martingale and thus by definition $\mathbb{E}(\widetilde{U}_{\tau^* \wedge n}) = \mathbb{E}(\widetilde{U}_0)$. By the Doob Decomposition, the LHS is given by,

$$\mathbb{E}(\widetilde{U}_{\tau^* \wedge n}) = \mathbb{E}(\widetilde{M}_{\tau^* \wedge n}) - \mathbb{E}(\widetilde{A}_{\tau^* \wedge n}) = \mathbb{E}(\widetilde{M}_0) - \mathbb{E}(\widetilde{A}_{\tau^* \wedge n}) = \mathbb{E}(\widetilde{U}_0) - \mathbb{E}(\widetilde{A}_{\tau^* \wedge n}). \tag{5.21}$$

Therefore, $\mathbb{E}(\widetilde{U}_0) - \mathbb{E}(\widetilde{A}_{\tau^* \wedge n}) = \mathbb{E}(\widetilde{U}_0)$ implying that for all n, $\mathbb{E}(\widetilde{A}_{\tau^* \wedge n}) = 0$. And since this result holds for all n then it is the case that $\mathbb{E}(\widetilde{A}_{\tau^*}) = 0$. Since A is monotone (non-decreasing), we have $\mathbb{P}(\widetilde{A}_{\tau^*} = 0) = 1$ [same result holds with $\mathbb{Q}$] thus concluding that $\tau^* \leq \nu_{\max}$ a.s. (i.e. $\nu_{\max}$ is the largest optimal stopping time).

□

5.2 Application to American Options

Recall that for a call option $Z_n = (S_n - K)^+$ and for a put option $Z_n = (K - S_n)^+$. We have identified (U_n) as being the value process (sequence) corresponding to an American option, defined by the process (Z_n), where U_n is defined by $U_N = Z_N$, and for all $n \leq N-1$

$$U_n = \max\left(Z_n, S_n^0 \mathbb{E}_{\mathbb{Q}}\left(\frac{U_{n+1}}{S_{n+1}^0}\Big|\mathcal{F}_n\right)\right). \tag{5.22}$$

Thus, the process $(\widetilde{U}_n)$ defined by $\frac{U_n}{S_n^0}$ ($\approx$ the discounted values of the option) is the Snell envelope under $\mathbb{Q}$ of the process $(\widetilde{Z}_n)$. From Corollary (5.1.5) we deduce that,

$$\widetilde{U}_0 = \sup_{\tau \in \mathcal{T}_{0,N}} \mathbb{E}_{\mathbb{Q}}(\widetilde{Z}_\tau) = \sup_{\tau \in \mathcal{T}_{0,N}} \mathbb{E}_{\mathbb{Q}}\left(\frac{Z_\tau}{S_\tau^0}\right). \tag{5.23}$$

In view of the Doob-Meyer Decomposition Theorem (3.1.13) we can write,

$$\widetilde{U_n} = \widetilde{M}_n - \widetilde{A}_n, \tag{5.24}$$

where $(\widetilde{M}_n)$ in a $\mathbb{Q}$-martingale and $(\widetilde{A}_n)$ is a predictable, non-decreasing process null at 0. Since the market is complete there is a self-financing strategy ϕ such that,

$$V_N(\phi) = S_N^0 \widetilde{M}_N. \tag{5.25}$$

Thus,

$$\mathbb{E}_{\mathbb{Q}}\left(\widetilde{V}_N(\phi)|\mathcal{F}_n\right) := \mathbb{E}_{\mathbb{Q}}\left(\frac{V_N(\phi)}{S_n^0}\Big|\mathcal{F}_n\right) = \mathbb{E}_{\mathbb{Q}}\left(\widetilde{M}_N|\mathcal{F}_n\right) = \widetilde{M}_n, \tag{5.26}$$

since $(\widetilde{M}_n)$ is a $\mathbb{Q}$-martingale (optimal stopping yields U_0). As a result, $\widetilde{U}_n = \widetilde{V}_n(\phi) - \widetilde{A}_n$ if and only if $U_n = V_n(\phi) - A_n$. In particular, from the point of view of a hedger, once he receives the premium $U_0 = V_0(\phi)$ he can get wealth equal to $V_n(\phi)$ at time n, which is bigger than U_n and a fortiori Z_n. Indeed, if we replicate $(\widetilde{M}_n)$ we obtain $\widetilde{U}_n = \widetilde{M}_n - \widetilde{A}_n = \widetilde{V}_n(\phi) - \widetilde{A}_n$ where $\widetilde{A}_n$ is non-decreasing (i.e. $-\widetilde{A}_n$ is non-increasing). Having in mind that $U_n = V_n(\phi) - A_n$, we find the following ordering $V_n(\phi) \geq U_n \geq Z_n$.

We finally need to find the optimal date to exercise the option. This date has to be chosen among the stopping times.

- Before ν, we have that $U_n > Z_n$ at time n and thus one trades an option worth U_n for Z_n, not optimal!

- Exercising after $\nu_{\max}$ is not optimal either since it would yield $U_{\nu_{\max}} = V_{\nu_{\max}}(\phi)$ and following the strategy from that time, with the money obtained by exercising the option, creates a portfolio whose value is strictly bigger that U_n at time $\nu_{\max} + \{1, ..., N\}$. In other words, since in general $U_n = V_n(\phi) - A_n$ it is better to follow the strategy ϕ and not wait any longer.

- An optimal date τ to exercise the option is given by $U_\tau = Z_\tau$. We also set as a second condition $\tau \leq \nu_{\max}$, which allows to say that $\widetilde{U}^\tau$ is a martingale. As a result, optimal dates of exercise are optimal stopping times τ^* for the process $(\widetilde{Z}_n)$ under the probability measure $\mathbb{Q}$ (i.e. stop at optimal $\tau^* \Longrightarrow \widetilde{U}^{\tau^*}$ is a martingale $\Longrightarrow$ no-arbitrage).

From the point of view of a hedger, if $U_\tau > Z_\tau$ or $A_\tau > 0$ one finds $V_\tau(\phi) - Z_\tau = U_\tau + A_\tau - Z_\tau$ which in both cases is strictly positive.

Chapter 6

The Black-Scholes Model, Pricing and Hedging

There are pros and cons of discrete time models relative to continuous time models highly depending on what we want to achieve. We have some computational advantages by considering discrete time models (e.g. related to American Options) however we can obtain nice closed form formulas based on some popular continuous time models for quite complicated derivatives. In this chapter we delve into the formalism of continuous time models by reviewing the so-called Geometric Brownian Motion and studying the so-famous Black-Scholes Model.

6.1 The Black-Scholes Model

The classical Black-Scholes Model contains two assets. The first one is a risk-free asset, such that its price process is denoted by $S^0 := (S_t^0)_{0\leq t\leq T}$. We assume that its dynamics is given by the ODE,

$$dS_t^0 = rS_t^0 dt \tag{6.1}$$

with $S_0^0 = 1$ and where r is a non-negative constant. Obviously this is solved by e^{rt}, noting that $r = r_c$. On the other hand, there is a risky-asset with price process denoted by $S := (S_t)_{0\leq t\leq T}$. We assume that its dynamics is given by the SDE,

$$dS_t = \mu S_t dt + \sigma S_t dB_t, \tag{6.2}$$

with $\sigma > 0$, μ being constants, $S_0 > 0$, and $(B_t)_{0\leq t\leq T}$ is a standard Brownian Motion. Equation (6.2) is said to be a **Geometric Brownian Motion (GBM)**. Remark that σ is called the **volatility**.

Example 6.1.1 (Standard model for the price)**.** *The unique (for given S_0) solution (S_t) of the SDE of the risky asset is given by,*

$$S_t = S_0 e^{\sigma B_t + \left(\mu - \frac{\sigma^2}{2}\right)t}, \tag{6.3}$$

where S_0 stands for the price at $t = 0$. Recall that the characteristic function for a Gaussian $X \sim \mathcal{N}(\mu, \sigma^2)$ is given by $\varphi_X(u) = \exp\left(iu\mu - \frac{1}{2}\sigma^2 u^2\right)$ where $u \in \mathbb{R}$. Note that this characteristic function may be extended to $\mathbb{C}$.

From there we compute,

$$\begin{aligned}\mathbb{E}(S_t) = \mathbb{E}\left(S_0 \exp\left(\mu t - \frac{1}{2}\sigma^2 t + \sigma B_t\right)\right) &= S_0 \exp\left(\mu t - \frac{1}{2}\sigma^2 t\right)\mathbb{E}\left(e^{i(-i)\sigma B_t}\right) \\ &= S_0 \exp\left(\mu t - \frac{1}{2}\sigma^2 t\right)\exp\left(-\frac{1}{2}(-i)^2\sigma^2 t\right) \\ &= S_0 \exp(\mu t).\end{aligned}$$

We now present four properties of the risky-asset.

Proposition 6.1.2. *The risky asset has continuous paths.*

Proposition 6.1.3. *For the risky-asset we have an independence of the relative increments, i.e. if $s \leq t$ we have that $\frac{S_t}{S_s}$ and $\frac{S_t - S_s}{S_s}$ are independent of $\mathcal{F}_s$.*

Proposition 6.1.4. *For the risky asset we have stationary relative increments, i.e. if $u \leq t$ we have that the law of $\frac{S_t - S_u}{S_u}$ shares the distribution with $\frac{S_{t-u} - S_0}{S_0}$.*

Proposition 6.1.5. *The risky-asset follows a log-normal distribution, where for all t,*

$$\log(S_t) \sim \mathcal{N}\left(\log(S_0) + \left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right). \tag{6.4}$$

Proof. Let $Y_t = \log(S_t)$ with $S_t = \mu S_t dt + \sigma S_t dB_t$ following a GBM. With a slight extension of Itô's Formula to open subsets of $\mathbb{R}$ and noting that $\frac{\partial f}{\partial x}(x) = \frac{1}{x}$, $\frac{\partial^2 f}{\partial x^2}(x) = -\frac{1}{x^2}$ one finds,

$$dY_t = \left(\frac{1}{S_t}\mu S_t + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)\sigma^2 S_t^2\right)dt + \frac{1}{S_t}\sigma S_t dB_t = \left(\mu - \frac{1}{2}\sigma^2\right)dt + \sigma dB_t. \tag{6.5}$$

In other words,

$$Y_t = Y_0 + \int_0^t \left(\mu - \frac{1}{2}\sigma^2\right)ds + \int_0^t \sigma dB_s = \log(S_0) + \left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t. \tag{6.6}$$

And as a result,

$$\log(S_t) \sim \mathcal{N}\left(\log(S_0) + \left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right). \tag{6.7}$$

□

Now, in view of the SDE for (S_t) for $\widetilde{S}_t = e^{-rt}S_t$ we have,

$$\begin{aligned}d\widetilde{S}_t = d(e^{-rt}S_t) = e^{-rt}dS_t + S_t(-re^{-rt})dt &= e^{-rt}\mu S_t dt + e^{-rt}\sigma S_t dB_t - re^{-rt}S_t dt \\ &= \widetilde{S}_t((\mu - r)dt + \sigma dB_t) \\ &= \widetilde{S}_t\sigma\left(\frac{\mu - r}{\sigma}dt + dB_t\right) = \widetilde{S}_t\sigma dW_t.\end{aligned}$$

Thus, by the Girsanov Theorem (3.3.10) for $\theta_t = \frac{\mu - r}{\sigma}$, it implies the existence of a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ under which $(W_t)_{t\in[0,T]}$ is a standard Brownian Motion.

Proposition 6.1.6. *The discounted asset price process $\widetilde{S}_t = e^{-rt}S_t$ is a $\mathbb{Q}$-martingale where under $\mathbb{Q}$ we have,*

$$\widetilde{S}_t = S_0 \exp\left(\sigma W_t - \frac{\sigma^2 t}{2}\right). \tag{6.8}$$

Proof. First note that $(\widetilde{S}_t)$ is adapted, that $\widetilde{S}_0 = S_0$ and that integrability will follow from the martingale property. Assume now that $s \leq t$ then,

$$\mathbb{E}_{\mathbb{Q}}\left(\widetilde{S}_t|\mathcal{F}_s\right) = \mathbb{E}_{\mathbb{Q}}\left(\widetilde{S}_0 e^{\sigma W_t + \sigma W_s - \sigma W_s - \frac{\sigma^2}{2}t}|\mathcal{F}_s\right) = S_0 e^{\sigma W_s - \frac{\sigma^2}{2}t}\mathbb{E}_{\mathbb{Q}}\left(e^{\sigma(W_t - W_s)}|\mathcal{F}_s\right). \tag{6.9}$$

Recall now that $W_{t-s} \sim \mathcal{N}(0, t-s)$ then,

$$\mathbb{E}_{\mathbb{Q}}\left(e^{\sigma(W_t - W_s)}|\mathcal{F}_s\right) = \mathbb{E}_{\mathbb{Q}}\left(e^{\sigma W_{t-s}}\right) = \mathbb{E}_{\mathbb{Q}}\left(e^{i(-i)W_{t-s}}\right) = \exp\left(-\frac{1}{2}(-i\sigma)^2(t-s)\right) = \exp\left(\frac{1}{2}\sigma^2(t-s)\right).$$

Overall, we obtain,

$$\mathbb{E}_{\mathbb{Q}}\left(\widetilde{S}_t|\mathcal{F}_s\right) = S_0 e^{\sigma W_s - \frac{\sigma^2}{2}t}\exp\left(\frac{1}{2}\sigma^2(t-s)\right) = S_0 e^{\sigma W_s - \frac{\sigma^2}{2}s} = \widetilde{S}_s. \tag{6.10}$$

□

6.2 Self-financing Strategies and Pricing

Definition 6.2.1. *A **strategy** is a process $\phi = (\phi_t)_{0\leq t\leq T} = \left(H_t^0, H_t\right)_{0\leq t\leq T}$ with values in $\mathbb{R}^2$ adapted to the filtration $(\mathcal{F}_t)_{0\leq t\leq T}$, where the components H_t^0 and H_t are the quantities of the risk-less asset and the risky asset respectively.*

Remark that the **value of the portfolio** at time t is given by,

$$V_t(\phi) = H_t^0 S_t^0 + H_t S_t. \tag{6.11}$$

Definition 6.2.2. *A **self-financing strategy** is a strategy satisfying the following two conditions:*

- *Integrability $\longrightarrow \int_0^T |H_t^0|dt + \int_0^T H_t^2 dt < \infty$ a.s.*
- *Self-financing $\longrightarrow dV_t = d(H_t^0 S_t^0) + d(H_t S_t) = H_t^0 dS_t^0 + H_t dS_t$, i.e. for all $t \in [0,T]$,*

$$V_t(\phi) := H_t^0 S_t^0 + H_t S_t = H_0^0 S_0^0 + H_0 S_0 + \int_0^t H_s^0 dS_s^0 + \int_0^t H_s dS_s \quad a.s. \tag{6.12}$$

Denote the present values of the risky asset and the portfolio respectively by $\widetilde{S}_t = e^{-rt}S_t$ and $\widetilde{V}_t = e^{-rt}V_t$.

Proposition 6.2.3. *Let $\phi = (\phi_t)_{0\le t\le T} = \left(H_t^0, H_t\right)_{0\le t\le T}$ be a strategy such that $\int_0^T |H_t^0|dt + \int_0^T H_t^2 dt < \infty$ a.s. Then, ϕ defines a self-financing strategy if and only if (for all $t \in [0,T]$),*

$$\widetilde{V}_t(\phi) = V_0(\phi) + \int_0^t H_s d\widetilde{S}_s \quad a.s. \tag{6.13}$$

Proof. • ($\Longrightarrow$) First recall the integration by parts formula for (X_t), (Y_t) two Itô processes with respect to the same standard Brownian Motion (B_t),

$$X_t = X_0 + \int_0^t a(\omega,s)ds + \int_0^t b(\omega,s)dB_s \tag{6.14}$$

$$Y_t = Y_0 + \int_0^t a'(\omega,s)ds + \int_0^t b'(\omega,s)dB_s \tag{6.15}$$

and then,

$$X_tY_t = X_0Y_0 + \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \langle X,Y\rangle_t = X_0Y_0 + \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \int_0^t b(\omega,s)b'(\omega,s)ds.$$

Now, suppose that ϕ is self-financing and consider $X_t := e^{-rt}$, $Y_t := V_t(\phi)$ and $\widetilde{V}_t(\phi) := e^{-rt}V_t(\phi)$. Using integration by parts yields,

$$d\widetilde{V}_t(\phi) = e^{-rt}dV_t(\phi) + V_t(\phi)d\big(e^{-rt}\big) = e^{-rt}dV_t(\phi) + V_t(\phi)\big(-re^{-rt}\big)dt.$$

Using the fact that $S_t^0 = e^{rt}$, equation (6.11), along with the self-financing condition, we obtain $e^{-rt}dV_t(\phi) = e^{-rt}H_t^0 d\big(e^{rt}\big) + e^{-rt}H_t dS_t = e^{-rt}H_t^0\big(re^{rt}\big)dt + e^{-rt}H_t dS_t$ and therefore,

$$\begin{aligned} d\widetilde{V}_t(\phi) = e^{-rt}H_t^0\big(re^{rt}\big)dt + e^{-rt}H_t dS_t + \big(H_t^0 e^{rt} + H_t S_t\big)\big(-re^{-rt}\big)dt &= H_t\big(e^{-rt}dS_t + \big(-re^{-rt}\big)S_t dt\big) \\ &= H_t d\big(e^{-rt}S_t\big) \\ &= H_t d\widetilde{S}_t. \end{aligned}$$

In other words, $\widetilde{V}_t(\phi) = V_0(\phi) + \int_0^t H_s d\widetilde{S}_s$ a.s.

• ($\Longleftarrow$) The converse is obtained in a similar way. Recall that $e^{rt}\widetilde{V}_t(\phi) = V_t(\phi)$, and then using the integration by parts yields,

$$\begin{aligned} dV_t(\phi) = e^{rt}d\widetilde{V}_t(\phi) + \widetilde{V}_t(\phi)d\big(e^{rt}\big) &= e^{rt}H_t d\widetilde{S}_t + \widetilde{V}_t(\phi)re^{rt}dt \\ &= e^{rt}H_t\big(e^{-rt}dS_t + S_t\big(-re^{-rt}\big)dt\big) + rV_t(\phi)dt \\ &= H_t dS_t - rH_t S_t dt + r\big(H_t^0 e^{rt} + H_t S_t\big)dt \\ &= H_t dS_t + rH_t^0 e^{rt}dt \\ &= H_t dS_t + H_t^0 d\big(e^{rt}\big) = H_t dS_t + H_t^0 d(S_t^0). \end{aligned}$$

In other words, for all $t \in [0,T]$,

$$V_t(\phi) = V_0(\phi) + \int_0^t H_u^0 dS_u^0 + \int_0^t H_u dS_u \quad \text{a.s.} \tag{6.16}$$

□

Remark 6.2.4. *Actually it is still possible to define a predictable process in continuous time, however, in the case of the Brownian filtration this does not restrict the class of adapted processes significantly.*

Definition 6.2.5 (Admissible strategy)**.** *A strategy $\phi = (H_t^0, H_t)_{0\le t\le T}$ is **admissible** if it is self-financing and if the discounted value $\widetilde{V}_t(\phi) = H_t^0 + H_t\widetilde{S}_t$ of the corresponding portfolio is non-negative for all t with $\mathbb{E}_{\mathbb{Q}}\left(\int_0^T H_t^2(\sigma\widetilde{S}_t)^2 dt\right) < \infty$.*

Theorem 6.2.6 (Pricing contingent claims)**.** *In the Black-Scholes model, any option defined by a non-negative $\mathcal{F}_T$-measurable random variable h which is square-integrable under the probability measure $\mathbb{Q}$, is replicable (attainable) and the value at time t of any replicating portfolio is given by,*

$$V_t = e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}\left(h|\mathcal{F}_t\right). \tag{6.17}$$

Hence, the option value at time t may be naturally defined by the expression $e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}(h|\mathcal{F}_t)$.

Proof. First assume that an admissible replicating strategy $\phi = (H_t^0, H_t)$ exists, then $V_t(\phi) = H_t^0 S_t^0 + H_t S_t$. By assumption, $V_T = h$. Now let $\widetilde{V}_t(\phi) = V_t(\phi)e^{-rt}$, i.e. $\widetilde{V}_t(\phi) = H_t^0 + H_t\widetilde{S}_t$. Then, by Proposition (6.2.3) giving $d\widetilde{S}_t = \sigma\widetilde{S}_t dW_t$ and the self-financing property of the strategy, we have,

$$\widetilde{V}_t(\phi) = V_0 + \int_0^t H_u d\widetilde{S}_u = V_0 + \int_0^t H_u\sigma\widetilde{S}_u dW_u. \tag{6.18}$$

In light of the definition of admissible strategies (6.2.5) we have that $\left(\widetilde{V}_t(\phi)\right)$ is a square-integrable $\mathbb{Q}$-martingale. Thus,

$$\widetilde{V}_t(\phi) = \mathbb{E}_{\mathbb{Q}}\left(\widetilde{V}_T(\phi)|\mathcal{F}_t\right) = \mathbb{E}_{\mathbb{Q}}\left(\frac{h}{e^{rT}}\Big|\mathcal{F}_t\right), \tag{6.19}$$

and consequently,

$$V_t(\phi) = \mathbb{E}_{\mathbb{Q}}\left(e^{-r(T-t)}h|\mathcal{F}_t\right) = e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}(h|\mathcal{F}_t). \tag{6.20}$$

Hence, if an admissible portfolio (H_t^0, H_t) replicates the option defined by h, then $V_t(\phi) = e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}(h|\mathcal{F}_t)$ is its value at t. Thus, it remains to show that h is indeed replicable, i.e. we need to find an admissible strategy such that,

$$V_t(\phi) = H_t^0 S_t^0 + H_t S_t = \mathbb{E}_{\mathbb{Q}}\left(e^{-r(T-t)}h|\mathcal{F}_t\right) = e^{rt}\mathbb{E}_{\mathbb{Q}}\left(e^{-rT}h|\mathcal{F}_t\right) \equiv e^{rt}M_t. \tag{6.21}$$

Note that, for all $t \in [0,T]$, one can use Jensen's Inequality to find,

$$\mathbb{E}_{\mathbb{Q}}(M_t^2) = \mathbb{E}_{\mathbb{Q}}\left(\mathbb{E}_{\mathbb{Q}}(e^{-rT}h|\mathcal{F}_t)^2\right) \le \mathbb{E}_{\mathbb{Q}}\left(\mathbb{E}_{\mathbb{Q}}(e^{-2rT}h^2|\mathcal{F}_t)\right) = e^{-2rT}\mathbb{E}_{\mathbb{Q}}(h^2) < \infty. \tag{6.22}$$

Thus, $(M_t)_{t\in[0,T]}$ is a square-integrable $\mathbb{Q}$-martingale and the augmented filtration of (B_t) is also the augmented filtration of (W_t). Hence, by the Martingale Representation Theorem (3.3.12), there exists an adapted process $(K_t)_{t\in[0,T]}$ such that $\mathbb{E}_{\mathbb{Q}}\left(\int_0^T K_s^2 ds\right) < \infty$ and for all $t \in [0,T]$,

$$M_t = M_0 + \int_0^t K_s dW_s \quad \text{a.s.} \tag{6.23}$$

The strategy $\phi = (H_t^0, H_t)$ with $H_t = \frac{K_t}{(\sigma \widetilde{S}_t)}$ and $H_t^0 = M_t - H_t \widetilde{S}_t$ is then self-financing. Indeed,

$$\widetilde{V}_t(\phi) = M_0 + \int_0^t K_s dW_s = M_0 + \int_0^t H_s \sigma \widetilde{S}_s dW_s = M_0 + \int_0^t H_s d\widetilde{S}_s. \tag{6.24}$$

And since $M_t = \widetilde{V}_t(\phi) = V_t(\phi)e^{-rt} \Longrightarrow M_0 = V_0(\phi)$, by Proposition (6.2.3) we have a self-financing strategy. Furthermore, note that $\widetilde{V}_t(\phi)$ is non-negative (consequence of h being non-negative and the properties of conditional expectations) and that $\mathbb{E}_{\mathbb{Q}}\left(\int_0^T K_s^2 ds\right) = \mathbb{E}_{\mathbb{Q}}\left(\int_0^T H_s^2 (\sigma \widetilde{S}_s)^2 ds\right) < \infty$. Clearly $V_T(\phi) = h$ and in conclusion, by Definition (6.2.5), there exists an admissible replication strategy.

□

We now go through the derivation for the price of a European call option for the Black-Scholes Model.

Proposition 6.2.7 (The Black-Scholes Formula - European call)**.** *In the Black-Scholes Model, the price of a European call option C_t with maturity $T > 0$ and strike $K > 0$ is given by,*

$$C_t = S_t \mathcal{N}(d_+) - K e^{-r(T-t)} \mathcal{N}(d_-), \tag{6.25}$$

where $\mathcal{N}(\cdot)$ is the CDF of a standard normally distributed random variable and $d_\pm$ are given by,

$$d_\pm(S_t) = \frac{1}{\sigma\sqrt{T-t}} \left(\log\left(\frac{S_t}{K}\right) + \left(r \pm \frac{\sigma^2}{2}\right)(T-t) \right). \tag{6.26}$$

Proof. For a European call option, $h = (S_T - K)_+$. Hence, by Theorem (6.2.6),

$$C_t = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}}((S_T - K)_+ | \mathcal{F}_t) = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left[\left(S_0 e^{\left(r - \frac{\sigma^2}{2}\right)t + \sigma W_t} e^{\left(r - \frac{\sigma^2}{2}\right)(T-t) + \sigma(W_T - W_t)} - K \right)_+ \middle| \mathcal{F}_t \right].$$

Remark that $S_t = S_0 e^{\left(r - \frac{\sigma^2}{2}\right)t + \sigma W_t}$ with is $\mathcal{F}_t$-measurable and $e^{\left(r - \frac{\sigma^2}{2}\right)(T-t) + \sigma(W_T - W_t)}$ is a price change factor which is independent of $\mathcal{F}_t$. Thus, $C_t = F(t, S_t)$ where,

$$F(t,x) = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left[\left(x e^{\left(r - \frac{\sigma^2}{2}\right)(T-t) + \sigma(W_T - W_t)} - K \right)_+ \right]. \tag{6.27}$$

Note that if $\sigma(W_T - W_t) + (r - \frac{\sigma^2}{2})(T-t) = Z + r(T-t)$ then $Z \sim \mathcal{N}(-\frac{\sigma^2}{2}(T-t), \sigma^2(T-t))$. From now on, let $\tau := T - t$. Remark now that $\left(x e^{z + r\tau} - K\right)_+ \geq 0$ if and only if $z \geq \log\left(\frac{K}{x}\right) - r\tau$. Thus,

$$F(t,x) = \frac{e^{-r\tau}}{\sqrt{2\pi\sigma^2\tau}} \int_{\mathbb{R}} \left(xe^{z+r\tau} - K\right)_+ e^{-\frac{\left(z+\frac{1}{2}\sigma^2\tau\right)^2}{2\sigma^2\tau}} dz$$
$$= \frac{1}{\sqrt{2\pi\sigma^2\tau}} \int_{\log\left(\frac{K}{x}\right)-r\tau}^{\infty} xe^z e^{-\frac{\left(z+\frac{1}{2}\sigma^2\tau\right)^2}{2\sigma^2\tau}} dz - \frac{Ke^{-r\tau}}{\sqrt{2\pi\sigma^2\tau}} \int_{\log\left(\frac{K}{x}\right)-r\tau}^{\infty} e^{-\frac{\left(z+\frac{1}{2}\sigma^2\tau\right)^2}{2\sigma^2\tau}} dz.$$

We perform a change of variable $v = -\frac{z+\frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}} \iff z = -\sigma\sqrt{\tau}v - \frac{1}{2}\sigma^2\tau$ where $\left|\frac{dz}{dv}\right| = \sigma\sqrt{\tau}$. Thus,

$$F(t,x) = \frac{\sigma\sqrt{\tau}}{\sqrt{2\pi\sigma^2\tau}} \int_{-\infty}^{\frac{\log\left(\frac{x}{K}\right)+\left(r-\frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}}} xe^{-\sigma\sqrt{\tau}v-\frac{1}{2}\sigma^2\tau-\frac{1}{2}v^2} dv - \frac{Ke^{-r\tau}\sigma\sqrt{\tau}}{\sqrt{2\pi\sigma^2\tau}} \int_{-\infty}^{\frac{\log\left(\frac{x}{K}\right)+\left(r-\frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}}} e^{-\frac{1}{2}v^2} dv$$
$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{d_-(x)} xe^{-\sigma\sqrt{\tau}v-\frac{1}{2}\sigma^2\tau-\frac{1}{2}v^2} dv - \frac{Ke^{-r\tau}}{\sqrt{2\pi}} \int_{-\infty}^{d_-(x)} e^{-\frac{1}{2}v^2} dv.$$

Note that $-\frac{1}{2}(v+\sigma\sqrt{\tau})^2 = -\frac{1}{2}v^2 - v\sigma\sqrt{\tau} - \frac{1}{2}\sigma^2\tau$ and we perform yet another change of variable $\omega = v + \sigma\sqrt{\tau}$ with $\left|\frac{dv}{d\omega}\right| = 1$. Thus,

$$F(t,x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{d_-(x)+\sigma\sqrt{\tau}} xe^{-\frac{1}{2}\omega^2} d\omega - \frac{e^{-r\tau}K}{\sqrt{2\pi}} \int_{-\infty}^{d_-(x)} e^{-\frac{1}{2}v^2} dv. \tag{6.28}$$

We finally write $d_+(x) = d_-(x) + \sigma\sqrt{\tau} = \frac{\log\left(\frac{x}{K}\right)+\left(r+\frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}}$ and $\mathcal{N}(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{y} e^{-\frac{1}{2}x^2} dx$. Therefore,

$$F(t,x) = x\mathcal{N}\left(\frac{\log\left(\frac{x}{K}\right) + \left(r+\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}\right) - Ke^{-r(T-t)}\mathcal{N}\left(\frac{\log\left(\frac{x}{K}\right) + \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}\right). \tag{6.29}$$

Hence, for $C_t = F(t,S_t)$ one finds,

$$C_t = F(t,S_t) = S_t\mathcal{N}\left(\frac{\log\left(\frac{S_t}{K}\right) + \left(r+\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}\right) - Ke^{-r(T-t)}\mathcal{N}\left(\frac{\log\left(\frac{S_t}{K}\right) + \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}\right)$$
$$= S_t\mathcal{N}(d_+) - Ke^{-r(T-t)}\mathcal{N}(d_-).$$

□

Corollary 6.2.8. *The hedging strategy in the case of a European call option with maturity $T > 0$ and strike $K > 0$ in the Black-Scholes model is given by,*

$$\phi_t = (H_t^0, H_t) = \left(-Ke^{-rT}\mathcal{N}(d_-), \mathcal{N}(d_+)\right). \tag{6.30}$$

Proof. Recall that $S_t^0 = e^{rt}$ and the value of a general option is in the form $C_t = H_t^0 S_t^0 + H_t S_t$ (6.11). By the previous Theorem (6.4.7) we know that,

$$C_t = S_t\mathcal{N}(d_+) - Ke^{-r(T-t)}\mathcal{N}(d_-) = S_t\mathcal{N}(d_+) - S_t^0 Ke^{-rT}\mathcal{N}(d_-), \tag{6.31}$$

thus matching the coefficients we find $H_t^0 = -Ke^{-rT}\mathcal{N}(d_-)$ and $H_t = \mathcal{N}(d_+)$, as required.

□

Proposition 6.2.9 (The Black-Scholes Formula - European put)**.** *In the Black-Scholes Model, the price of a European put option P_t with maturity $T>0$ and strike $K>0$ is given by,*

$$P_t = Ke^{-r(T-t)}\mathcal{N}(-d_-) - S_t\mathcal{N}(-d_+), \tag{6.32}$$

where $\mathcal{N}(\cdot)$ is the CDF of a standard normally distributed random variable and $d_\pm$ are given by,

$$d_\pm(S_t) = \frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{S_t}{K}\right) + \left(r \pm \frac{\sigma^2}{2}\right)(T-t)\right). \tag{6.33}$$

Proof. One could prove this formula by similar steps as in the case of a call option, however a simpler way is to use the martingale property of $(\widetilde{S}_t)$ along with the call option formula. Now, recall that a function f may be expressed in terms of its positive part f_+ and negative part f_- by $f = f_+ - f_-$, where $f_+ := \max(f(x),0)$ and $f_- = \max(-f(x),0) = -\min(f(x),0) = (-f)_+$. Thus,

$$\begin{aligned} e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}(S_T - K|\mathcal{F}_t) &= e^{rt}\mathbb{E}_{\mathbb{Q}}(\widetilde{S}_T - Ke^{-rT}|\mathcal{F}_t) \\ &= e^{rt}\left[\mathbb{E}_{\mathbb{Q}}\left(\left(\widetilde{S}_T - Ke^{-rT}\right)_+|\mathcal{F}_t\right) - \mathbb{E}_{\mathbb{Q}}\left(\left(\widetilde{S}_T - Ke^{-rT}\right)_-|\mathcal{F}_t\right)\right] \\ &= e^{rt}\left[\mathbb{E}_{\mathbb{Q}}\left(\left(\widetilde{S}_T - Ke^{-rT}\right)_+|\mathcal{F}_t\right) - \mathbb{E}_{\mathbb{Q}}\left(\left(Ke^{-rT} - \widetilde{S}_T\right)_+|\mathcal{F}_t\right)\right]. \end{aligned}$$

On the other hand, using the fact that $(\widetilde{S}_t)$ is a martingale, we obtain,

$$e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}(S_T - K|\mathcal{F}_t) = e^{rt}\mathbb{E}_{\mathbb{Q}}(\widetilde{S}_T - Ke^{-rT}|\mathcal{F}_t) = e^{rt}\widetilde{S}_t - Ke^{-r(T-t)} = S_t - Ke^{-r(T-t)}.$$

Thus, rearranging the two previous equations yields,

$$\begin{aligned} &e^{rt}\mathbb{E}_{\mathbb{Q}}\left(\left(Ke^{-rT} - \widetilde{S}_T\right)_+|\mathcal{F}_t\right) = e^{rt}\mathbb{E}_{\mathbb{Q}}\left(\left(\widetilde{S}_T - Ke^{-rT}\right)_+|\mathcal{F}_t\right) + Ke^{-r(T-t)} - S_t \\ &\iff \underbrace{e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}\left(\left(K - S_T\right)_+|\mathcal{F}_t\right)}_{\text{Put price } P_t} = \underbrace{e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}\left(\left(S_T - K\right)_+|\mathcal{F}_t\right)}_{\text{Call price } C_t \text{ (6.4.7)}} + Ke^{-r(T-t)} - S_t. \end{aligned}$$

We may now input the price C_t of a call and use the symmetry in a standard normally distributed random variable, i.e. $\mathcal{N}(-x) = 1 - \mathcal{N}(x)$, to find,

$$\begin{aligned} P_t = S_t\mathcal{N}(d_+) - Ke^{-r(T-t)}\mathcal{N}(d_-) + Ke^{-r(T-t)} - S_t &= Ke^{-r(T-t)}(1-\mathcal{N}(d_-)) - S_t(1-\mathcal{N}(d_+)) \\ &= Ke^{-r(T-t)}\mathcal{N}(-d_-) - S_t\mathcal{N}(-d_+). \end{aligned}$$

□

Corollary 6.2.10. *The hedging strategy in the case of a European put option with maturity $T>0$ and strike $K>0$ in the Black-Scholes model is given by,*

$$\phi_t = (H_t^0, H_t) = (Ke^{-rT}\mathcal{N}(-d_-), -\mathcal{N}(-d_+)). \tag{6.34}$$

Proof. Recall that $S_t^0 = e^{rt}$ and the value of a general option is in the form $C_t = H_t^0 S_t^0 + H_t S_t$ (6.11). By the previous Theorem (6.2.9) we know that,

$$P_t = Ke^{-r(T-t)}\mathcal{N}(-d_-) - S_t\mathcal{N}(-d_+) = S_t^0 Ke^{-rT}\mathcal{N}(-d_-) - S_t\mathcal{N}(-d_+), \tag{6.35}$$

thus matching the coefficients we find $H_t^0 = Ke^{-rT}\mathcal{N}(-d_-)$ and $H_t = -\mathcal{N}(-d_+)$, as required. □

6.3 Hedging

In the proof of Theorem (6.2.6) we referred to the Martingale Representation Theorem (3.3.12) to show the existence of a replicating portfolio/strategy. In practice, an existence theorem is not satisfactory for application and it is essential to be able to get an explicit real replication portfolio to hedge an option. In the case where the option is defined by a random variable $h = f(S_T)$, it is often possible to find an explicit hedging portfolio/replication portfolio. We will demonstrate two (very similar) ways to find this hedging portfolio.

Proposition 6.3.1. *The hedging strategy in the case of a European claim with maturity $T > 0$ defined by a random variable of the form $h = f(S_T)$ is given by,*

$$\phi_t = (H_t^0, H_t) = \left(e^{-rt}\left(F(t,S_t) - \frac{\partial F}{\partial x}(t,S_t)S_t\right), \frac{\partial F}{\partial x}(t,S_t)\right), \tag{6.36}$$

where $F(t,x)$ is defined by,

$$F(t,x) = e^{-r(T-t)}\int_{\mathbb{R}} f\left(xe^{\left(r-\frac{\sigma^2}{2}\right)(T-t)+\sigma z\sqrt{T-t}}\right)\frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}}dz. \tag{6.37}$$

Proof. (Method 1) Consider a European claim given by Theorem (6.2.6) as $V_t = e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}(f(S_T)|\mathcal{F}_t)$. Then,

$$\begin{aligned} V_t = e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}(f(S_T)|\mathcal{F}_t) &= e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}\left[f\left(S_0 e^{\left(r-\frac{\sigma^2}{2}\right)T+\sigma W_T}\right)\Big|\mathcal{F}_t\right] \\ &= e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}\left[f\left(S_0 e^{\left(r-\frac{\sigma^2}{2}\right)t+\sigma W_t}e^{\left(r-\frac{\sigma^2}{2}\right)(T-t)+\sigma(W_T-W_t)}\right)\Big|\mathcal{F}_t\right] \\ &= e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}\left[f\left(S_t e^{\left(r-\frac{\sigma^2}{2}\right)(T-t)+\sigma(W_T-W_t)}\right)\Big|\mathcal{F}_t\right]. \end{aligned}$$

We note that S_t is $\mathcal{F}_t$-measurable and $e^{\left(r-\frac{\sigma^2}{2}\right)(T-t)+\sigma(W_T-W_t)}$ is $\mathcal{F}_t$-independent. As a result, V_t is a function of t and S_t in the form $V_t = F(t,S_t)$ with,

$$F(t,x) = e^{-r(T-t)}\int_{\mathbb{R}} f\left(xe^{\left(r-\frac{\sigma^2}{2}\right)(T-t)+\sigma z\sqrt{T-t}}\right)\frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}}dz. \tag{6.38}$$

Note that by integrating the payoff function multiplied by the above density we obtain a **regularization effect**. Furthermore, recall that $S_t = S_0\exp\left((r-\frac{\sigma^2}{2})t+\sigma W_t\right)$ for a $\mathbb{Q}$-standard Brownian

Motion so that (Itô Formula) $dS_t = rS_t dt + \sigma S_t dW_t$. By the Itô Formula on $F(t, S_t)$ we obtain,

$$dF(t,S_t) = \left(\frac{\partial F}{\partial t}(t,S_t) + rS_t\frac{\partial F}{\partial x}(t,S_t) + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 F}{\partial x^2}(t,S_t)\right)dt + \sigma S_t\frac{\partial F}{\partial x}(t,S_t)dW_t. \tag{6.39}$$

On the other hand, since ϕ is self-financing,

$$\begin{aligned} dV_t(\phi) = H_t^0 dS_t^0 + H_t dS_t = H_t^0 d(e^{rt}) + H_t dS_t &= H_t^0 re^{rt}dt + H_t(rS_t dt + \sigma S_t dW_t) \\ &= H_t^0 re^{rt}dt + H_t rS_t dt + H_t\sigma S_t dW_t \\ &= (H_t^0 e^{rt} + H_t S_t)rdt + \sigma S_t H_t dW_t. \end{aligned}$$

Now, since the drift and diffusion uniquely determine an Itô process in $F(t, S_t) = V_t(\phi)$, we in particular have that their diffusion must agree [more generally, decompositions of special semi-martingales are unique]. Thus, we obtain the so-called **Δ-hedging**,

$$H_t = \frac{\partial F}{\partial x}(t,S_t). \tag{6.40}$$

Hence, we may cleverly rewrite $F(t, S_t)$ using $\widetilde{S}_t = e^{-rt}S_t$ and $S_t^0 = e^{rt}$ as,

$$V_t(\phi) = F(t,S_t) = e^{rt}(e^{-rt}F(t,S_t) - H_t\widetilde{S}_t) + H_t S_t = S_t^0\underbrace{\left(e^{-rt}\left(F(t,S_t) - \frac{\partial F}{\partial x}(t,S_t)S_t\right)\right)}_{=H_t^0} + \underbrace{\frac{\partial F}{\partial x}(t,S_t)}_{=H_t}S_t.$$

As a result, the hedging strategy in the case of a European claim with maturity $T > 0$ defined by a random variable of the form $h = f(S_T)$ is given by,

$$\phi_t = (H_t^0, H_t) = \left(e^{-rt}\left(F(t,S_t) - \frac{\partial F}{\partial x}(t,S_t)S_t\right), \frac{\partial F}{\partial x}(t,S_t)\right). \tag{6.41}$$

□

Proof. (Method 2) Note that a replicating portfolio must have, at any time t, a discounted value equal to $M_t = \widetilde{V}_t = e^{-rt}V_t = e^{-rt}F(t,S_t)$ where F is defined as,

$$F(t,x) = e^{-r(T-t)}\int_{\mathbb{R}} f\left(xe^{\left(r-\frac{\sigma^2}{2}\right)(T-t)+\sigma z\sqrt{T-t}}\right)\frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}}dz. \tag{6.42}$$

If we set $\widetilde{F}(t,x) = e^{-rt}F(t,xe^{rt})$ we have,

$$\widetilde{V}_t = e^{-rt}F(t,S_t) = e^{-rt}F\left(t, \frac{S_t}{e^{rt}}e^{rt}\right) = \widetilde{F}(t,\widetilde{S}_t), \tag{6.43}$$

and for $t < T$ we get from the Itô Formula,

$$\widetilde{V}_t = \widetilde{F}(t,\widetilde{S}_t) = \widetilde{F}(0,\widetilde{S}_0) + \int_0^t \underbrace{\left(\frac{\partial \widetilde{F}}{\partial t}(u,\widetilde{S}_u) + \frac{1}{2}\sigma^2\widetilde{S}_u^2\frac{\partial^2\widetilde{F}}{\partial x^2}(u,\widetilde{S}_u)\right)}_{=K_u}du + \int_0^t \sigma\widetilde{S}_u\frac{\partial\widetilde{F}}{\partial x}(u,\widetilde{S}_u)dW_u.$$

The martingale property of $\widetilde{F}(t,\widetilde{S}_t) = \widetilde{V}_t = M_t$ under $\mathbb{Q}$ implies that $K_u = 0$. Hence,

$$\widetilde{V}_t = \widetilde{F}(t,\widetilde{S}_t) = \widetilde{F}(0,\widetilde{S}_0) + \int_0^t \sigma \widetilde{S}_u \frac{\partial \widetilde{F}}{\partial x}(u,\widetilde{S}_u) dW_u. \tag{6.44}$$

Now, for $\widetilde{F}(t,x) = e^{-rt}F(t,xe^{rt})$ we have that,

$$\frac{\partial \widetilde{F}}{\partial x}(t,x) = e^{-rt}\frac{\partial F}{\partial x}(t,xe^{rt})e^{rt} = \frac{\partial F}{\partial x}(t,xe^{rt}), \tag{6.45}$$

and as a result,

$$H_t = \frac{\partial \widetilde{F}}{\partial x}(t,\widetilde{S}_t) = \frac{\partial F}{\partial x}(t,\widetilde{S}_t e^{rt}) = \frac{\partial F}{\partial x}(t,S_t). \tag{6.46}$$

Therefore, $V_t(\phi) = e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}(h|\mathcal{F}_t)$ may be rewritten by,

$$\begin{aligned} V_t(\phi) = e^{rt}(e^{-rT}\mathbb{E}_{\mathbb{Q}}(h|\mathcal{F}_t) - H_t\widetilde{S}_t) + H_t S_t &= S_t^0(e^{-rt}(F(t,S_t) - H_t S_t)) + H_t S_t \\ &= S_t^0 \underbrace{\left(e^{-rt}\left(F(t,S_t) - \frac{\partial F}{\partial x}(t,S_t)S_t\right)\right)}_{=H_t^0} + \underbrace{\frac{\partial F}{\partial x}(t,S_t)}_{=H_t} S_t. \end{aligned}$$

As a result, the hedging strategy in the case of a European claim with maturity $T > 0$ defined by a random variable of the form $h = f(S_T)$ is given by,

$$\phi_t = (H_t^0, H_t) = \left(e^{-rt}\left(F(t,S_t) - \frac{\partial F}{\partial x}(t,S_t)S_t\right), \frac{\partial F}{\partial x}(t,S_t)\right). \tag{6.47}$$

For $\phi_t = (H_t^0, H_t)$ we have that ϕ is self-financing and that $V_t = F(t,S_t)$ [non-negative for $\mathbb{R}_+$-valued payoff functions f].

□

6.4 Analysis of the Model - Greeks and Implicit Volatility

The Black-Scholes Model assumes several factors: (i) log-normal diffusion, (ii) no bid-ask spreads, (iii) no taxes, (iv) no (other) transaction costs, (v) borrowing and lending rates are equal, (vi) underlying instruments (e.g. stocks) are unlimited divisible, (vii) own transactions have no influence on the price (very liquid markets), (viii) complete information which is reflected in the traded prices, etc... Recall that the Black-Scholes Model $S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t}$, i.e. $dS_t = S_t(\mu dt + \sigma dB_t)$, with a constant volatility σ has the weakness that really observable data tells us that σ is **NOT** a constant (volatility skew, volatility smile, non-constant volatility surface).
Some obvious attempts may be done to solve the problems, and we will see some of them in the last chapter of these notes. We will in particular discuss some models based on Brownian Motion consisting of the Exponential Integral Brownian Model and the Exponential Time-Changed Brownian Model.

6.4.1 The Greeks

For European call and put options, the Black-Scholes setting leads to a closed form formulae depending on $S_t, K, r, T-t$ and σ. In view of the hypotheses one could (and should) be interested in the effect obtained from variations in the parameters.

Remark 6.4.1. *An important remark is that the Black-Scholes formulae have been obtained under the hypothesis that the parameters are constant, i.e. this type of analysis is heuristic (with the exception of the derivation of the Δ).*

We now give the definitions of the Greeks that are part of the analysis of financial models, in terms of the pricing formula of the option V_t along with the Black-Scholes counterpart.

Definition 6.4.2. *The **Delta** corresponds to the partial derivative of the pricing formula V_t with respect to the underlying,*

$$\Delta := \frac{\partial V_t}{\partial S_t} \equiv \frac{\partial V_t}{\partial x}(T-t, x, K, r, \sigma)\Big|_{x=S_t}. \tag{6.48}$$

Definition 6.4.3. *The **Gamma** corresponds to the second partial derivative of the pricing formula V_t with respect to the underlying,*

$$\Gamma := \frac{\partial^2 V_t}{\partial S_t^2} \equiv \frac{\partial^2 V_t}{\partial x^2}(T-t, x, K, r, \sigma)\Big|_{x=S_t}. \tag{6.49}$$

Definition 6.4.4. *The **Theta** corresponds to $(-1)\times$ the partial derivative of the pricing formula V_t with respect to the time to maturity $\tau := T-t$,*

$$\Theta := -\frac{\partial V_t}{\partial \tau} \equiv -\frac{\partial V_t}{\partial s}(s, S_t, K, r, \sigma)\Big|_{s=\tau}. \tag{6.50}$$

Definition 6.4.5. *The **Rho** corresponds to the partial derivative of the pricing formula V_t with respect to the interest rate r,*

$$\rho := \frac{\partial V_t}{\partial r} \equiv \frac{\partial V_t}{\partial p}(T-t, S_t, K, p, \sigma)\Big|_{p=r}. \tag{6.51}$$

Definition 6.4.6. *The **Vega** corresponds to the partial derivative of the pricing formula V_t with respect to the volatility σ,*

$$\text{Vega} := \frac{\partial V_t}{\partial \sigma} \equiv \frac{\partial V_t}{\partial x}(T-t, S_t, K, r, x)\Big|_{x=\sigma}. \tag{6.52}$$

• In the case of a European call option, i.e. with $C_t = S_t\mathcal{N}(d_+) - Ke^{-r(T-t)}\mathcal{N}(d_-)$ and $d_\pm(S_t) = \frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{S_t}{K}\right) + \left(r \pm \frac{\sigma^2}{2}\right)(T-t)\right)$, it is not difficult (but tedious) to show the following properties for the sensitivities (the Greeks):

$$\Delta_C = \mathcal{N}(d_+) > 0. \tag{6.53}$$

$$\Gamma_C = \frac{1}{\sqrt{2\pi}}\frac{1}{S_t\sigma\sqrt{T-t}}e^{-\frac{1}{2}d_+^2} > 0. \tag{6.54}$$

$$\Theta_C = -\frac{S_t\sigma}{2\sqrt{2\pi(T-t)}}e^{-\frac{1}{2}d_+^2} - rKe^{-r(T-t)}\mathcal{N}(d_-) < 0. \tag{6.55}$$

$$\rho_C = K(T-t)e^{-r(T-t)}\mathcal{N}(d_-) > 0. \tag{6.56}$$

$$\text{Vega}_C = \frac{1}{\sqrt{2\pi}} S_t e^{-\frac{1}{2}d_+^2}\sqrt{T-t} > 0. \tag{6.57}$$

• In the case of a European put option, i.e. with $P_t = Ke^{-r(T-t)}\mathcal{N}(-d_-) - S_t\mathcal{N}(-d_+)$ and $d_\pm(S_t) = \frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{S_t}{K}\right) + \left(r \pm \frac{\sigma^2}{2}\right)(T-t)\right)$, it is not difficult (but tedious) to show the following properties for the sensitivities (the Greeks):

$$\Delta_P = \mathcal{N}(d_+) - 1. \tag{6.58}$$

$$\Gamma_P = \frac{1}{\sqrt{2\pi}} \frac{1}{S_t\sigma\sqrt{T-t}} e^{-\frac{1}{2}d_+^2}. \tag{6.59}$$

$$\Theta_P = rKe^{-r(T-t)}(1-\mathcal{N}(d_-)) - \frac{S_t\sigma}{2\sqrt{2\pi(T-t)}} e^{-\frac{1}{2}d_+^2}. \tag{6.60}$$

$$\rho_P = -K(T-t)e^{-r(T-t)}(1-\mathcal{N}(d_-)). \tag{6.61}$$

$$\text{Vega}_P = \frac{1}{\sqrt{2\pi}} S_t e^{-\frac{1}{2}d_+^2}\sqrt{T-t}. \tag{6.62}$$

A very useful formula to compute these quantities and for general financial applications is the relation between a put and a call option, given by the put-call parity (which we state without proof).

Proposition 6.4.7 (Put-call parity)**.** *The European put-call parity is given by:*

$$C_t - P_t = S_t - Ke^{-r(T-t)}. \tag{6.63}$$

6.4.2 Implicit Volatility

In this subsection, an option is said to be (i) **at the money** if the underlying is (around) the strike, (ii) **in the money** if the intrinsic value is positive, and (iii) **out of the money** if the intrinsic value is negative. The price of an option is often split in two parts: the **intrinsic value** would be obtained if the option would be exercised immediately and the **time value** is the difference between the price and the intrinsic value.

Recall that the partial derivative of the pricing formula with respect to the volatility σ for a European call is strictly positive. Hence the function $\sigma \mapsto C_t(S_t, K, T-t, r, \sigma)$ admits an inverse function.

In several markets, vanilla options have been *liquidly* traded over the last decades (e.g. European options written on certain indices). Hence, we can compute the **implicit volatility** "obtained from the market", i.e. the volatility we have to plug in the Black-Scholes formula in order to arrive at the market price. Let us now see how the value of a call option at time $t = 0$ behaves as the volatility $\sigma \to 0$ and as $\sigma \to \infty$. Let $F = S_0 e^{rt}$ and the corresponding call option be,

$$C_0 = S_0\mathcal{N}\left(\frac{\log\left(\frac{F}{K}\right)}{\sigma\sqrt{T}} + \frac{1}{2}\sigma\sqrt{T}\right) - Ke^{-rT}\mathcal{N}\left(\frac{\log\left(\frac{F}{K}\right)}{\sigma\sqrt{T}} - \frac{1}{2}\sigma\sqrt{T}\right). \tag{6.64}$$

We consider three options depending on the value of F (sign of the $\log(\cdot)$) for the case $\sigma \to 0$.

• $C_0 \xrightarrow[F>k]{\sigma\to 0} \frac{S_0}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2}dy - \frac{Ke^{-rT}}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2}dy = S_0 - Ke^{-rT}$ [**in the money** without volatility].

• $C_0 \xrightarrow[F=k]{\sigma\to 0} \frac{S_0}{\sqrt{2\pi}}\int_{-\infty}^{0} e^{-\frac{1}{2}y^2}dy - \frac{Ke^{-rT}}{\sqrt{2\pi}}\int_{-\infty}^{0} e^{-\frac{1}{2}y^2}dy = \frac{1}{2}(S_0 - Ke^{-rT})$ [**at the money** without volatility].

• $C_0 \xrightarrow[F<k]{\sigma\to 0} \frac{S_0}{\sqrt{2\pi}}\int_{-\infty}^{-\infty} e^{-\frac{1}{2}y^2}dy - \frac{Ke^{-rT}}{\sqrt{2\pi}}\int_{-\infty}^{-\infty} e^{-\frac{1}{2}y^2}dy = 0$ [**out of the money** without volatility].

We note that we have obtained the lower static arbitrage bound. For **in/at/out of the money** they are here used including discounted effects (slightly non-standard).

We now consider three options for the case $\sigma \to \infty$, where here we remark that $\mathcal{N}(\cdot)$ are independent of the sign of $\log\left(\frac{F}{K}\right)$. Thus, for all F,

$$C_0 \xrightarrow{\sigma\to\infty} \frac{S_0}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2}dy - \frac{Ke^{-rT}}{\sqrt{2\pi}}\int_{-\infty}^{-\infty} e^{-\frac{1}{2}y^2}dy = S_0 \quad [\textbf{upper static arbitrage bound}].$$

Recall that $\max\left(S_0 - Ke^{-rT}, 0\right) \leq C \leq S_0$. Hence, we have obtained the **lower static arbitrage bound** for $\sigma \to 0$ and the **upper static arbitrage bound** for $\sigma \to \infty$. Thus $Vega > 0$ as long as the market price of the European call is between the static arbitrage bounds, and we can find a unique implicit volatility. This idea can be used/seen as a simple case of model calibration to traded derivatives, i.e. we derive the market parameter σ from *liquidly* traded derivatives in order to price new, not yet *liquidly* traded derivatives (risk neutral setting).

Hence, we have two approaches in order to estimate the model parameters (in the Black-Scholes setting, uniquely the volatility):
- Statistic analysis, i.e. statistical approach based on historic data (time series in the real world).
- Model calibration to traded derivatives (risk neutral setting).
Recall that the volatility does not change if we change the measure by applying the Girsanov Theorem! Hence we should have $\widehat{\sigma} \approx \sigma_{\text{IMP}}$, however in the market we usually observe $\sigma_{\text{IMP}} >> \widehat{\sigma}$.
In view of this inconsistency, we have to choose which volatility we use (if we want to price and/or hedge on the basis of a GBM). Usually, the implicit volatility is used since:
- One wants to ensure that the model leads to prices being close to the market prices.
- $\widehat{\sigma}$ is obtained from the history, where σ_{IMP} is derived from the expectations of the market participants about the future, being reflected in the market quotes.

The implicit volatility is obtained by (numerical) **Black-Scholes inversion**. Hence σ_{IMP} is a function depending on other variables of the Black-Scholes formula, in particular the price of the underlying, the time to maturity and the strike.

$$C_t \mapsto \sigma_{\text{IMP}}(S_t, K, T-t) \tag{6.65}$$

We may fix the time(s) to maturity by fixing t. At t, S_t will be known, but often, we will have different maturity dates and for each maturity, different strike prices (5-10 strikes for a fixed maturity is easily possible). If the model would be consistent with the market data σ_{IMP} would be constant for all maturities and strikes. Unfortunately, this is not the case, σ_{IMP} varies depending on the time to maturity and also on the strike (leads to a complicated shape of the volatility surface).

6.4.3 Inverse Problems and Model Calibration

In this last subsection, we briefly describe the setup for inverse problems and model calibration mentioned at the end of the previous subsection.

Consider a risk-neutral model with parameters $\theta \in \Theta$-set of parameters (not necessarily only σ). Given observed prices $\{C_i\}_i$ of call options for maturities $\{T_i\}_i$ and strikes $\{K_i\}_i$, find θ such that the discounted asset price $(\widetilde{S}_t)$ is a martingale and observed option prices are given by their risk-neutral expectations, where for all $i \in I$,

$$C_i = e^{-rT_i}\mathbb{E}_{\mathbb{Q}^\theta}\big((S_{T_i} - K_i)_+\big). \tag{6.66}$$

This problem is usually ill-posed:
- No solution at all may exist.
- An infinite number of solutions may exist.
- Even with additional criterion to choose one solution, the dependence of the solution on input prices may be discontinuous. This implies a numerical instability of the calibration algorithm.

In these problems, regularization often helps. The overall idea is to find $\theta \in \Theta$ which minimizes a quantity of the form,

$$\sum_{i=1}^{N} \omega_i \big| e^{-rT_i}\mathbb{E}_{\mathbb{Q}^\theta}\big((S_{T_i} - K_i)_+\big) - C_i\big|^2 + F(\mathbb{Q}^\theta, \mathbb{P}_0, \alpha), \tag{6.67}$$

where $\{\omega_i\}_i$ are weights and $F(\mathbb{Q}^\theta, \mathbb{P}_0, \alpha)$ is an additional penalty function depending on θ, $\mathbb{P}_0$ and a parameter(s) α; subject to $\mathbb{Q}^\theta$ being equivalent to $\mathbb{P}_0$ (and a martingale measure). In this context, $\mathbb{P}_0$ is the estimated real-world measure and α is a parameter to fix the quantity to regularize.

Chapter 7

Theory of Arbitrage for Continuous Time Models

Recall that the FTAP roughly states that a model of a financial market is free of arbitrage if and only if there is a probability measure $\mathbb{Q}$, equivalent to the real-world measure $\mathbb{P}$ (i.e. $\mathbb{P}$ and $\mathbb{Q}$ vanish on the same events), such that the discounted asset price processes are martingales under $\mathbb{Q}$. The measure $\mathbb{Q}$ is then called the **equivalent martingale measure**. In this case, taking discounted expectations with respect to $\mathbb{Q}$ is order to price contingent claims yields the arbitrage-free pricing rules, where $\mathbb{Q}$ runs through all equivalent martingale measures.
Furthermore, if $\mathbb{Q}$ is unique then the discounted expected payoff (with respect to $\mathbb{Q}$) yields the unique arbitrage-free price. We have already proved this in the case where the underlying probability space is finite, however the FTAP has different formulations depending on the level of generality. This paves the way for using martingale methods in mathematical finance.

It turned out that in the continuous-time setting, a very simple no-arbitrage condition does **not** guarantee the existence of an equivalent martingale measure and not even of a more general equivalent local martingale measure. Due to this, not completely unreasonable modifications of the no-arbitrage property have been introduced by Delbaen and Schachermayer (1994) in order to guarantee at least the existence of an even more general equivalent σ-martingale measure. However, in recent works by Platen and other researchers indicate that modifications due to Delbaen and Schachermayer exclude some models with economically very interesting properties which are not excluded under other definitions of absence of arbitrage.

To characterize the ideas in continuous-time financial markets, we model uncertainty with a filtered probability space $\left(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in\mathbb{T}}, \mathbb{P}\right)$ where $\mathbb{T} = [0, T]$ for a fixed $T > 0$ or $\mathbb{T} = \mathbb{R}_+ = [0, \infty)$. In view of the possibility to embed different times sets into $\mathbb{R}_+$, the results in this section will be stated for $\mathbb{R}_+$. Furthermore, the process of traded asset prices $S = (S_{t0}, S_{t1}, ..., S_{tn})_{t\geq 0}$ where $(S_{t0})_{t\geq 0}$ often stands for a money market account, is usually assumed to be an $\mathbb{R}^{n+1}$-valued semimartingale defined over and adapted to the filtered probability space $\left(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P}\right)$.
The use of semimartingales in mathematical finance is justified by several reasons, one of which is that this class is wide enough to consider several situations and a second one is that for semimartingales the theory of stochastic integration is well developed which suits well for the construction of arbitrage theory.

7.1 Strategies and Arbitrage in Continuous Time

In this section, we present different types of strategies and give an account of arbitrage theory in context of continuous time models. For this, we will use some tools described in Section (3.4) including the vector stochastic integral formalism.

Definition 7.1.1. *A **self-financing strategy** π is a pair (x, H) where $x \in \mathbb{R}$ and a predictable process $H = (H_{t1}, ..., H_{tn})_{t\geq 0}$ is $\widetilde{S}$-integrable (i.e. there exists a vector stochastic integral $H \cdot \widetilde{S}$), where $\widetilde{S}$ stands for the discounted price processes of the risky-assets, i.e. $S_{t0} = B_t$ (assumed to be a.s. strictly positive) and for $t \geq 0$,*

$$\widetilde{S}_t = \left(\frac{S_{t1}}{B_t}, ..., \frac{S_{tn}}{B_t} \right). \tag{7.1}$$

Definition 7.1.2. *The **discounted capital process** of the strategy $\pi = (x, H)$ is given by,*

$$V_t^\pi = x + \left(H \cdot \widetilde{S}\right)_t, \quad t \geq 0 \tag{7.2}$$

where $\left(H \cdot \widetilde{S}\right)_t$ represents the discounted return of the strategy π over time $t \geq 0$.

Note that, for every $t \geq 0$, we also have,

$$V_t^\pi = \langle H_t, \widetilde{S}_t \rangle + H_{t0}, \tag{7.3}$$

where H_{t0} is the amount invested in asset 0 at time t (where $B_t/B_t = 1$ for every $t \geq 0$).

Definition 7.1.3. *The **cumulative (discounted) costs** up to time t incurred by a (not necessarily self-financing) strategy is,*

$$C_t = V_t^\pi - \left(H \cdot \widetilde{S}\right)_t. \tag{7.4}$$

In the self-financing case, $C_t = x$ for every t (i.e. a self-financing strategy is completely described by x and H, determining H_{0t} for every $t \geq 0$).

Definition 7.1.4. *A strategy $\pi = (x, H)$ realises **arbitrage** if:*
(i) $x = 0$.
(ii) There exists a constant $a \geq 0$ such that $\mathbb{P}(V_t^\pi \geq -a \text{ for all } t \geq 0) = 1$.
(iii) The limit $V_\infty^\pi := \lim_{t\to\infty} V_t^\pi$ exists $\mathbb{P}$-a.s.
(iv) $V_\infty^\pi \geq 0$ $\mathbb{P} - a.s.$
(v) $\mathbb{P}(V_\infty^\pi > 0) > 0$.

A model satisfies the **No-Arbitrage Condition (NA)** if such a strategy does **not** exist.

Definition 7.1.5. *A sequence of strategies $\pi_k = (x_k, H_k)$ realises* ***free lunch with vanishing risk*** *if for all $k \geq 0$:*
(i) $x_k = 0$.
(ii) There exists a constant $a_k > 0$ such that $\mathbb{P}(V_t^{\pi_k} \geq -a_k \text{ for all } t \geq 0) = 1$.
(iii) The limit $V_\infty^{\pi_k} := \lim_{t\to\infty} V_t^{\pi_k}$ exists $\mathbb{P}$-a.s.
(iv) $V_\infty^{\pi_k} \geq -\frac{1}{k}$ $\mathbb{P}-a.s.$
(v) There exists constants $\delta_1, \delta_2 > 0$ such that $\mathbb{P}(V_\infty^{\pi_k} > \delta_1) > \delta_2$ (for δ_1,δ_2 independent of k).

A model satisfies the **No Free Lunch with Vanishing Risk Condition (NFLVR)** if such a sequence of strategies does **not** exist.

Theorem 7.1.6 (First Fundamental Theorem of Asset Pricing - First FTAP)**.** *The following assertions are equivalent for an $\mathbb{R}^n$-valued semimartingale model $\widetilde{S}$ of a financial market:*
(i) There is a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ such that $\widetilde{S}$ is a σ-martingale under $\mathbb{Q}$.
(ii) $\widetilde{S}$ satisfies the NFLVR condition.

Due to a result by Ansel and Stricker (1994) we have that if the components of $\widetilde{S}$ are non-negative, NFLVR is equivalent to the existence of an equivalent local martingale measure (i.e. a measure $\mathbb{Q}$ under which $\widetilde{S}$ is a local martingale).

The so-called Second Fundamental Theorem of Asset Pricing roughly tells us that an arbitrage-free model is complete if the equivalent martingale measure is unique, where completeness means a possibility to hedge any contingent claim. It should be pointed out that all models employed in "classical" mathematical finance are free of arbitrage, but only a few of them are complete.

Definition 7.1.7. *A model of a market described by the semimartingale $\widetilde{S}$ is* ***complete*** *if for any bounded $\mathcal{F}$-measurable random variable Y one can find a strategy π such that:*
(i) For some constants a and b, $\mathbb{P}(a \leq V_t^\pi \leq b \text{ for all } t \geq 0) = 1$.
(ii) The limit $V_\infty^\pi := \lim_{t\to\infty} V_t^\pi$ exists $\mathbb{P}$-a.s.
(iii) $Y = V_\infty^\pi$ $\mathbb{P}$-a.s.

Definition 7.1.8. *Let $\widetilde{S} = (\widetilde{S}_{t1}, ..., \widetilde{S}_{tn})_{t\geq 0}$ be a semimartingale given on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$. One says that a local martingale M given on this probability space admits* $\widetilde{\boldsymbol{S}}$***-representation*** *if one can find an integrable, predictable process $H = (H_{t1}, ..., H_{tn})_{t\geq 0}$ such that for all $t \geq 0$,*

$$M_t = M_0 + \left(H \cdot \widetilde{S}\right)_t \quad a.s. \tag{7.5}$$

Theorem 7.1.9 (Second Fundamental Theorem of Asset Pricing - Second FTAP)**.** .
• *(a) Assume that the family of equivalent σ-martingale measures is non-empty. Then the following conditions are equivalent:*
(i) The model described by the semimartingale $\widetilde{S}$ is complete.
(ii) The family of equivalent σ-martingale measures consists of a single measure.

(iii) In the family of equivalent σ-martingale measures there exists a measure $\mathbb{Q}$ such that any local martingale M admits the $\widetilde{S}$-representation (with respect to this measure).

- *(b) Assume that the components of $\widetilde{S}$ are non-negative and that the family of equivalent local martingale measures is non-empty. Then the following conditions are equivalent:*
(i) The model described by the semimartingale $\widetilde{S}$ is complete.
(ii) The family of equivalent local martingale measures consists of a single measure.
(iii) In the family of equivalent local martingales there exists a measure $\mathbb{Q}$ such that any local martingale M admits the $\widetilde{S}$-representation (with respect to this measure).

As a result, in case of semimartingales with non-negative components, we in particular have that the existence of exactly one local martingale measure ensures completeness of the model. It should be emphasised that the Second FTAP does not hold if the vector stochastic integrals are replaced by the componentwise stochastic integrals. Furthermore, a model can be complete whereas the family of equivalent σ-martingales is empty.

7.2 Risk-Neutral Valuation

The results in the previous section may now be applied to construct the so-called *risk-neutral* valuation. We consider a finite time horizon $T > 0$ and a $\mathcal{F}_T$-measurable non-negative random variable C, hence a contingent claim, i.e. a payoff at time T but the amount to be paid may depend on the whole information contained in $\mathcal{F}_T$. Assume that $Y = B_T^{-1}C$ is $\mathbb{Q}$-integrable for some $\mathbb{Q}$ being equivalent to $\mathbb{P}$ such that all components of $\widetilde{S}$ are $\mathbb{Q}$-martingales with a put price,

$$P_t = B_t \mathbb{E}_{\mathbb{Q}}(B_T^{-1}C|\mathcal{F}_t), \quad t \in [0,T]. \tag{7.6}$$

If we consider $(P_t)_{t\in[0,T]}$ to be the price process of an asset, then the market extended with this asset is still free of arbitrage since the discounted price process of this asset is a $\mathbb{Q}$-martingale. Thus, $(P_t)_{t\in[0,T]}$ (satisfying $P_T = C$) is a candidate for a *fair* price process of C. However, the definition of *fair* depends on the choice of the equivalent martingale measure.

If there is an equivalent martingale measure, the corresponding market satisfies NFLVR implying NA. Furthermore, assume that in this arbitrage free market the equivalent martingale measure is unique in the class of equivalent local martingale measures. Then, for suitably defined (e.g. as in the Second FTAP) contingent claims, there exists a self-financing dynamic hedging strategy consisting of continuously trading in the assets (including the bond), having value,

$$B_t \mathbb{E}_{\mathbb{Q}}(B_T^{-1}C|\mathcal{F}_t), \quad \text{for all } t \in [0,T]. \tag{7.7}$$

In general, contingent claims which can be replicated in this way are often called **attainable**. The advantage of complete markets stems from the fact that it allows the pricing and hedging of contingent claims in a preference-independent way. However, completeness is a quite delicate property which is typically destroyed as soon as one considers even minor modifications of a basic complete model. For instance, we know that the classical Black-Scholes model based on a one-dimensional Geometric Brownian Motion, is complete, but becomes incomplete if the volatility is influenced by a second stochastic

factor or if one adds a jump component to the model. If one insists on the preference-free approach under incompleteness, one can study the range of possible prices for C which are consistent with the absence of arbitrage in a market.

Another approach is to introduce subjective criteria according to which strategies for the martingale measures are chosen. Hence, for a given semimartingale model satisfying NFLVR, the problem of pricing financial derivatives essentially boils down to choosing an appropriate equivalent martingale measure and then computing the conditional expectations,

$$B_t \mathbb{E}_{\mathbb{Q}}(B_T^{-1} C | \mathcal{F}_t), \quad \text{for all } t \in [0,T]. \tag{7.8}$$

Application of the FTAP I - We present a concise method to find an arbitrage-free pricing rule. Consider a claim C and let $(\widetilde{S}_{t1}, ..., \widetilde{S}_{tn})$ be a $\mathbb{Q}$-martingale with a put price,

$$P_t = S_t^0 \mathbb{E}_{\mathbb{Q}} \left(\frac{C}{S_T^0} \Big| \mathcal{F}_t \right), \quad t \in [0,T]. \tag{7.9}$$

Consider now the discounted put price defined by $\widetilde{P}_t = \mathbb{E}_{\mathbb{Q}} \left(\frac{C}{S_T^0} | \mathcal{F}_t \right)$. For all $t \geq s$, we use the tower property to find,

$$\mathbb{E}_{\mathbb{Q}}(\widetilde{P}_t | \mathcal{F}_s) = \mathbb{E}_{\mathbb{Q}} \left(\mathbb{E}_{\mathbb{Q}} \left(\frac{C}{S_T^0} \Big| \mathcal{F}_t \right) \Big| \mathcal{F}_s \right) = \mathbb{E}_{\mathbb{Q}} \left(\frac{C}{S_T^0} \Big| \mathcal{F}_s \right) = \widetilde{P}_s. \tag{7.10}$$

Thus, $(\widetilde{S}_{t1}, ..., \widetilde{S}_{tn}, \widetilde{P}_t)$ is a $\mathbb{Q}$-martingale (note that $\mathbb{Q}$ is fixed in the first equality). Thus, for all equivalent σ-martingale measures,

$$S_t^0 \mathbb{E}_{\mathbb{Q}} \left(\frac{C}{S_T^0} \Big| \mathcal{F}_t \right) \tag{7.11}$$

yields an arbitrage-free pricing rule.

During more than the last twenty-five years many different strategies for choosing an appropriate martingale measure have been developed, e.g. in the sense of minimising a distance from the measure $\mathbb{P}$ (entropic distance, L^2-distance, general f-distance, Hellinger distance, etc...) or in the sense of constructing the simplest possible measure (e.g. steaming from the Esscher transform). From the point of view of the practitioners, the choice of this measure should be the result of a calibration of the model to the market price of plain vanilla options (the corresponding inverse problems are often ill-posed so that regularization should be used). Applying an incomplete market model yields for many claims that there are no perfect hedging strategies by investing solely in the underlying and the cash-bond.

If certain derivatives of the underlyings are traded on the market, it is possible to hold a dynamic portfolio containing these instruments, typically consisting of cash-bonds, a position in the univariate underlying, and a collection of calls and puts. If one allows this dynamic trading in options, the market can be completed in some situations by adding a suitable collection of derivatives (e.g. European calls or variance swaps) to the liquidly traded underlying asset.

Chapter 8

Model Extensions

As on finite probability spaces, the theory on the FTAP in continuous time lead to an important but quite abstract theory about the absence of arbitrage and about complete markets, a notion being closely related to hedging. However, one of the key topics in both theory of mathematical finance and practice of financial trading consists in realizing which process simulates the prices S, respectively $\widetilde{S}$. In the discrete case, we have considered for example the trinomial model (incomplete market) and in particular the CRR model (complete market). On the other hand, in the continuous case, we have already discussed the Black-Scholes model (complete market) for the concrete behaviour of share price processes.

Recall that the Black-Scholes model is given by the SDE $dS_t = S_t(\mu dt + \sigma dB_t)$ for which the solution is $S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t}$. However in this case, the constant volatility σ has the weakness that really observable data tells us that σ is **not** a constant (i.e. volatility skew, volatility smile, non-constant volatility surface). A first order correction $\sigma \rightsquigarrow \sigma(t)$ was provided by R. Merton in 1973, while a second order correction $\sigma \rightsquigarrow \sigma(t, S_t)$ was provided by B. Dupire in 1994.

• More concretely, in the framework of models with non-stochastic volatility, one can obtain smile effects by assuming $\sigma \equiv \sigma(t, x)$ for a deterministic function σ. Then consider the SDE,

$$dS_t = \mu S_t dt + S_t \sigma(t, S_t) dB_t, \tag{8.1}$$

for which the advanced theory of existence and uniqueness of strong and weak solutions is developed. An annoying circumstance in this case is that the prices and the volatility turn out to be perfectly correlated and this contradicts statistical observations showing that the correlation should be negative but not equal to -1.

• We may go further into the generalization, where we may assume that the volatility depends not only on t and S_t but also on all preceding values S_u for $u \leq t$ (i.e. the history of the process) by considering the SDE,

$$dS_t = \mu S_t dt + S_t \sigma(t; S_u, u \leq t) dB_t, \tag{8.2}$$

where in this scenario, $\sigma \equiv \sigma(t; x_u, u \leq t)$. For these models, markets are complete and their study is not advanced because of analytical difficulties.

• Assume that the volatility to be itself *volatile*. In other words, that the volatility is generated by a source of randomness which is different from the driving Brownian Motion B. In the framework of Brownian Models, consider another Brownian Motion $\widetilde{B}$ such that for a suitable $\rho \in (-1,1)$,

$$d\langle B, \widetilde{B}\rangle_t = \rho dt. \tag{8.3}$$

As to the volatility $\sigma(t)$, it is usually convenient to assume that $\sigma(t) = f(Y_t)$ where $f(y)$ is a non-negative function (e.g. e^y or $\sqrt{|y|}$) and Y is a diffusion satisfying,

$$dY_t = a(t,Y_t)dt + b(t,Y_t)d\widetilde{B}_t. \tag{8.4}$$

Or, even one could assume that Y belongs to the larger class of Itô-processes,

$$dY_t = a(t,\omega)dt + b(t,\omega)d\widetilde{B}_t, \tag{8.5}$$

with $a(t,\omega)$ and $b(t,\omega)$ being $\mathcal{F}_t$-measurable functions for every t and such that a.s.,

$$\int_0^t |a(s,\omega)|ds < \infty, \qquad \int_0^t b^2(s,\omega)ds < \infty, \qquad t > 0. \tag{8.6}$$

However, some difficulty arises since the coefficients $a(t,\omega)$ and $b(t,\omega)$ must be adjusted to the behaviour of volatility, which is not directly observable. Nevertheless, indirect observations can allow one to make important conclusions about properties of volatility itself. Perhaps, the most important is to consider *mean reversion* (i.e. return of the process towards the mean) - we will study some examples later on.

8.1 Numerical Algorithms and Feynman-Kac

In this section we investigate some numerical methods for general SDEs along with the statement of the Feynman-Kac formula in different settings.

Definition 8.1.1. *An adapted measurable process (X_t) is said to have the **Markov property** if for all $t \geq s \geq 0$ and f-bounded measurable we have,*

$$\mathbb{E}\big(f(X_t)|\mathcal{F}_s\big) = \mathbb{E}\big(f(X_t)|\sigma(X_s)\big). \tag{8.7}$$

In this section, we consider an SDE of the form,

$$dX_t = \beta(t,X_t)dt + \gamma(t,X_t)dW_t \tag{8.8}$$

where the drift and diffusion are such that for a constant c they satisfy a Lipschitz condition in the second variable $|\beta(t,x)-\beta(t,y)| + |\gamma(t,x)-\gamma(t,y)| \leq c|x-y|$, and also a linear-growth bound condition $|\beta(t,x)| + |\gamma(t,x)| \leq c(1+|x|)$ for all $t \geq 0$ and $x, y \in \mathbb{R}$.
The problem at hand is to find a process (X_T) for $T \geq t \geq 0$ such that $X_t = x \in \mathbb{R}$ and

$$X_T = X_t + \int_t^T \beta(s,X_s)ds + \int_t^T \gamma(s,X_s)dW_s. \tag{8.9}$$

For h a Borel measurable function, denote by $g(t,x) = \mathbb{E}^{t,x}(h(X_T))$ where (X_T) is the solution to (8.8) with initial condition $X_t = x$. It is assumed that $\mathbb{E}^{t,x}(h(X_T)) < \infty$ for all x,t and $g(\cdot,\cdot)$ is a Borel measurable deterministic function. If we do not have an explicit expression of X_T, we could compute $g(t,x)$ numerically by beginning at $X_t = x$ and simulating the SDE.
One of the most simple way of simulating such an SDE is to use a *Euler method* which is a particular type of Monte Carlo method. The idea is to choose a small positive step size δ and set,

$$X_{t+\delta} = x + \beta(t,x)\delta + \gamma(t,x)\sqrt{\delta}\epsilon_1, \tag{8.10}$$

where $\epsilon_1 \sim \mathcal{N}(0,1)$. Then, for the step 2δ one has,

$$X_{t+2\delta} = X_{t+\delta} + \beta(t+\delta, X_{t+\delta})\delta + \gamma(t+\delta, X_{t+\delta})\sqrt{\delta}\epsilon_2, \tag{8.11}$$

with $\epsilon_2 \sim \mathcal{N}(0,1)$ such that ϵ_1 is independent of ϵ_2. Iterating this process for any δ, assuming that $\frac{T-t}{\delta} \in \mathbb{N}$, we can find the value for X_T. This gives a realisation of X_T, corresponding to one ω. Now repeat this process many times and compute the average of $h(X_T)$ over all these simulations to get an approximate value for $g(t,x)$. Note that if one would begin with a different time t and initial value x, one would get a different answer (i.e. the answer is a function of t and x; with the notation $\mathbb{E}^{t,x}$).

Proposition 8.1.2. *Let $(X_t)_{t\geq 0}$ be a solution of (8.8) with initial condition given at time $t = 0$. Then, for $0 \leq t \leq T$,*

$$\mathbb{E}(h(X_T)|\mathcal{F}_t) = g(t, X_t). \tag{8.12}$$

The problem of the Euler method is that this procedure of approximation converges slowly and gives the function value only for a pair (t,x). Hence, we translate the problem into a PDE.

Theorem 8.1.3 (Feynman-Kac Formula)**.** *Consider the SDE $dX_t = \beta(t,X_t)dt + \gamma(t,X_t)dW_t$. Let $h(y)$ be a Borel-measurable function. Fix $T > 0$ and let $t \in [0,T]$ be given. Define the function $g(t,x) = \mathbb{E}^{t,x}(h(X_T))$ where we assume that $\mathbb{E}^{t,x}(h(X_T)) < \infty$ for all t,x. Then, $g(t,x)$ satisfies the PDE,*

$$\frac{\partial g}{\partial t}(t,x) + \beta(t,x)\frac{\partial g}{\partial x}(t,x) + \frac{1}{2}\gamma^2(t,x)\frac{\partial^2 g}{\partial x^2}(t,x) = 0, \tag{8.13}$$

with terminal condition $g(T,x) = h(x)$ for all x.

Numerical algorithms for solving (8.13) converge quickly in the case of one-dimensional x considered here and gives the function $g(t,x)$ for all values of (t,x) simultaneously.

Theorem 8.1.4 (Discounted Feynman-Kac Formula)**.** *Consider the SDE $dX_t = \beta(t,X_t)dt + \gamma(t,X_t)dW_t$. Let $h(y)$ be a Borel-measurable function and r be a constant. Fix $T > 0$ and let $t \in [0,T]$ be given. Define the function $f(t,x) = \mathbb{E}^{t,x}\left(e^{-r(T-t)}h(X_T)\right)$ [i.e. discounted conditional expectation] where we assume that $\mathbb{E}^{t,x}(h(X_T)) < \infty$ for all t,x. Then, $f(t,x)$ satisfies the PDE,*

$$\frac{\partial f}{\partial t}(t,x) + \beta(t,x)\frac{\partial f}{\partial x}(t,x) + \frac{1}{2}\gamma^2(t,x)\frac{\partial^2 f}{\partial x^2}(t,x) = rf(t,x), \tag{8.14}$$

with terminal condition $f(T,x) = h(x)$ for all x.

Remark 8.1.5. *For a concrete set of assumptions for the three previous results see D.Duffie, Security Markets (1988). The details of the proofs are quite technical, we thus skip the proofs.*

8.2 Stochastic Volatility Models and Short Rates

In this section we investigate some examples of stochastic volatility models and explore short rates as stochastic processes.

• Assume that $S = (S_t)_{t\geq 0}$ satisfies $dS_t = \mu S_t dt + \sigma(t) S_t dB_t$ with $\sigma(t) = e^{Y_t}$ and where Y_t is defined by a Ornstein-Uhlenbeck process (with mean reversion),

$$dY_t = (\alpha - \beta Y_t)dt + \gamma d\widetilde{B}_t, \tag{8.15}$$

for $Y_0 = y \in \mathbb{R}$ and $\alpha, \beta, \gamma > 0$. Noting that $f(y) = e^y = \frac{\partial f}{\partial y} = \frac{\partial^2 f}{\partial y^2}$, Itô's formula yields,

$$d\sigma(t) = \sigma(t)\left(\alpha - \beta Y_t + \frac{1}{2}\gamma^2\right)dt + \sigma(t)\gamma d\widetilde{B}_t. \tag{8.16}$$

Thus, using (8.16) and $dS_t = \mu S_t dt + \sigma(t) S_t dB_t$ one has,

$$d\langle S, \sigma\rangle_t = \sigma(t)^2 S_t \gamma d\langle B, \widetilde{B}\rangle_t. \tag{8.17}$$

Finally, with (8.3) given by $d\langle B, \widetilde{B}\rangle_t = \rho dt$ we finally find,

$$d\langle S, \sigma\rangle_t = \sigma(t)^2 S_t \gamma \rho dt. \tag{8.18}$$

Hence, $\rho < 0$ is needed in order to ensure negative covariation between prices and volatility.

• Another application of the Ornstein-Uhlenbeck process is the so-called **Vasiček model** where,

$$dr_t = (\alpha - \beta r_t)dt + \sigma dW_t, \tag{8.19}$$

for (W_t) a standard Brownian Motion, and such that $S_t^0 = B_t = \exp\left(\int_0^t r_s ds\right)$.

• A more general model is defined in terms of k, θ, σ which are bounded Borel functions,

$$dr_t = k_t(\theta_t - r_t)dt + \sigma_t dW_t. \tag{8.20}$$

In the case where k is a positive bounded Borel function we have the **Hull-White model**. Using the method of Variation of Constants, the solution is (r_0-given),

$$r_t = e^{-K_t}\left(r_0 + \int_0^t e^{K_s} k_s \theta_s ds + \int_0^t e^{K_s}\sigma_s dW_s\right), \tag{8.21}$$

where $K_t = \int_0^t k_s ds$. Remark that (r_t) is a Gaussian process, and it is called the **short rate**. Let us now verify that (8.21) is indeed a solution of (8.20). For this, set $r_t = F(t, X_t) = e^{-K_t} X_t$ where $X_t = r_0 + \int_0^t e^{K_s} k_s \theta_s ds + \int_0^t e^{K_s}\sigma_s dW_s$. In other words, $dX_t = e^{K_t} k_t \theta_t dt + e^{K_t}\sigma_t dW_t$. Now, since the function of the process is $F(t,x) = e^{-K_t}x$ we compute the partial derivatives to find, $\frac{\partial F}{\partial x}(t,x) = e^{-K_t}$,

$\frac{\partial^2 F}{\partial x^2}(t,x) = 0$ and $\frac{\partial F}{\partial t}(t,x) = -k_t e^{-K_t} x$. We now apply the Itô formula and recover (8.20),

$$dr_t = \left(k_t\theta_t - k_t e^{-K_t} X_t\right) dt + \sigma_t dW_t = k_t(\theta_t - r_t)dt + \sigma_t dW_t. \tag{8.22}$$

• The Cox-Ingersoll-Ross model (CIR process) is also used for short rate modelling (note that r_t in this case has mean reversion and is non-negative):

$$dr_t = (\alpha - \beta r_t)dt + \sigma\sqrt{r_t}dW_t, \tag{8.23}$$

for a r_0 given and $\alpha, \beta, \sigma > 0$. The solution to this SDE may be approximated by Monte Carlo simulations and many of its properties can be determined even though we do not have an explicit formula for it. The choice of the CIR process as the driving process for the instantaneous volatility leads to closed form expressions for characteristic functions of various quantities of interest in this model enabling, for example, the use of Fourier transform methods for option pricing. The case where $\sigma(t) = \sqrt{Y_t}$ is the so-called **Heston model**, which is itself based on the CIR model.

Example 8.2.1 (Valuation with stochastic (r_t))**.** *Let X be a T-claim such that $B_t = S_t^0 = \exp\left(\int_0^t r_s ds\right)$ with $\mathbb{E}_{\mathbb{Q}}\left(\frac{|X|}{S_T^0}\right) < \infty$. The **most** important example is in the case where $X = 1$ (i.e. zero bond). The price process is given by,*

$$P(t,T) = S_t^0 \mathbb{E}_{\mathbb{Q}}\left(\frac{1}{S_T^0}\Big|\mathcal{F}_t\right) = \mathbb{E}_{\mathbb{Q}}\left(e^{-\int_t^T r_s ds}\Big|\mathcal{F}_t\right), \tag{8.24}$$

where $\left(e^{-\int_t^T r_s ds}\right)$ represents one random variable. For a more complex X,

$$P_t = S_t^0 \mathbb{E}_{\mathbb{Q}}\left(\frac{X}{S_T^0}\Big|\mathcal{F}_t\right), \tag{8.25}$$

where in this case, X and S_T^0 are two random variables. Now, $\frac{d\mathbb{Q}^T}{d\mathbb{Q}} = \frac{P(T,T)}{P(0,T)S_T^0} = \frac{1}{P(0,T)S_T^0}$ is assumed to be $\mathbb{Q}$-integrable. We further assume that $\left(\frac{P(t,T)}{P(0,T)S_t^0}\right)_t$ is a true martingale with respect to $\mathbb{Q}$. Now, for $t \leq T$,

$$Z_t = \frac{d\mathbb{Q}^T}{d\mathbb{Q}}\Big|_{\mathcal{F}_t} = \mathbb{E}_{\mathbb{Q}}\left(\frac{d\mathbb{Q}^T}{d\mathbb{Q}}\Big|\mathcal{F}_t\right) = \frac{P(t,T)}{P(0,T)S_t^0}, \tag{8.26}$$

where we have used the fact that $\mathbb{E}_{\mathbb{Q}}\left(\frac{d\mathbb{Q}^T}{d\mathbb{Q}}\Big|\mathcal{F}_t\right)$ is a martingale with respect to $\mathbb{Q}$ by the tower property. Thus, we have that,

$$P_t = S_t^0 \mathbb{E}_{\mathbb{Q}}\left(\frac{X}{S_T^0}\Big|\mathcal{F}_t\right) = S_t^0 \mathbb{E}_{\mathbb{Q}^T}\left(\frac{X}{S_T^0}\frac{Z_t}{Z_T}\Big|\mathcal{F}_t\right) = S_t^0 \mathbb{E}_{\mathbb{Q}^T}\left(\frac{X}{S_T^0}\frac{P(t,T)}{P(0,T)S_t^0}\frac{P(0,T)S_T^0}{P(T,T)}\Big|\mathcal{F}_t\right).$$

Thus, using the facts that $P(T,T) = 1$ and that $P(t,T)$ is $\mathcal{F}_t$-measurable we conclude,

$$P_t = P(t,T)\mathbb{E}_{\mathbb{Q}^T}(X|\mathcal{F}_t). \tag{8.27}$$

Thus, P_t factorises as the product of two single random variables.

8.3 Models based on Brownian Motion

Until now, we have mainly seen some standard models in Stochastic Calculus. However, we can define some more exotic ones operating in different ways.
As a first important case, one may consider an **exponential integral Brownian model** given by,

$$S_t = S_0 \exp\left(\int_0^t \mu_s ds + \int_0^t \sigma_s dB_s\right) \tag{8.28}$$

Another example is to consider an **exponential time-changed Brownian model** where we consider $(B_{T(t)})$ to be a time change in a (standard) Brownian Motion. Thus we may consider a model of the form,

$$S_t = S_0 \exp\left(\mu T(t) + B_{T(t)}\right). \tag{8.29}$$

In the later case, we may consider a filtered probability space (or a stochastic basis) $(\Omega, \mathcal{F}, (\mathcal{F}_\theta)_{\theta\geq 0}, \mathbb{P})$ satisfying the usual conditions. It is usually convenient, when defining the notion of **change of time**, to distinguish between the *old* (physical, calendar) θ-time and a *new* (operational, business) t-time. Then, the following definition is useful if we need to construct, starting from the initial process $X = (X_\theta)_{\theta\geq 0}$ adapted to the filtration $(\mathcal{F}_\theta)$, a new process $\widehat{X} = \left(\widehat{X}_t\right)_{t\geq 0}$ evolving in t-time and having certain desired properties.

Definition 8.3.1. *A family of random variables $T = (T(t))_{t\geq 0}$ is said to be a **random change of time** if:*
(a) $(T(t))_{t\geq 0}$ is a non-decreasing right-continuous family of $[0,\infty]$-valued random variables $T(t)$.
(b) For all $t \geq 0$, the random variables $T(t)$ are stopping times with respect to the filtration $(\mathcal{F}_\theta)_{\theta\geq 0}$, i.e. for $t, \theta \geq 0$,

$$\{T(t) \leq \theta\} \in \mathcal{F}_\theta. \tag{8.30}$$

One out of others exceptional roles of Brownian Motion is the possibility to represent a wide class of processes (in fact, semimartingales) X in the form $X \overset{\text{Law}}{=} B \circ T$ (perhaps on different probability spaces) for a certain Brownian Motion B and change of time T [see Monroe's Theorem (1978)].
A particularly nice subfamily of semimartingales is given by the well-known Lévy Processes, which we investigate in the next section.

8.4 Lévy Processes - An Introduction and some Properties

In this section, we will present a class of semimartingales known as Lévy Processes along with some properties. We will state the characteristic function of such a process and a representation into a continuous/discontinuous decomposition.

Definition 8.4.1. *A process (X_t) is said to have **independent increments** if $X_t - X_s$ is independent of $\mathcal{F}_s$ (i.e. $X_t - X_s \perp \mathcal{F}_s$) for all $0 \leq s < t < \infty$.*

Definition 8.4.2. *A process (X_t) is said to have **stationary increments** if the distribution of $X_t - X_s$ depends only on $t - s$ and thus $\text{Law}(X_{t+h} - X_{s+h}) = \text{Law}(X_t - X_s)$ for all $h > 0$.*

Definition 8.4.3. *A process (X_t) is said to be **continuous in probability** if for every $\epsilon > 0$ one has,*

$$\lim_{s \to t} \mathbb{P}(|X_s - X_t| > \epsilon) = 0. \tag{8.31}$$

Remark 8.4.4. *Note that if a process X has independent increments it does **not** imply that X is a semimartingale. However, X has the decomposition $X = D + S$ where D is a deterministic process (possibly of finite variation) and S is a semimartingale.*

Definition 8.4.5. *A **Lévy process** $L = (L_t)_{t \geq 0}$ is a càdlàg process with independent and stationary increments, which is also continuous in probability and is such that $L_0 = 0$.*

Note that, if (L_t) is Lévy process such that $S_t = S_0 e^{L_t}$ and assuming $\mathbb{E}_{\mathbb{Q}}(|f(S_T)|) < \infty$ then,

$$V_t = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left(f(S_T) | \mathcal{F}_t \right) = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left(f(S_0 e^{L_T}) | \mathcal{F}_t \right) = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left(f(S_0 e^{L_t} e^{L_T - L_t}) | \mathcal{F}_t \right) = \varphi(S_t),$$

where $\varphi(s) = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left(f(s e^{L_T - L_t}) \right)$ for all $s > 0$. Hence, this gives a nice framework for valuation properties. We now give, without proof, the characteristic function of a Lévy process.

Proposition 8.4.6 (Kolmogorov-Lévy-Khinchin's Formula)**.** *The characteristic function of a Lévy process (L_t) is given by,*

$$\mathbb{E}\left(e^{iuL_t} \right) = \exp\left(t \left[iub - \frac{u^2}{2} c + \int \left(e^{iux} - 1 - iuh(x) \right) F(dx) \right] \right), \tag{8.32}$$

where the classical truncation function is $h(x) = x \mathbb{1}_{\{|x| \leq 1\}}$ and:
- $F(dx)$ is a measure on $\mathbb{R} \backslash \{0\}$ such that $\int \min(1, x^2) F(dx) < \infty$.
- $b \in \mathbb{R}$ and $c \geq 0$.
- (b, c, F) is the Lévy triplet (or the triplet of local characteristics) of L, which characterizes the Lévy process.

Definition 8.4.7. *A **subordinator** is a real-valued stochastic process $T = (T_t)_{t \geq 0}$ that it non-negative and a Lévy process.*

In particular, let (T_t) be a subordinator with Lévy triplet $(\widetilde{b}, 0, \rho)$. As a result of Proposition (8.4.6), the moment generating function of (T_t) is $\mathbb{E}(e^{uT_t}) = e^{t\psi(u)}$ for all u such that the moment generating function exists and where,

$$\psi(u) = \widetilde{b} u + \int_0^\infty (e^{ux} - 1) \rho(dx), \tag{8.33}$$

such that $\rho((-\infty, 0]) = 0$, $\int_0^\infty \bullet \, \rho(dx) < \infty$ and $\widetilde{b} \geq 0$. Define now (Y_t) by $Y_t(\omega) = X(T_t(\omega), \omega)$. Then, for X, T- independent and $\mathbb{E}(e^{iuL_t}) = e^{t\phi(u)}$, one finds the following characteristic function for (Y_t),

$$\mathbb{E}\left(e^{iuY_t} \right) = \mathbb{E}\left(e^{iuX(T(t))} \right) = \mathbb{E}\left(\mathbb{E}\left(e^{iuX(T(t))} \middle| \mathcal{F}_\infty^T \right) \right) = \mathbb{E}\left(e^{T_t \phi(u)} \right) = e^{t\psi(\phi(u))}, \tag{8.34}$$

where $\mathbb{E}\left(e^{iuX(T(t))} | \mathcal{F}_\infty^T \right)$ is the characteristic function of $X(T_t)$ given T_t at u.

Let us now find a decomposition/representation for Lévy Processes. In the general case of a semimartingale $(X_t)_{t\geq 0}$ with $X_0 = 0$,

$$X_t - \sum_{s\leq t} \Delta X_s \mathbb{1}_{\{|\Delta X_s|>1\}}, \tag{8.35}$$

is a process with bounded jumps. This process is in fact a special semimartingale, and thus it has a unique decomposition into a local martingale and a predictable process of bounded variation $A+M$. Thus we find,

$$X_t = A_t + M_t + \sum_{s\leq t} \Delta X_s \mathbb{1}_{\{|\Delta X_s|>1\}}, \tag{8.36}$$

where all the terms depend on the truncation function $h(x) = x\mathbb{1}_{\{|x|\leq 1\}}$ and thus $x - h(x) = x\mathbb{1}_{\{|x|>1\}}$. Now, M_t may be decomposed further as $M_t = M_t^c + M_t^d$ where M_t^c is continuous and M_t^d is purely discontinuous (i.e. $M_t^d N_t$-local martingale for all N_t being continuous local martingales).

Remark 8.4.8. *It turns out that the continuous martingale component M_t^c of a semimartingale does not depend on the function $h(x)$.*

In the Lévy-case, $A_t = bt$ and $M_t^c = \sqrt{c}W_t$ where (W_t) is a standard Brownian Motion. Let μ be the random measure of jumps of the Lévy Process $L = (L_t)_{t\geq 0}$ such that $\mu(\omega; (0,t]\times A)$ counts how many jumps of size A occur for the path ω from 0 to t. Furthermore, let $\nu((0,t]\times A) = tF(A) = \mathbb{E}(\mu(\,\bullet\,;(0,t]\times A))$ where $F(A) = \int_A F(dx) = \mathbb{E}\left(\#\{t\in(0,1] : \Delta L_t \neq 0, \Delta L_t \in A\}\right)$ is the **Lévy measure** which counts the expected number of jumps per unit time whose size belongs to A (does not depend on the function $h(x)$). Here we have used the notation $\Delta L_s := L_s - L_{s-1}$. Thus, for the purely discontinuous (local) martingale M^d we find,

$$M_t^d = \int_0^t \int_{\mathbb{R}} x\mathbb{1}_{\{|x|\leq 1\}}(\mu(ds,dx) - dsF(dx)) \equiv \int_0^t \int_{\mathbb{R}} x\mathbb{1}_{\{|x|\leq 1\}} d(\mu-\nu). \tag{8.37}$$

Then we obtain,

$$L_t = bt + \sqrt{c}W_t + \int_0^t \int_{\mathbb{R}} x\mathbb{1}_{\{|x|\leq 1\}}(\mu(ds,dx) - dsF(dx)) + \underbrace{\int_0^t \int_{\mathbb{R}} x\mathbb{1}_{\{|x|>1\}}\mu(ds,dx)}_{\text{big jumps}} \tag{8.38}$$

We can thus regroup everything into a single proposition.

Proposition 8.4.9 (Lévy-Itô Representation)**.** *The representation for the trajectories of $L = (L_t)_{t\geq 0}$ is,*

$$L_t = bt + \sqrt{c}W_t + \int_0^t \int h(x)d(\mu-\nu) + \int_0^t \int (x-h(x))d\mu, \tag{8.39}$$

where:
- $A_t = bt$ and $L_t^c = \sqrt{c}W_t$, for (W_t) a standard Brownian Motion, are continuous components of L.
- μ is a measure of jumps: for $A\in\mathcal{B}(\mathbb{R}\backslash\{0\})$, $\mu(\omega;(0,t]\times A) = \sum_{0<s\leq t}\mathbb{1}_A(\Delta L_s)$ where $\Delta L_s = L_s - L_{s-1}$.
- ν is the compensator of the measure of jumps μ: for the Lévy measure $F(A) = \int_A F(dx)$,

$$\nu((0,t]\times A) = tF(A) = \mathbb{E}(\mu(\,\bullet\,;(0,t]\times A)). \tag{8.40}$$

Remark 8.4.10. *We can finally make the following four remarks:*
(i) The four terms in the decomposition are independent.
(ii) $F(dx)$ and c do not depend on the truncation.
(iii) The following convergence result holds a.s. and is uniform in $t \in [0,T]$,

$$M_t^d = \lim_{\epsilon \to 0} \left(\sum_{s \leq t} \Delta L_s \mathbb{1}_{\{\epsilon \leq |\Delta L_s| \leq 1\}} - t \int x \mathbb{1}_{\{\epsilon \leq |x| \leq 1\}} F(dx) \right). \tag{8.41}$$

(iv) The process representing the big jumps may be written as,

$$\int_\bullet^t \int_{\mathbb{R}} x \mathbb{1}_{\{|x|>1\}} \mu(ds,dx) = \sum_{s \leq t} \Delta L_s \mathbb{1}_{\{|\Delta L_s|>1\}}. \tag{8.42}$$

Example 8.4.11 (Examples of Lévy Processes)**.** *All the following are Lévy Processes:*
(i) Brownian Motion.
(ii) Poisson Process: a process $(N_t)_{t \geq 0}$ is a Poisson process with intensity parameter λ if it is a counting process with $N_0 = 0$ a.s., stationary and independent increments, where $N_t \sim Poi(\lambda t)$.
(iii) A Compound Poisson process defined by,

$$L_t = \sum_{k=0}^{N_t} \xi_k, \tag{8.43}$$

where $(N_t)_{t \geq 0}$ is a Poisson process, and $(\xi_k)_{k \geq 1}$ is a sequence of i.i.d. random variables and also independent of (N_t). Furthermore, $\xi_0 = 0$.

8.5 Hyperbolic Lévy Processes

In connection with Financial Econometrics, Hyperbolic Lévy processes are of great interest as they model well the really observable processes $H = (H_t)_{t \geq 0}$ for many underlying financial instruments (exchange rates, stocks, etc...). In this section, the aim will be to construct these processes.

For a Lévy process $(H_t)_{t \geq 0}$ we have that $\mathbb{E}(e^{iuH_t}) = \mathbb{E}(e^{iuH_1})^t$. The properties of Lévy's processes imply that the random variable $h = H_1$ is **infinitely divisible**, i.e. for any n one can find i.i.d. random variables $\xi_1, ..., \xi_n$ such that $\text{Law}(h) = \text{Law}(\xi_1 + ... + \xi_n)$. We will in particular look for infinitely divisible random variables h having the form,

$$h = \mu + \beta\sigma^2 + \sigma\epsilon, \tag{8.44}$$

where $\epsilon \sim \mathcal{N}(0,1)$, $\sigma = \sigma(\omega)$ is the volatility (independent of ϵ) for whose square, σ^2, we will construct a special distribution: the **Generalized Inverse Gaussian distribution - $\mathbb{G}$IG**. Strikingly, the distribution (on $\mathbb{R}_+$) is infinitely divisible and also the distribution of $h = \mu + \beta\sigma^2 + \sigma\epsilon$ (on $\mathbb{R}$) is infinitely divisible. Due to this fact, there exist Lévy processes $T = (T(t))_{t \geq 0}$ and $H^* = (H_t^*)_{t \geq 0}$ such that $\text{Law}(T(1)) = \text{Law}(\sigma^2)$ and $\text{Law}(H_1^*) = \text{Law}(h)$. As a realization of $H^* = (H_t^*)_{t \geq 0}$ one can take,

$$H_t = \mu t + \beta T(t) + B_{T(t)}, \tag{8.45}$$

where the time change $T = (T(t))_{t\geq 0}$ and the Brownian motion $B = (B_\theta)_{\theta\geq 0}$ are independent. In the sequel, we do not distinguish between the processes H and H^*. This process H, remarkable in many respect, bears the name **Generalized Hyperbolic Lévy process - L($\mathbb{G}$H)**.

8.5.1 Construction of the $\mathbb{G}$IG-Distributions for σ^2

Let $W = (W_t)_{t\geq 0}$ be a standard Brownian Motion. For $A \geq 0$ and $B > 0$ introduce,

$$T^A(B) = \inf\{s \geq 0 : As + W_s \geq B\}. \tag{8.46}$$

Proposition 8.5.1. *The density of $T^A(B)$ is given by,*

$$p_{T^A(B)}(s) := \frac{\partial}{\partial s}\mathbb{P}(T^A(B) \leq s) = \frac{B}{s\sqrt{2\pi s}} e^{-\frac{(B-As)^2}{2s}}. \tag{8.47}$$

Proof. For the purpose of this proof we will use the fact that, for $\widehat{W} = (\alpha t + W_t)_{t\in[0,T]}$ with (W_t) a standard Brownian Motion and setting $\widehat{M}_T = \sup_{t\in[0,T]} \widehat{W}_t$, the density $f(m,w)$ of $(\widehat{M}_T, \widehat{W}_T)$ is given by,

$$f(m,w) = \frac{2(2m-w)}{T\sqrt{2\pi T}} e^{\alpha w - \frac{1}{2}\alpha^2 T - \frac{1}{2T}(2m-w)^2} \mathbb{1}_{\{w\leq m\}} \mathbb{1}_{\{m>0\}}. \tag{8.48}$$

Thus, for $\alpha = A$,

$$\begin{aligned}
\mathbb{P}(\widehat{M}_T \leq B) &= \int_{-\infty}^{B}\int_{w^+}^{B} \frac{2(2m-w)}{T\sqrt{2\pi T}} e^{Aw - \frac{1}{2}A^2T - \frac{1}{2T}(2m-w)^2} dm dw \\
&= \int_{-\infty}^{B} \frac{2}{T\sqrt{2\pi T}} e^{Aw - \frac{1}{2}A^2T} \left(\int_{w^+}^{B} (2m-w) e^{-\frac{1}{2T}(2m-w)^2} dm \right) dw \\
&= \frac{1}{\sqrt{2\pi T}} \int_{-\infty}^{B} e^{Aw - \frac{1}{2}A^2T - \frac{1}{2T}w^2} dw - \frac{1}{\sqrt{2\pi T}} \int_{-\infty}^{B} e^{Aw - \frac{1}{2}A^2T - \frac{1}{2T}(2B-w)^2} dw.
\end{aligned}$$

Observe that for the first integral $-\frac{1}{2T}(w - AT)^2 = -\frac{1}{2T}w^2 + Aw - \frac{1}{2}A^2T$, and for the second integral $-\frac{1}{2T}((w-2B) - AT)^2 = [-\frac{1}{2T}(w-2B)^2 + Aw - \frac{1}{2}A^2T] - 2AB$. Thus, completing the square yields,

$$\mathbb{P}(\widehat{M}_T \leq B) = \frac{1}{\sqrt{2\pi T}} \int_{-\infty}^{B} e^{-\frac{1}{2T}(w-AT)^2} dw - \frac{e^{2AB}}{\sqrt{2\pi T}} \int_{-\infty}^{B} e^{-\frac{1}{2T}(w-2B-AT)^2} dw. \tag{8.49}$$

In the first integral, perform now the substitution $y = \frac{w-AT}{\sqrt{T}}$ so that $\left|\frac{dy}{dw}\right|^{-1} = \sqrt{T}$ and for the second integral perform the substitution $z = \frac{w-2B-AT}{\sqrt{T}}$ so that $\left|\frac{dz}{dw}\right|^{-1} = \sqrt{T}$. Thus,

$$\mathbb{P}(\widehat{M}_T \leq B) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{B-AT}{\sqrt{T}}} e^{-\frac{1}{2}y^2} dy - \frac{e^{2AB}}{\sqrt{2\pi}} \int_{-\infty}^{\frac{-B-AT}{\sqrt{T}}} e^{-\frac{1}{2}z^2} dz = \mathcal{N}\left(\frac{B-AT}{\sqrt{T}}\right) - e^{2AB}\mathcal{N}\left(\frac{-B-AT}{\sqrt{T}}\right).$$

Now, we have that,

$$\mathbb{P}(T^A(B) \leq s) = \mathbb{P}(\widehat{M}_s \geq B) = 1 - \mathbb{P}(\widehat{M}_s \leq B) = 1 + e^{2AB}\mathcal{N}\left(\frac{-B-As}{\sqrt{s}}\right) - \mathcal{N}\left(\frac{B-As}{\sqrt{s}}\right).$$

Hence, by differentiating the previous expression, we obtain the density,

$$\begin{aligned} p_{T^A(B)}(s) = \frac{\partial}{\partial s}\mathbb{P}(T^A(B) \leq s) &= \frac{\partial}{\partial s}\left(1 + \frac{e^{2AB}}{\sqrt{2\pi}}\int_{-\infty}^{\frac{-B-As}{\sqrt{s}}} e^{-\frac{1}{2}z^2}dz - \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\frac{B-As}{\sqrt{s}}} e^{-\frac{1}{2}y^2}dy\right) \\ &= \frac{e^{2AB}e^{-\frac{(B+As)^2}{2s}}}{\sqrt{2\pi}}\left(-\frac{A}{\sqrt{s}} - \frac{(-B-As)}{2s^{\frac{3}{2}}}\right) - \frac{e^{-\frac{(B-As)^2}{2s}}}{\sqrt{2\pi}}\left(-\frac{A}{\sqrt{s}} - \frac{(B-As)}{2s^{\frac{3}{2}}}\right) \\ &= \frac{B}{s\sqrt{2\pi s}}e^{-\frac{(B-As)^2}{2s}}. \end{aligned}$$

□

Now letting $b = B^2 > 0$ and $a = A^2 > 0$ we find, from the previous Proposition, the following formula for $p(s;a,b) \equiv p_{T^{\sqrt{a}}(\sqrt{b})}(s)$:

$$p(s;a,b) = \frac{\sqrt{b}e^{\sqrt{ab}}}{\sqrt{2\pi}}s^{-\frac{3}{2}}e^{-\frac{(as+b/s)}{2}}. \tag{8.50}$$

This density is in the form $p(s;a,b) = c_1(a,b)s^{\nu-1}e^{-\frac{(as+b/s)}{2}}$, where in our case $c_1(a,b) = \frac{\sqrt{b}e^{\sqrt{ab}}}{\sqrt{2\pi}}$ and $\nu = -\frac{1}{2}$. The distribution $p(s;a,b)$ is named **Inverse Gaussian distribution - IG**$(\mathbf{a},\mathbf{b})$.

The next important step for this situation, is to define the function,

$$p(s;a,b,\nu) = c_2(a,b,\nu)s^{\nu-1}e^{-\frac{(as+b/s)}{2}}, \tag{8.51}$$

where the parameters $a,b,\nu \in \mathbb{R}$ are chosen in such a way that $p(s;a,b;\nu)$ is a probability density on $\mathbb{R}_+$. The three sets of possibilities for those constants are given by: $\{a \geq 0, b > 0, \nu < 0\}$, $\{a > 0, b > 0, \nu = 0\}$ and $\{a > 0, b \geq 0, \nu > 0\}$ so that,

$$\int_0^\infty s^{\nu-1}e^{-\frac{(as+b/s)}{2}}ds < \infty. \tag{8.52}$$

It is well-known that $K_\nu(y) = \frac{1}{2}\int_0^\infty s^{\nu-1}e^{-\frac{y(s+1/s)}{2}}ds$ is the third-kind **Bessel function of order ν**, which for $y > 0$ solves,

$$y^2f''(y) + yf'(y) - (y^2+\nu^2)f(y) = 0. \tag{8.53}$$

The constant in (8.51) has the form $c_2(a,b,\nu) = \frac{(a/b)^{\nu/2}}{2K_\nu(\sqrt{ab})}$ and where the distribution on $\mathbb{R}_+$ with density,

$$p(s;a,b,\nu) = \frac{(a/b)^{\nu/2}}{2K_\nu(\sqrt{ab})}s^{\nu-1}e^{-\frac{(as+b/s)}{2}}, \tag{8.54}$$

for $s > 0$ bears the name **Generalized Inverse Gaussian distribution - $\mathbb{G}$IG**.

Proposition 8.5.2 (Important properties of $\mathbb{G}$IG-distributions (for σ^2) - without proof)**.** .
(i) This distribution is infinitely divisible.
(ii) The density $p(s;a,b,\nu)$ is unimodal (i.e. a density with a single peak) with mode (i.e. peak),

$$m = \begin{cases} \frac{b}{2(1-\nu)}, & \text{if } a = 0 \\ \frac{\nu-1+\sqrt{ab+(\nu-1)^2}}{a}, & \text{if } a > 0 \end{cases}. \tag{8.55}$$

(iii) The Laplace Transform $\mathcal{L}(\lambda) := \int_0^\infty e^{-\lambda s}p(s;a,b,\nu)ds$ is given by,

$$\mathcal{L}(\lambda) = \left(1+\frac{2\lambda}{a}\right)^{-\frac{\nu}{2}} \frac{K_\nu\left(\sqrt{ab(1+\frac{2\lambda}{a})}\right)}{K_\nu(\sqrt{ab})}. \tag{8.56}$$

Some important special cases of $\mathbb{G}$IG-distributions:

(i) For $a \geq 0$, $b > 0$ and $\nu = -\frac{1}{2}$ gives the **Inverse Gaussian distribution** $\mathbb{G}\text{IG}(a,b,-1/2) = \text{IG}(a,b)$. As we have seen, its density is given by,

$$p(s;a,b) = c_1(a,b)s^{-\frac{3}{2}}e^{-\frac{(as+b/s)}{2}} \tag{8.57}$$

where $c_1(a,b) = \sqrt{\frac{b}{2\pi}}e^{\sqrt{ab}}$. The associated density of the Lévy measure is given by,

$$f(y) = \sqrt{\frac{b}{2\pi}}\frac{e^{-\frac{ay}{2}}}{y^{\frac{3}{2}}}. \tag{8.58}$$

(ii) For $a > 0$, $b = 0$ and $\nu > 0$ gives the **Gamma distribution** $\mathbb{G}\text{IG}(a,0,\nu) = \text{Gamma}(a/2,\nu)$. Its density is given by,

$$p(s;a,0,\nu) = \frac{\left(\frac{a}{2}\right)^\nu}{\Gamma(\nu)}s^{\nu-1}e^{-\frac{as}{2}} \tag{8.59}$$

where $\Gamma(\nu)$ is the Gamma function evaluated at ν. The associated density of the Lévy measure is given by,

$$f(y) = y^{-1}\nu e^{-\frac{ay}{2}}. \tag{8.60}$$

(iii) For $a > 0$, $b > 0$ and $\nu = 1$ gives the **Positive Hyperbolic distribution - PH** (or the **$\mathbf{H}^+$-distribution**). Its density is given by,

$$p(s;a,b,1) = \frac{\sqrt{\frac{a}{b}}}{2K_1(\sqrt{ab})}e^{-\frac{(as+b/s)}{2}}. \tag{8.61}$$

Now, since a $\mathbb{G}$IG-distribution is infinitely divisible, we have that if one takes it as the distribution of σ^2, $\text{Law}(\sigma^2) = \mathbb{G}\text{IG}$, then one may construct a non-negative and non-decreasing Lévy process $T = (T(t))_{t\geq 0}$ (a subordinator) such that,

$$\text{Law}(T(1)) = \text{Law}(\sigma^2) = \mathbb{G}\text{IG}. \tag{8.62}$$

In the subsequent constructions, this process plays the role of: change of time, operational time and business time. As we explained above, the next step is the construction of the (normal) log of the normalized asset price process $H = (H_t)_{t\geq 0}$.

Proposition 8.5.3. *For the variable $h = \mu + \beta\sigma^2 + \sigma\epsilon$ where $\epsilon \sim \mathcal{N}(0,1)$, and from the independence of σ^2 and ϵ, the distribution of h is given by:*

$$p_h(x) = \int_0^\infty \frac{1}{\sqrt{2\pi y}}\exp\left(-\frac{\left(x-(\mu+\beta y)\right)^2}{2y}\right)p_{\mathbb{G}IG}(y)dy. \tag{8.63}$$

Proof. We find the characteristic function of $h = \mu + \beta\sigma^2 + \sigma\epsilon$, remembering that $\sigma^2 \sim \mathbb{G}\text{IG}$ is a random variable and also using the σ-measurability,

$$\begin{aligned}
\mathbb{E}(e^{iuh}) = \mathbb{E}\left(e^{iu(\mu+\beta\sigma^2+\sigma\epsilon)}\right) &= \mathbb{E}\left(\mathbb{E}\left(e^{iu(\mu+\beta\sigma^2+\sigma\epsilon)}\Big|\sigma\right)\right) \\
&= \mathbb{E}_{\sigma^2\sim\mathbb{G}\text{IG}}\left(\int_{\mathbb{R}} e^{iux}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{\left(x-(\mu+\beta\sigma^2)\right)^2}{2\sigma^2}\right)dx\right) \\
&= \int_0^\infty\int_{\mathbb{R}} e^{iux}\frac{1}{\sqrt{2\pi y}}\exp\left(-\frac{\left(x-(\mu+\beta y)\right)^2}{2y}\right)p_{\mathbb{G}\text{IG}}(y)dxdy \\
&= \int_{\mathbb{R}} e^{iux}\left(\int_0^\infty \frac{1}{\sqrt{2\pi y}}\exp\left(-\frac{\left(x-(\mu+\beta y)\right)^2}{2y}\right)p_{\mathbb{G}\text{IG}}(y)dy\right)dx \\
&=: \int_{\mathbb{R}} e^{iux}p_h(x)dx.
\end{aligned}$$

From the characteristic function of h, we read off the desired density $p_h(x)$ for h.

□

By denoting $p_h(x)$ as $p^*(x;a,b,\mu,\beta,\nu)$, the density of h may also be written as:

$$p^*(x;a,b,\mu,\beta,\nu) = c_3(a,b,\beta,\nu)\frac{K_{\nu-\frac{1}{2}}\left(\alpha\sqrt{b+(x-\mu)^2}\right)}{\left(\sqrt{b+(x-\mu)^2}\right)^{\frac{1}{2}-\nu}}e^{\beta(x-\mu)}, \tag{8.64}$$

where $\alpha := \sqrt{a+\beta^2}$ and $c_3(a,b,\beta,\nu) := \frac{\left(\frac{a}{b}\right)^{\frac{\nu}{2}}\alpha^{\frac{1}{2}-\nu}}{\sqrt{2\pi}K_\nu(\sqrt{ab})}$. The distribution given by Law(h) with density $p^*(x;a,b,\mu,\beta,\nu)$ bears the name **Generalized Hyperbolic distribution - $\mathbb{G}\text{H}$=$\mathbb{G}\text{H}(a,b,\mu,\beta,\nu)$**.

Proposition 8.5.4 (Important properties of $\mathbb{G}$H-distribution (for h) - without proof)**.** .
(i) This distribution is infinitely divisible.
(ii) If $\beta = 0$, then the distribution is unimodal (i.e. a density with a single peak) with mode (i.e. peak) $m = \mu$. In general, m is determined as a root of a certain transcendental equation.
(iii) The Lévy-Itô representation is known. It contains no centered Gaussian term and the Lévy measure has a (quite complicated) density.

Some important special cases of $\mathbb{G}$H-distributions:

(I) For $a \geq 0$, $b > 0$ and $\nu = -\frac{1}{2}$ gives the **Inverse Gaussian distribution** $\mathbb{G}\text{IG}(a,b,-1/2) = \text{IG}(a,b)$. The corresponding $\mathbb{G}$H-distribution is commonly named as a **Normal Inverse Gaussian - $\mathbb{N}\circ\mathbf{IG}$**.

(II) For $a > 0$, $b = 0$ and $\nu > 0$ gives the **Gamma distribution** $\mathbb{G}\text{IG}(a,0,\nu) = \text{Gamma}(a/2,\nu)$. The corresponding $\mathbb{G}$H-distribution is commonly named as a **Normal Gamma distribution - $\mathbb{N}\circ$Gamma** or also by **VG-distribution - $\mathbb{V}\mathbf{G}$** (Variance Gamma distribution).

(III) For $a > 0$, $b > 0$ and $\nu = 1$ gives the **Positive Hyperbolic distribution - PH** (or the **$\mathbf{H}^+$-distribution**). The corresponding $\mathbb{G}$H-distribution is commonly named as a **Normal Positive Hyperbolic distribution - $\mathbb{N}\circ\mathbf{H}^+$** or also by $\mathbb{H}$.

8.5.2 Construction of Lévy Processes

Having $\mathbb{G}$IG-distributions for σ^2 and $\mathbb{G}$H-distributions for h, we can turn to the construction of the Lévy process $H = (H_t)_{t\geq 0}$ used for modelling the prices $S_t = S_0 e^{H_t}$, $t \geq 0$. There are two possibilities:

(i) The fact that h is an infinitely divisible distribution allows one to construct, using the general theory, the Lévy process $H^* = (H_t^*)_{t\geq 0}$ such that,

$$\text{Law}(H_1^*) = \text{Law}(h). \tag{8.65}$$

(ii) Using the constructed process $T = (T(t))_{t\geq 0}$, one forms the process $H = (H_t)_{t\geq 0}$,

$$H_t = \mu t + \beta T(t) + B_{T(t)}, \tag{8.66}$$

where the Brownian Motion B and the process T are taken to be independent.

The processes $H = (H_t)_{t\geq 0}$ bears the name **Generalized Hyperbolic Lévy processes - L($\mathbb{G}$H)**. In the special cases of the $\mathbb{G}$H-distributions mentioned above (i.e. (I), (II) and (III)), they take the following special names: (I) L($\mathbb{N}\circ$IG)-process, (II) L($\mathbb{N}\circ$Gamma)-process or L($\mathbb{V}$G)-process, and (III) L($\mathbb{N}\circ \text{H}^+$)-process or L(H)-process. It is interesting to note that L($\mathbb{N}\circ$IG) and L($\mathbb{N}\circ$Gamma) have the important property that: Law(H_t) belongs to the same type of distributions as Law(H_1).

Concluding remarks to L($\mathbb{G}$H) and some other Lévy processes

- Densities of distributions of h ($= H_1$) are determined by five parameters (a, b, μ, β, ν) that gives a great freedom in determining parameters which would fit well the empirical data.
- The approach via independently time-changing Brownian motions has advantages related to simulation.
- In statistics there exist other methods in order to construct densities of distributions which would also fit well distributions of empirical data. The density $p^*(x; a, b, \mu, \beta, \nu)$ of $\mathbb{G}$H-distribution of (consecutively built) variables $h = \mu + \beta\sigma^2 + \sigma\epsilon$ has the essential advantage that $\mathbb{G}$H-distributions are **infinitely divisible** which enables us to construct processes $H = (H_t)_{t\geq 0}$ which describe adequately the time dynamics of logarithmic return of the prices $S = (S_t)_{t\geq 0}$.

Of course, also other well-known Lévy processes are applied in Mathematical Finance. Among them:

- The classical **CGMY** (Carr, Geman, Madan, Yor) model, where we again have no centered Gaussian term and the Lévy measure (describing the jump parts) has a density given by,

$$f(x) = \frac{C}{|x|^{1+Y}} e^{-G|x|} \mathbb{1}_{\{x<0\}} + \frac{C}{|x|^{1+Y}} e^{-M|x|} \mathbb{1}_{\{x>0\}}, \tag{8.67}$$

with $C, G, M > 0$ and $Y < 2$. For $Y = 0$ we obtain a different parametrization of a L($\mathbb{V}$G)-process.

- The **Meixner** process, again without centered Gaussian term and the Lévy measure has again a density given by,

$$f(x) = \delta \frac{\exp\left(\frac{\beta x}{\alpha}\right)}{x \sinh\left(\frac{\pi x}{\alpha}\right)}, \tag{8.68}$$

where $\alpha, \delta > 0$, and $-\pi < \beta < \pi$.

Remark 8.5.5. *As a final remark, it is important to stress that when working with exponential Lévy models, integrability of the asset price is not always guaranteed. A corresponding criteria for the existence of the first (and other) exponential moments can, for example, be derived from K. Sato, Lévy Processes and Infinitely Divisible Distributions (1999) [Theorem 25.17].*